



V93000 Specifications

V93000 SOC Device Power Supplies and VI Cards

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DC Scale Device Power Supplies and VI cards

Further information

For more information about the Advantest V93000 SoC Series, please visit www.advantest.com.

Contact information

For more information about the Advantest V93000 SoC cards, please contact your local Advantest sales representative.

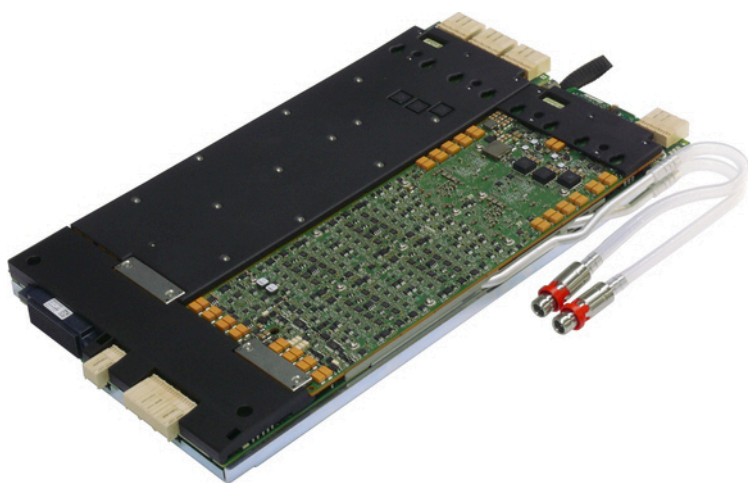
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DC Scale AVI64 Universal Pin VI (E8025AVI)

The AVI64 is a general purpose V/I card that features analog and high-voltage digital capabilities and is optimized for providing a true universal analog pin, covering a wide range of test application needs. The card belongs to the DC Scale family and can reside in any channel card slot.

Scope of specifications

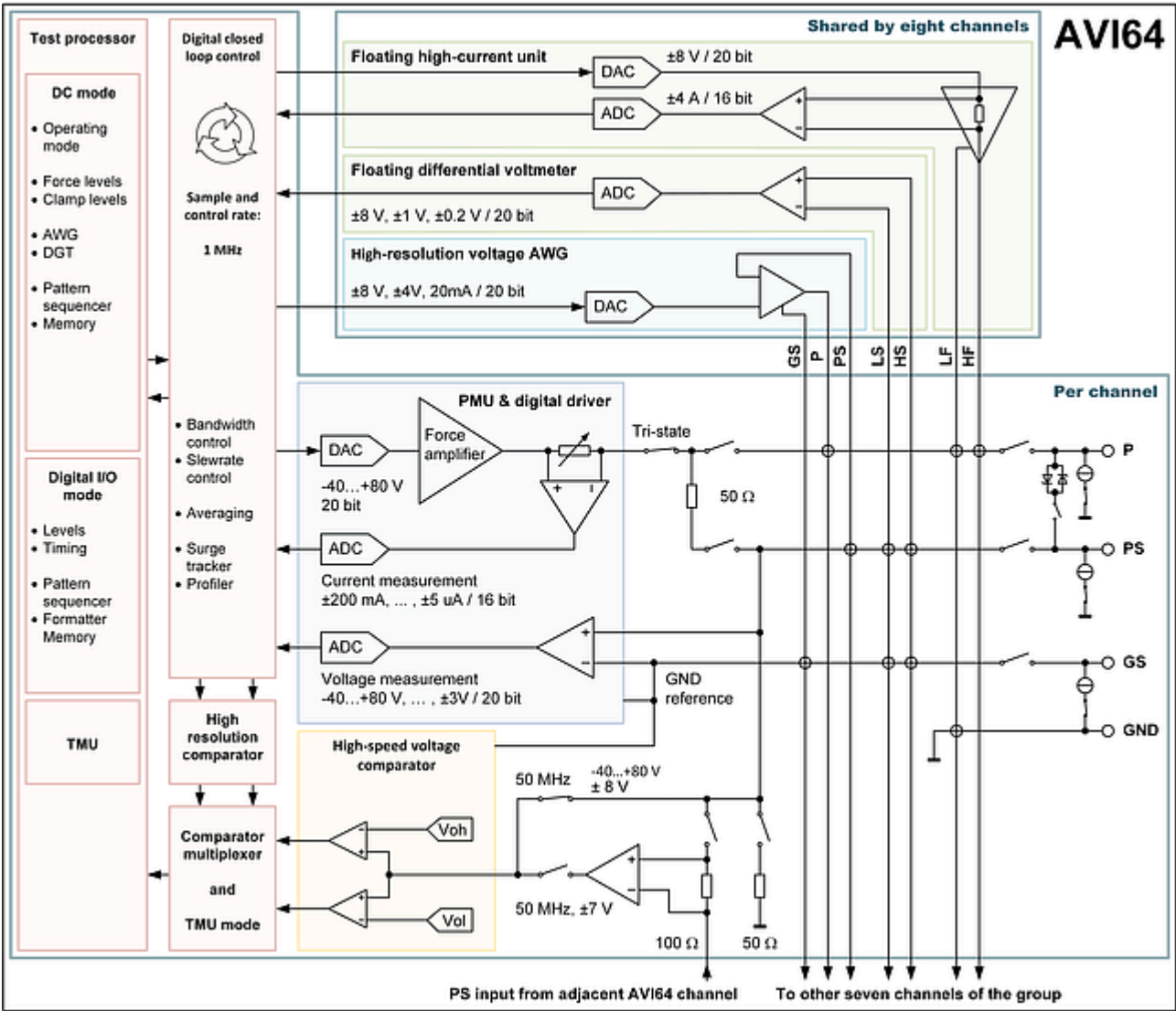
Advantest specifies and verifies the specifications at the DUT interface pogo level. Using an adequate DUT board and valid system calibration, these specifications are valid at the device under test as well. Specifications describe warranted product performance. Characteristics are included as typical values to provide additional useful information by describing typical non-warranted performance.



AVI64 card photo

AVI64 block diagram and configuration

Block diagram



Configuration

Number of VI channels per card	64 (8 groups, 8 channels each)			
Maximum source / sink current per card	Voltage ranges		Maximum current per card in range	
	-40 V ... +80 V		0.8 A	
	±30 V		3.2 A	
	±10 V (±3 V)		12.8 A ^[1]	
	Maximum total current for any given range is fully independent of total current used for other voltage ranges on the same card.			
Maximum number of cards and channels	A -test head	Compact test head	Small test head	Large test head

^[1] Total current for all channels in both ranges.

Configuration				
Maximum number of cards per system	8	16	32	64
Maximum number of VI channels per system	512	1024	2048 ^[2]	4096

Other configuration data

- AVI64 is slot compatible to AVI64, DPS64, DPS128, PVI8, and Pure Clock cards.
- Diagnostics requires the Diagnostics Interconnect Boards E7010E, E7010F, or E7010MUX.

AVI64 PMU voltage force and measurement specifications

Voltage force ^{[3][4]}	Range ^[5]	Resolution	Accuracy
	±3 V	10 µV	± (25 µV + 0.003 % of relative setting) ^{[6][7]} ± (100 µV + 0.003 % of setting) ^[6] ± (500 µV + 0.01 % of setting)
	±10 V	25 µV	± (80 µV + 0.003 % of relative setting) ^{[6][7]} ± (300 µV + 0.003 % of setting) ^[6] ± (1.0 mV + 0.01 % of setting)
	±30 V	70 µV	± (350 µV + 0.01 % of relative setting) ^{[6][7]} ± (3.5 mV + 0.01 % of setting)
	-40 V ... +80 V	200 µV	± (700 µV + 0.01 % of relative setting) ^{[6][7]} ± (10 mV + 0.01 % of setting)

^[2] Using more than 1024 channels requires the DUT Scale Interface.

^[3] GND sense can be ±5 V around system ground. GND sense voltage reduces usable voltage range except for ±3 V range.

^[4] Maximum allowed voltage drop on force line: ±1 V.

^[5] Voltage ranges can be selected independent of programmed current range.

^[6] DC update (SmarTest 7) / Enhanced Accuracy Adjustment (SmarTest 8) every 7 days.

^[7] Specifications are valid within 6 hours. Requires hardware revision 008 or later.

Voltage measurement ^[3]	Range ^[8]	Resolution	Accuracy ^[9]
	±3 V	10 µV	± (25 µV + 0.003 % of relative reading) ^{[10][11]} ± (100 µV + 0.003 % of reading) ^[10] ± (500 µV + 0.01 % of reading)
	±10 V	25 µV	± (80 µV + 0.003 % of relative reading) ^{[10][11]} ± (300 µV + 0.003 % of reading) ^[10] ± (1.0 mV + 0.01 % of reading)
	±30 V	70 µV	± (350 µV + 0.01 % of relative reading) ^{[10][11]} ± (3.5 mV + 0.01 % of reading)
	-40 V ... +80 V	200 µV	± (700 µV + 0.01 % of relative reading) ^{[10][11]} ± (10 mV + 0.01 % of reading)
Current clamp	Range	Resolution	Accuracy ^[12]
	±200 mA	10 µA	± (500 µA + 0.2 % of setting + 1 µA / V * V _{act})
	±50 mA	2 µA	± (50 µA + 0.2 % of setting + 0.2 µA / V * V _{act})
	±5 mA	200 nA	± (5 µA + 0.2 % of setting + 20 nA / V * V _{act})
	±500 µA	20 nA	± (0.5 µA + 0.2 % of setting + 2 nA / V * V _{act})
	±50 µA	2 nA	± (50 nA + 0.2 % of setting + 1 nA / V * V _{act})
	±5 µA	200 pA	± (10 nA + 0.2 % of setting + 1 nA / V * V _{act})
n channels ganged ^[13]	±n * 200 mA	n * 10 µA	Sum of single channel errors

^[8] Voltage measurement range is controlled by voltage force settings (or voltage clamp, if active).

^[9] Requires 1000 averaging samples.

^[10] DC update (SmarTest 7) / Enhanced Accuracy Adjustment (SmarTest 8) every 7 days.

Specifications are for averaging over 1 power line cycle.

^[11] Specifications are valid for measurements taken within 6 hours. Requires hardware revision 008 or later.

^[12] V_{act} is the actual voltage of the channel, which is either the programmed force voltage or the voltage forced by the DUT, depending on the mode the AVI64 channel is in (forcing or clamping).

^[13] Unlimited ganging of all channels on a single card requires software version 8.1.0 or later. Otherwise, ganging of up to 8 consecutive channels in same channel group is supported by software.

AVI64 PMU current force and measurement specifications

Current force	Range ^[14]	Resolution	Accuracy ^[15]
	±200 mA	10 µA	$\pm (100 \mu\text{A} + 0.05 \% \text{ of setting} + 1 \mu\text{A} / V * V_{\text{act}})$
	±50 mA	2 µA	$\pm (25 \mu\text{A} + 0.05 \% \text{ of setting} + 0.2 \mu\text{A} / V * V_{\text{act}})$
	±5 mA	200 nA	$\pm (2.5 \mu\text{A} + 0.05 \% \text{ of setting} + 20 \text{ nA} / V * V_{\text{act}})$
	±500 µA	20 nA	$\pm (0.25 \mu\text{A} + 0.05 \% \text{ of setting} + 2 \text{ nA} / V * V_{\text{act}})$
	±50 µA	2 nA	$\pm (25 \text{ nA} + 0.05 \% \text{ of setting} + 1 \text{ nA} / V * V_{\text{act}})$
	±5 µA	200 pA	$\pm (5 \text{ nA} + 0.05 \% \text{ of setting} + 1 \text{ nA} / V * V_{\text{act}})$
n channels ganged ^[16]	±n * 200 mA	n * 10 µA	Sum of single channel errors

Current measurement	Range ^[17]	Resolution	Accuracy ^{[15][18]}
	±200 mA	10 µA	$\pm (100 \mu\text{A} + 0.05 \% \text{ of reading} + 1 \mu\text{A} / V * V_{\text{act}})$
	±50 mA	2 µA	$\pm (25 \mu\text{A} + 0.05 \% \text{ of reading} + 0.2 \mu\text{A} / V * V_{\text{act}})$
	±5 mA	200 nA	$\pm (2.5 \mu\text{A} + 0.05 \% \text{ of reading} + 20 \text{ nA} / V * V_{\text{act}})$
	±500 µA	20 nA	$\pm (0.25 \mu\text{A} + 0.05 \% \text{ of reading} + 2 \text{ nA} / V * V_{\text{act}})$
	±50 µA	2 nA	$\pm (25 \text{ nA} + 0.05 \% \text{ of reading} + 1 \text{ nA} / V * V_{\text{act}})$
	±5 µA	200 pA	$\pm (5 \text{ nA} + 0.05 \% \text{ of reading} + 1 \text{ nA} / V * V_{\text{act}})$
n channels ganged ^[16]	±n * 200 mA	n * 10 µA	Sum of single channel errors

Voltage clamp	Range	Resolution	Accuracy
	±3 V	10 µV	$\pm (3 \text{ mV} + 0.1 \% \text{ of setting})$
	±10 V	25 µV	$\pm (10 \text{ mV} + 0.1 \% \text{ of setting})$

^[14] Current ranges can be selected independent of programmed voltage range.

^[15] V_{act} is the actual voltage of the channel, which is either the programmed force voltage or the voltage forced by the DUT, depending on the mode the channel is in (forcing or clamping).

^[16] Unlimited ganging of all channels on a single card requires software version 8.1.0 or later. Otherwise, ganging of up to 8 consecutive channels in same channel group is supported by software.

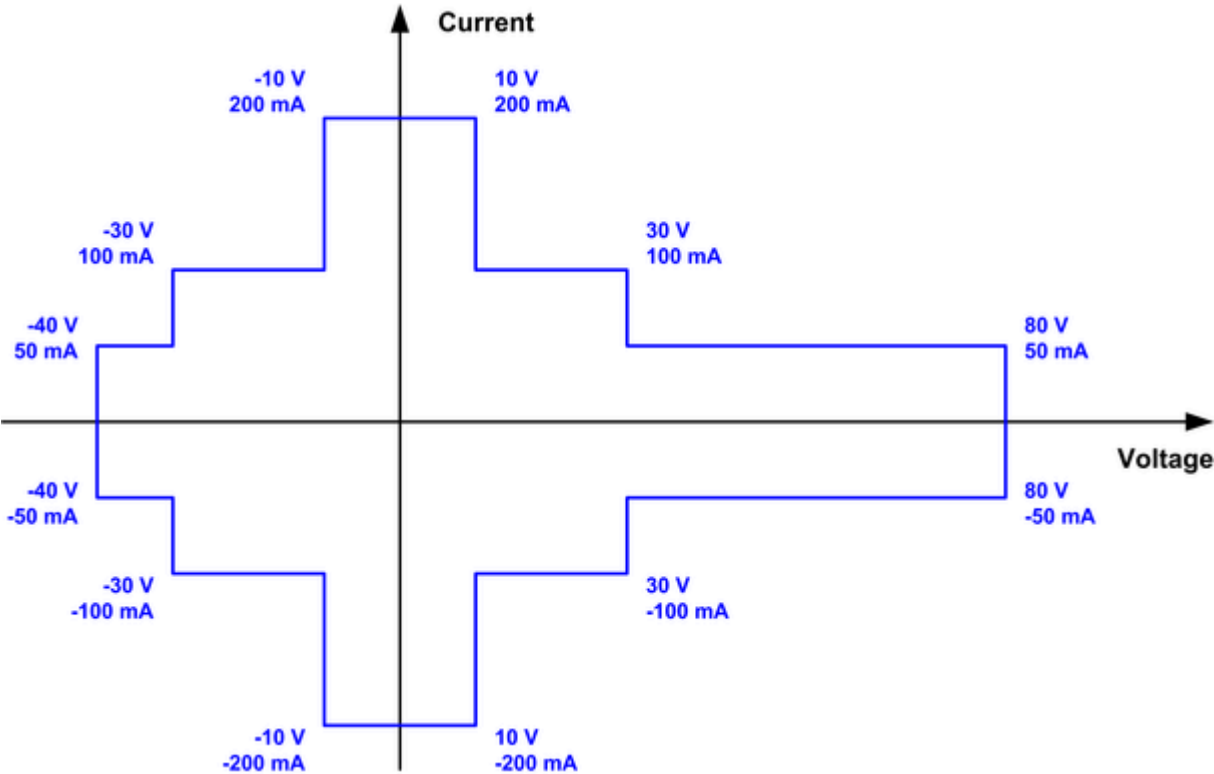
^[17] Current measurement range is controlled by current force settings (or current clamp, if active).

^[18] Requires 1000 averaging samples.

Voltage clamp	Range	Resolution	Accuracy
	±30 V	70 µV	± (30 mV + 0.1 % of setting)
	-40 V ... +80 V	200 µV	± (80 mV + 0.1 % of setting)

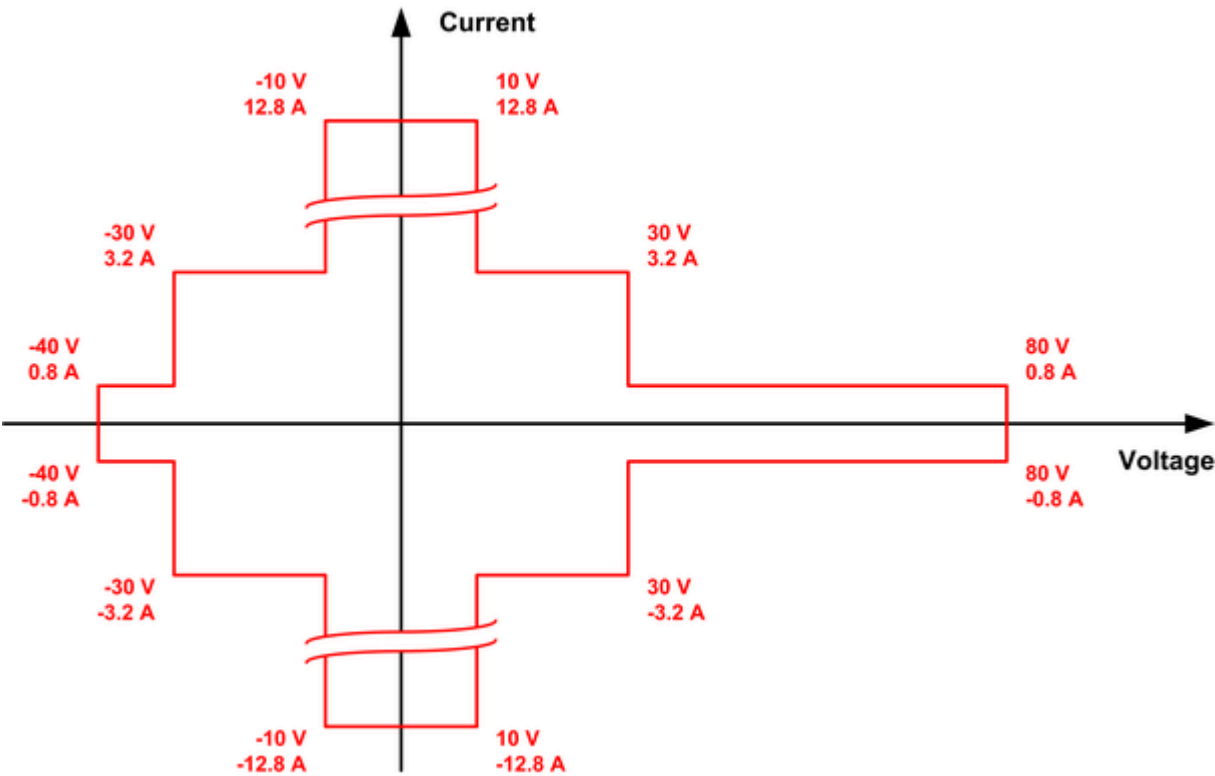
AVI64 PMU voltage and current capability specifications

Voltage and current capabilities per channel



AVI64 PMU voltage and current capability per channel

Maximum current per card and voltage range



AVI64 power budget per card and voltage range

AVI64 digitizer specifications

Per-pin voltage digitizer ^[19]				
Voltage ranges	± 3 V	± 10 V	± 30 V	-40 V ... +80 V
Resolution	20 bit (10 µV)	20 bit (25 µV)	20 bit (70 µV)	20 bit (200 µV)
DC accuracy ^[20]	±(0.5 mV + 100 ppm of reading)	±(1 mV + 100 ppm of reading)	±(3.5 mV + 100 ppm of reading)	±(10 mV + 100 ppm of reading)
INL (typical)	± 24 µV (± 4 ppm FSR)	± 40 µV (± 2 ppm FSR)	± 500 µV (± 8 ppm FSR)	± 1 mV (± 8 ppm FSR)
DNL (typical)	± 1 ppm FSR			
Ground sense range ^[21]	± 5 V ^[22]			
Input impedance (typical)	> 1 GΩ			
Input capacitance (typical)	165 pF			

^[19] Voltage and current digitizer are available for every channel and can be used in parallel. Software selectable range settings are the same as for PMU modes.

^[20] 1 kHz digital filter.

^[21] Independent GND sense per channel.

^[22] GND sense voltage reduces usable voltage range except for ± 3 V range.

Per-pin voltage digitizer^[19]

Sampling rate	0 ... 1 Msps (million samples per second)		
Analog bandwidth (typical, -3 dB)	250 kHz		
Filters	Programmable digital filter 100 Hz ... 250 kHz		
Waveform capture memory (per pin)	Software configurable, not exceeding total available memory size per channel (107 MByte) ^[23] , 32 Bytes required per sample.		
Noise (typical)		± 3 V range	± 10 V range
	22 Hz ... 22 kHz	30 μV_{RMS}	50 μV_{RMS}
	22 Hz ... 100 kHz	35 μV_{RMS}	150 μV_{RMS}
	22 Hz ... 250 kHz	45 μV_{RMS}	180 μV_{RMS}
SNR (typical)	1 kHz ^[24]	98 dB @ 5 V _{PP}	94 dB @ 8 V _{PP}
	10 kHz ^[25]	94 dB @ 5 V _{PP}	89 dB @ 8 V _{PP}
	20 kHz ^[25]	93 dB @ 5 V _{PP}	88 dB @ 8 V _{PP}
THD (typical)	1 kHz ^[26]	-105 dB @ 5 V _{PP}	-102 dB @ 8 V _{PP}
	10 kHz ^[27]	-98 dB @ 5 V _{PP}	-92 dB @ 8 V _{PP}
	20 kHz ^[27]	-97 dB @ 5 V _{PP}	-92 dB @ 8 V _{PP}

Current digitizer^[19]

Current ranges ^[28]	± 5 μA , ± 50 μA , ± 500 μA , ± 5 mA, ± 50 mA, ± 200 mA		
Resolution	16 bit		
Sampling rate	0 ... 1 Msps (million samples per second)		
Analog bandwidth (-3 dB)	250 kHz (typical)		
Programmable digital filter	100 Hz to 250 kHz		
Waveform capture memory (per pin)	Software configurable, not exceeding total available memory size per channel (107 MByte) ^[23] , 32 Bytes required per sample.		
DC Accuracy	Range ^[29]	Resolution	Accuracy ^{[20][30]}

^[19] Voltage and current digitizer are available for every channel and can be used in parallel. Software selectable range settings are the same as for PMU modes.

^[23] Memory pooling is supported for every eight channels in the same channel group.

^[24] 22 kHz digital filter, bandwidth 22 Hz ... 22 kHz, 1 Msps, initial discard 120 samples.

^[25] 100 kHz digital filter, bandwidth 22 Hz ... 100 kHz, 1 Msps, initial discard 30 samples.

^[26] 22 kHz digital filter, H2 ... H5, 1 Msps, initial discard 120 samples.

^[27] 100 kHz digital filter, H2 ... H5, 1 Msps, initial discard 30 samples.

^[28] Current measure range is used in digitizer mode.

^[29] Current measurement range is controlled by current force settings (or current clamp, if active).

Current digitizer^[19]

	±200 mA	10 µA	$\pm (100 \mu\text{A} + 0.05 \% \text{ of reading} + 1 \mu\text{A} / V * V_{\text{act}})$
	±50 mA	2 µA	$\pm (25 \mu\text{A} + 0.05 \% \text{ of reading} + 0.2 \mu\text{A} / V * V_{\text{act}})$
	±5 mA	200 nA	$\pm (2.5 \mu\text{A} + 0.05 \% \text{ of reading} + 20 \text{ nA} / V * V_{\text{act}})$
	±500 µA	20 nA	$\pm (0.25 \mu\text{A} + 0.05 \% \text{ of reading} + 2 \text{ nA} / V * V_{\text{act}})$
	±50 µA	2 nA	$\pm (25 \text{ nA} + 0.05 \% \text{ of reading} + 1 \text{ nA} / V * V_{\text{act}})$
	±5 µA	200 pA	$\pm (5 \text{ nA} + 0.05 \% \text{ of reading} + 1 \text{ nA} / V * V_{\text{act}})$
n channels ganged ^[31]	±n * 200 mA	n * 10 µA	Sum of single channel errors

AVI64 AWG specifications**Per-pin voltage AWG**

The voltage and current AWG are available for every channel. Range settings are software selectable.

Voltage ranges^[32] ±3 V, ±10 V, ±30 V, -40 V ... +80 V

Resolution 20 bit

DC Accuracy See Voltage force accuracy

Maximum current See Voltage / Current capabilities

Ground sense range^[33] ± 5 V^[34]

Sampling rate > 0 ... 1 Msps (million samples per second)

Analog bandwidth (typical) 20 kHz

Waveform memory (per pin) SmarTest 7: 1 Msamples. Waveform memory shared between all AWG modes and DC test points.
SmarTest 8: Waveform memory shared between all actions. Eight bytes are required per sample, the total memory size available per channel (107 Mbytes) must not be exceeded.

	± 3 V range	± 10 V range
Stability (typical)	± 18 µV (± 3 ppm FSR)	± 40 µV (± 2 ppm FSR)
INL (typical)	± 30 µV (± 5 ppm FSR)	± 100 µV (± 5 ppm FSR)
DNL (typical)	± 0.5 ppm FSR	± 0.5 ppm FSR

^[30] V_{act} is the actual voltage of the channel, which is either the programmed force voltage or the voltage forced by the DUT, depending on the mode the channel is in (forcing or clamping).

^[31] Unlimited ganging of all channels on a single card requires software version 8.1.0 or later. Otherwise, ganging of up to 8 consecutive channels in same channel group is supported by software.

^[32] Voltage force range is used in AWG mode.

^[33] Independent GND sense per channel.

^[34] GND sense voltage reduces usable voltage range except for ± 3 V range.

Per-pin voltage AWG

Noise (typical)	22 Hz ... 22 kHz ^[35]	420 μV_{RMS}	450 μV_{RMS}
	22 Hz ... 100 kHz ^[35]	680 μV_{RMS}	700 μV_{RMS}
SNR (typical)	1 kHz ^[36]	65 dB @ 5 V _{PP}	70 dB @ 8 V _{PP}
THD (typical)	1 kHz ^[37]	-60 dBc @ 5 V _{PP}	-65 dBc @ 8 V _{PP}

Per-pin current AWG

The voltage and current AWG are available for every channel. Range settings are software selectable.

Current ranges ^[38]	$\pm 5 \mu\text{A}$, $\pm 50 \mu\text{A}$, $\pm 500 \mu\text{A}$, $\pm 5 \text{ mA}$, $\pm 50 \text{ mA}$, $\pm 200 \text{ mA}$
Resolution	16 bit
Accuracy	See Current force accuracy
Sampling rate	0 ... 1 Msps (million samples per second)
Analog bandwidth (typical)	20 kHz (50 μA ... 200 mA range)
Waveform memory (per pin)	SmarTest 7: 1 Msamples. Waveform memory shared between all AWG modes and DC test points. SmarTest 8: Waveform memory shared between all actions. Eight bytes are required per sample, the total memory size available per channel (107 Mbytes) must not be exceeded.

High-speed voltage AWG

The high-speed voltage AWG is available for every channel. Range settings are software selectable.

Voltage ranges	$\pm 8 \text{ V}$, $\pm 30 \text{ V}$, $-40 \text{ V} \dots +80 \text{ V}$
Resolution	14 bit (1 mV @ $\pm 8 \text{ V}$, 10 mV @ $\pm 30 \text{ V}$, 10 mV @ $-40 \text{ V} \dots +80 \text{ V}$)
Accuracy	$\pm (50 \text{ mV} + 0.1 \% \text{ of setting})$ ^[39]
Sampling rate	0 ... 10 Msps (million samples per second)
Analog bandwidth (typical)	1 MHz ^[39]
Maximum current	200 mA @ $\pm 8 \text{ V}$, 100 mA @ $\pm 30 \text{ V}$, 50 mA @ $-40 \text{ V} \dots +80 \text{ V}$
Source impedance (typical)	10 Ω / 50 Ω ^{[39][40]}

^[35] 200 mA current range selected.

^[36] Bandwidth 22 Hz ... 22 kHz, 1000 ksps.

^[37] H2 ... H5, 1000 ksps

^[38] Current force range is used in AWG mode.

^[39] Valid in 200 mA range.

^[40] 10 Ω when connected over force line (P#), 50 Ω when connected over sense line (PS#).

High-speed voltage AWG

Waveform memory (per pin) SmarTest 7: 1 Msamples. Waveform memory shared between all AWG modes and DC test points.

SmarTest 8: Waveform memory shared between all actions. Eight bytes are required per sample, the total memory size available per channel (107 Mbytes) must not be exceeded.

High-resolution voltage AWG

The high-resolution voltage AWG is available once for every group of eight channels. It can be enabled by software for any channel of the group and replaces the PMU of the channel when enabled.

		± 4 V range	± 8 V range
Resolution		20 bit (10 μ V)	20 bit (20 μ V)
Accuracy		$\pm (50 \mu\text{V} + 30 \text{ ppm of relative setting})^{[41][42]}$	$\pm (80 \mu\text{V} + 30 \text{ ppm of relative setting})^{[41][42]}$
		$\pm (100 \mu\text{V} + 30 \text{ ppm of setting})^{[41]}$	$\pm (200 \mu\text{V} + 30 \text{ ppm of setting})^{[41]}$
		$\pm (150 \mu\text{V} + 100 \text{ ppm of setting})$	$\pm (300 \mu\text{V} + 100 \text{ ppm of setting})$
Stability (typical)		$\pm 10 \mu\text{V} (\pm 1.25 \text{ ppm FSR})$	$\pm 20 \mu\text{V} (\pm 1.25 \text{ ppm FSR})$
Maximum current		$\pm 20 \text{ mA}$	$\pm 20 \text{ mA}$
Ground sense range ^[33]		$\pm 0.5 \text{ V}$	$\pm 0.5 \text{ V}$
INL (typical)		$\pm 32 \mu\text{V} (\pm 4 \text{ ppm FSR})$	$\pm 64 \mu\text{V} (\pm 4 \text{ ppm FSR})$
DNL (typical)		$\pm 1 \text{ ppm FSR}$	$\pm 1 \text{ ppm FSR}$
Noise (typical)	22 Hz ... 22 kHz	10 μV_{RMS}	10 μV_{RMS}
	22 Hz ... 100 kHz	20 μV_{RMS}	20 μV_{RMS}
SNR (typical)	50 / 60 Hz ^[43]	114 dB @ 7 V _{pp}	117 dB @ 14 V _{pp}
	1 kHz ^[44]	106 dB @ 7 V _{pp}	112 dB @ 14 V _{pp}
	1 kHz ^[44]	110 dB @ 7 V _{pp}	116 dB @ 14 V _{pp}
	(EIAJ A-weighted)		
	10 kHz ^[45]	98 dB @ 7 V _{pp}	103 dB @ 14 V _{pp}
	20 kHz ^[45]	98 dB @ 7 V _{pp}	98 dB @ 14 V _{pp}

^[41] DC update (SmarTest 7) / Enhanced Accuracy Adjustment (SmarTest 8) every 7 days.

^[42] Specifications are valid within 6 hours. Requires hardware revision 008 or later.

^[43] 1.5 kHz hardware filter selected, bandwidth 22 Hz ... 2 kHz, 10 ksps.

^[44] 1.5 kHz hardware filter selected, bandwidth 22 Hz ... 22 kHz, 200 ksps.

^[45] 25 kHz hardware filter selected, bandwidth 22 Hz ... 100 kHz, 1000 ksps.

High-resolution voltage AWG

THD	50 / 60 Hz ^[46]	-110 dBc @ 7 V _{pp}	-113 dBc @ 14 V _{pp}
(typical)	1 kHz ^[47]	-96 dBc @ 7 V _{pp}	-94 dBc @ 14 V _{pp}
	10 kHz ^[48]	-73 dBc @ 7 V _{pp}	-70 dBc @ 14 V _{pp}
	20 kHz ^[48]	-73 dBc @ 7 V _{pp}	-64 dBc @ 14 V _{pp}
Sampling rate	0 ... 1 Msps (million samples per second)		
Analog bandwidth (typical)	125 kHz		
Selectable hardware filters	1.5 kHz / 25 kHz / bypass		
Waveform memory (per pin)	SmarTest 7: 1 Msamples. Waveform memory shared between all AWG modes and DC test points. SmarTest 8: Waveform memory shared between all actions. Eight bytes are required per sample, the total memory size available per channel (107 Mbytes) must not be exceeded.		

AVI64 floating differential voltmeter specifications

There are eight floating differential voltmeters for every card (one per group of eight channels). Floating differential voltmeters can be used for (software selectable) differential measurements on any channel of that group or between two (software selectable) channels of such a group. The floating differential voltmeters can also be used as floating differential digitizer.

When using a floating high-current unit, the floating differential voltmeter of that group will be used for measuring voltages at the floating channel and cannot be used for other measurements on the same group of eight channels.

Floating differential voltmeter / digitizer

Voltage ranges	±200 mV, ±1 V, ±8 V
Common mode range ^[49]	-40 V ... +80 V
Input current (maximum)	± (1 nA + 1 nA / V * V _{act}) ^[50]
Input capacitance (typical)	530 pF (differential), 750 pF (single ended to GND)

^[46] 1.5 kHz hardware filter selected, H2 ... H5, 10 ksps.

^[47] 1.5 kHz hardware filter selected, H2 ... H5, 200 ksps.

^[48] 25 kHz hardware filter selected, H2 ... H5, 1000 ksps.

^[49] The voltage at either input of the differential voltmeter must not exceed common mode range.

^[50] V_{act} is the actual voltage at either input of the differential voltmeter.

Voltage measurement	Range	Resolution	Accuracy ^[51]
	±200 mV	0.4 µV	± (10 µV + 0.003 % of relative reading) ^{[52][53]} ± (60 µV + 0.003 % of reading) ^[52] ± (200 µV + 0.05 % of reading)
	±1 V	2 µV	± (15 µV + 0.003 % of relative reading) ^{[52][53]} ± (100 µV + 0.003 % of reading) ^[52] ± (300 µV + 0.01 % of reading)
	±8 V	20 µV	± (50 µV + 0.003 % of relative reading) ^{[52][53]} ± (300 µV + 0.003 % of reading) ^[52] ± (1.0 mV + 0.01 % of reading)

Floating differential digitizer			
	± 200 mV range	± 1 V range	± 8 V range
Resolution	20 bit (0.4 µV)	20 bit (2 µV)	20 bit (20 µV)
DC Accuracy ^[54]	±(0.2 mV + 500 ppm of reading)	±(0.3 mV + 100 ppm of reading)	±(1 mV + 100 ppm of reading)
INL (typical)	± 3 µV (± 7 ppm FSR)	± 6 µV (± 3 ppm FSR)	± 40 µV (± 3 ppm FSR)
DNL (typical)	± 1 ppm FSR		
Sampling rate	0 ... 1 Msps (million samples per second)		
Analog bandwidth (typical, -3 dB)	250 kHz		
Filters	Programmable digital filter 100 Hz ... 250 kHz		
Waveform capture memory (per pin)	Software configurable, not exceeding total available memory size per channel (107 MByte) ^[55] , 32 Bytes required per sample.		

^[51] Requires 1000 averaging samples.

^[52] DC update (SmarTest 7) / Enhanced Accuracy Adjustment (SmarTest 8) every 7 days.
Specifications are for averaging over 1 power line cycle.

^[53] Specifications are valid for measurements taken within 6 hours. Requires hardware revision 008 or later.

^[54] 1 kHz digital filter.

^[55] Memory pooling is supported for every eight channels in the same channel group.

AVI64 floating high-current unit (HCU) specifications

There are eight floating high-current units for every card (one per group of eight channels). Floating high-current capabilities can be added to any pin using software selection.

Floating high-current unit (pulsed operation)	
Voltage range	±8 V
Voltage measure resolution	20 µV
Voltage measure accuracy	± (1 mV + 0.01 % of reading) ^[56]
Voltage force resolution	300 µV
Voltage force accuracy	± (3.5 mV + 0.03 % of setting)
Voltage clamp accuracy	± (10 mV + 0.1 % of setting)
Floating range	-30 V ... +80 V ^[57]
Current range	±4 A (pulsed operation)
Current measure resolution	140 µA
Current measure accuracy	± (2 mA + 0.1 % of reading) ^{[56][58]}
Current force resolution	200 µA
Current force accuracy	± (2 mA + 0.1 % of setting) ^[58]
Current clamp accuracy	± (4 mA + 0.5 % of setting)
Minimum pulse width	100 µs
Maximum pulse width	See diagrams below

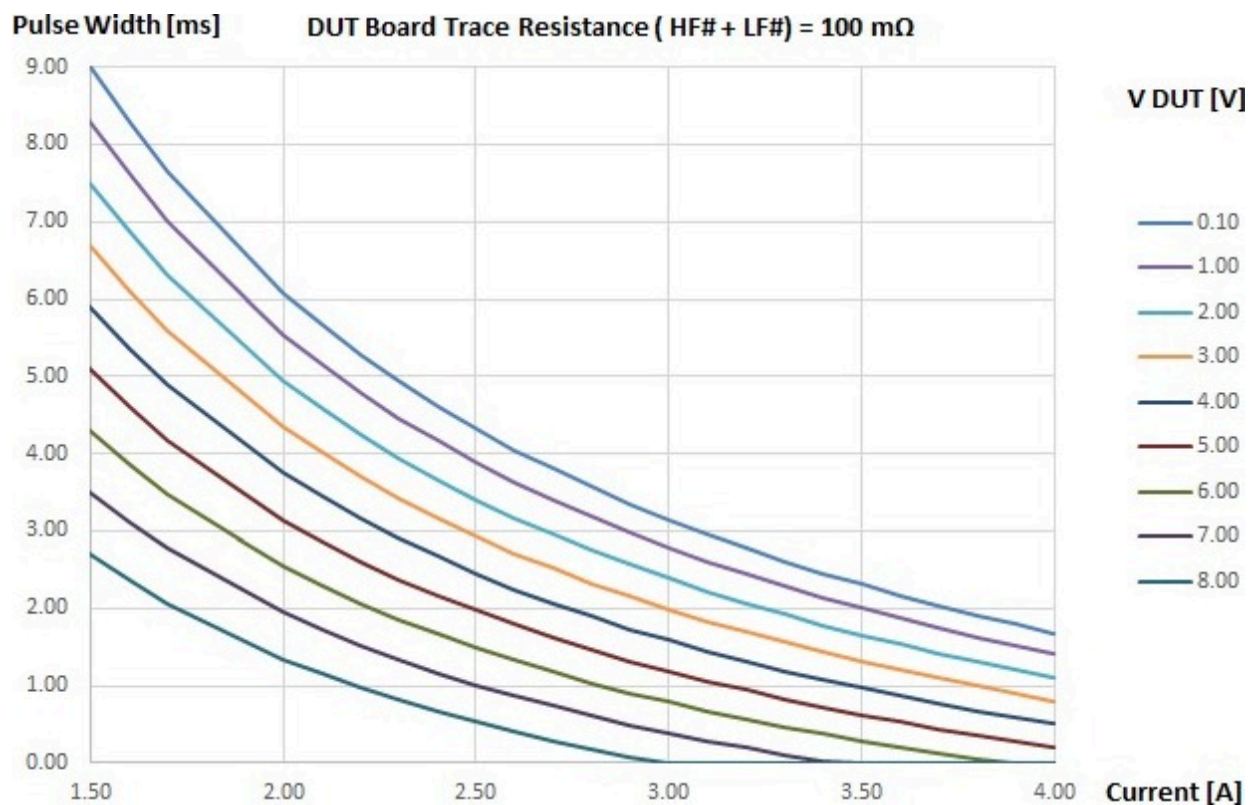
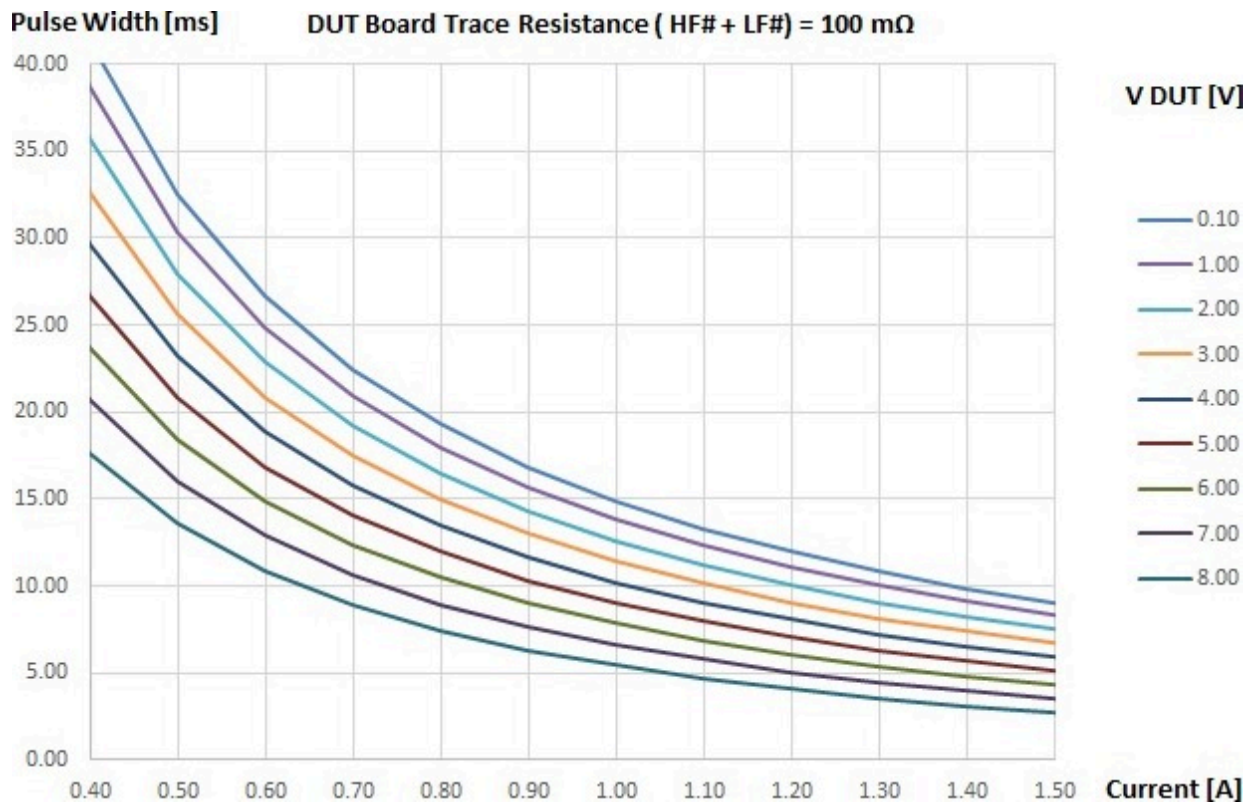
Pulse width vs. load current - 100 mΩ DUT board resistance

The following two diagrams show the relationship between the current, the voltage at the DUT, and the maximum pulse length for a total DUT board trace resistance of 100 mΩ. Note: The total DUT board trace resistance includes the resistance of the HF# traces, the LF# traces, and the contact resistance at the DUT.

^[56] 20 kHz low pass filter. Requires 100 averaging samples.

^[57] The voltage at either of the force and sense pins must not exceed floating range.

^[58] Valid if calibrated with software version 7.4.1 or later.



Pulse width vs. load current - 1000 mΩ DUT board resistance

The following two diagrams show the relationship between the current, the voltage at the DUT, and the maximum pulse length for a total DUT board trace resistance of 1000 mΩ. Note: The total DUT board trace resistance includes the resistance of the HF# traces, the LF# traces, and the contact resistance at the DUT.



AVI64 digital driver, comparator, and TMU specifications

Digital I/O operation mode

The Digital I/O operation mode can be selected on a per channel granularity. Changing the mode between the "Digital I/O operating mode" and the "PMU operating mode" (and vice versa) within a test program flow is supported independently for every channel.

Digital I/O operation mode

Vector period range	2.5 ns ... 31250 ns
Vector period accuracy	±15 ppm of period setting
Vector period resolution	0.001 fs
Edge placement range ^[59]	-1 ... 32 vector periods
Edge placement resolution	1 ps
Total available memory per channel	107 MByte

Digital driver^[60]

	$-8\text{ V} \leq V_{IL}, V_{IH}, V_T \leq +8\text{ V}$	$-40\text{ V} \leq V_{IL}, V_{IH}, V_T \leq +80\text{ V}$
Driver states	V_{IL}, V_{IH}, V_T ^[61] , High-Z	V_{IL}, V_{IH}, V_T ^[61] , High-Z
Level accuracy	± (30 mV + 0.1 % of setting)	± (50 mV + 0.1 % of setting)
Overshoot / undershoot	10 % of programmed level	10 % of programmed level
Maximum drive current	± 50 mA	± 50 mA
Driver slew rate (typical)	50 V / μs @ 3 V swing	80 V / μs @ 12 V swing
Minimum pulse width	100 ns at 3 V swing	400 ns at 12 V swing
Maximum driver swing	16 V	32 V
Maximum data rate (20% / 80%); $t_r, t_f < (T_{cycle} / 2)$: ^[62]		
3V swing	10 Mbps	5 Mbps
5V swing	5 Mbps	5 Mbps
12V swing	2.5 Mbps	2.5 Mbps
24V swing	n/a	1 Mbps
Delay time active # high-Z	10 μs (maximum)	10 μs (maximum)

^[59] Maximum edge placement range is -6250 ns to +19000 ns. For vector periods > 752 ns the edge placement range is -1 to 12 vector periods. For vector periods < 30 ns the edge placement range is +60 ns to 32 vector periods.

^[60] Specifications are given for having "sense line" connection only.

^[61] V_T is the third level which can be used instead of "High-Z". The third level is software programmable within the maximum driver level range independent of V_{IL} and V_{IH} .

^[62] Programmable up to 20 Mbit/s.

Digital driver^[60]

Edge placement accuracy (EPA)	$\pm 10 \text{ ns}^{[63]}$	$\pm 10 \text{ ns}^{[63]}$
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Digital compare (high-speed mode)

Programmable thresholds ^[64]	V_{OL}, V_{OH}	
Input voltage and threshold ranges (single ended) ^[65]	$-8 \text{ V} \leq V_{OL}, V_{OH} \leq +8 \text{ V}$:	$\pm (30 \text{ mV} + 0.1 \% \text{ of setting})$
	$-40 \text{ V} \leq V_{OL}, V_{OH} \leq +80 \text{ V}$:	$\pm (100 \text{ mV} + 0.1 \% \text{ of setting})$
Threshold hysteresis (single ended)	25 mV (typical) ^[66]	
Input voltage range (differential)	$-8 \text{ V} \leq V_{DUT} \leq +8 \text{ V}^{[67]}$	
Threshold range (differential)	$-7 \text{ V} \leq V_{DIFF} \leq +7 \text{ V}$:	$\pm (30 \text{ mV} + 5 \% \text{ of setting})$
Threshold hysteresis (differential)	25 mV (typical) ^[66]	
Input termination ^[68]	High impedance, ^[69]	
	50 Ω single ended,	
	100 Ω differential ^[70]	
Signal path bandwidth	50 MHz (typical) ^[71]	
Maximum data rate	100 Mbps (typical) ^[71]	
Intrinsic transition time (20% / 80%)	2 ns (typical) ^[72]	
Compare timing accuracy (EPA)	$\pm 5 \text{ ns}^{[63]}$	
Compare timing resolution	1 ps	

Digital compare (high-resolution mode)^[73]

Programmable thresholds ^[74]	$V_{OL}, V_{OH}, I_{CL}, I_{CH}$ (with programmable hysteresis)
Threshold ranges (voltage) ^[75]	$\pm 3 \text{ V}, \pm 10 \text{ V}, \pm 30 \text{ V}, -40 \text{ V} \dots +80 \text{ V}$

^[60] Specifications are given for having "sense line" connection only.

^[63] Calibrated at 3 V swing, EPA verified at 1.5 V and 4.5 V swing. Specification is valid across full voltage range (-40 V ... +80 V).

^[64] Dual level comparator (Low, High, Intermediate, X).

^[65] Software selectable.

^[66] Software selectable on / off

^[67] Protected up to -40 V ... +80 V.

^[68] Software selectable, single ended termination to GND.

^[69] 1 G Ω (typical)

^[70] 20 k Ω (typical) input impedance to ground

^[71] 3 V swing (0 V ... 3 V)

^[72] 3 V swing (0 V ... 3 V, 50 Ω termination)

^[73] Software selectable for voltage / current real time P/F compare, digital capture, and TMU mode.

^[74] Dual level comparator (Low, High, Intermediate, X)

^[75] Voltage measure range is used in compare mode.

Digital compare (high-resolution mode)^[73]

Threshold accuracy (voltage)	$\pm 3 \text{ V}$	$\pm(5 \text{ mV} + 0.03 \% \text{ of setting})$
	$\pm 10 \text{ V}$	$\pm(10 \text{ mV} + 0.03 \% \text{ of setting})$
	$\pm 30 \text{ V}$	$\pm(30 \text{ mV} + 0.03 \% \text{ of setting})$
	$-40 \text{ V} \dots +80 \text{ V}$	$\pm(80 \text{ mV} + 0.03 \% \text{ of setting})$
Threshold hysteresis (voltage)	Programmable within full range	
Threshold ranges (current) ^[76]	$\pm 5 \mu\text{A}, \pm 50 \mu\text{A}, \pm 500 \mu\text{A}, \pm 5 \text{ mA}, \pm 50 \text{ mA}, \pm 200 \text{ mA}$	
Threshold accuracy (current) ^[77]	$\pm 5 \mu\text{A}$	$\pm(20 \text{ nA} + 0.05\% \text{ of setting} + 1 \text{ nA} / \text{V} * V_{\text{act}})$
	$\pm 50 \mu\text{A}$	$\pm(150 \text{ nA} + 0.05\% \text{ of setting} + 1 \text{ nA} / \text{V} * V_{\text{act}})$
	$\pm 500 \mu\text{A}$	$\pm(1.5 \mu\text{A} + 0.05\% \text{ of setting} + 2 \text{ nA} / \text{V} * V_{\text{act}})$
	$\pm 5 \text{ mA}$	$\pm(15 \mu\text{A} + 0.05\% \text{ of setting} + 20 \text{ nA} / \text{V} * V_{\text{act}})$
	$\pm 50 \text{ mA}$	$\pm(150 \mu\text{A} + 0.05\% \text{ of setting} + 200 \text{ nA} / \text{V} * V_{\text{act}})$
	$\pm 200 \text{ mA}$	$\pm(600 \mu\text{A} + 0.05\% \text{ of setting} + 1 \mu\text{A} / \text{V} * V_{\text{act}})$
Threshold hysteresis (current)	Programmable within full range	
Compare timing accuracy (EPA)	$\pm 500 \text{ ns}$ ^[78]	
Compare timing resolution	5 ns (interpolated) ^[79]	
Timing skew (V_{OL} vs. V_{OH} , I_{CL} vs. I_{CH})	$\pm 5 \text{ ns}$ (maximum)	
Signal path bandwidth ^[80]	250 kHz (typical)	

Timing Measurement Unit (TMU)

Timing measurements are done using the digital comparator functionality available per channel (see above).

The TMU allows for time delay / skew measurements ($V_{\text{OH}} / I_{\text{CH}} \# V_{\text{OH}} / I_{\text{CH}}, V_{\text{OL}} / I_{\text{CL}} \# V_{\text{OL}} / I_{\text{CL}}$) also across different instrument types (i.e. PinScale1600), frequency / pulse width measurements, and rise / fall time measurements ($V_{\text{OL}} / I_{\text{CL}} \# V_{\text{OH}} / I_{\text{CH}}, V_{\text{OH}} / I_{\text{CH}} \# V_{\text{OL}} / I_{\text{CL}}$).

The TMU can be used with all compare modes above. The following table shows available compare modes for channels in "Digital I/O operating mode" or in "PMU operating mode":

^[73] Software selectable for voltage / current real time P/F compare, digital capture, and TMU mode.

^[76] When used in "Digital I/O" operating mode, threshold range is fixed to $\pm 200 \text{ mA}$.

^[77] V_{act} is the actual voltage of the channel, which is either the programmed force voltage or the voltage forced by the DUT, depending on the mode the channel is in (forcing or clamping).

^[78] Additional fixed channel delay in high resolution mode of 3 μs .

^[79] Sampling rate = 1 μs

^[80] In "Digital I/O" operating mode

Compare modes	Digital I/O operating mode	PMU operating mode
High speed (single ended)	supported	supported
High speed (differential)	supported	n/a
High resolution (voltage)	supported	supported
High resolution (differential voltmeter)	n/a	supported
High resolution (current)	supported	supported
High resolution (high current unit)	n/a	supported

Timing Measurement Unit (TMU)

DC / AC Performance	See "Digital compare" above.
Accuracy for rise / fall time measurements (single shot) ^[81]	± 5 ns
Accuracy for skew / propagation delay measurements (single shot)	± (2 * EPA)
Accuracy for single timestamp to ARM (time zero)	± 35 ps
Maximum sampling rate	50 Msps

AVI64 other specifications

Interface characteristics	Force	Sense	GND sense
Compliance range if disconnected	-44 V ... +84 V	-44 V ... +84 V	-44 V ... +84 V
Input leakage current (typical) ^[82]	± (5 nA + 1 nA / V * V _{act}) ^[83]	± (1 nA + 1 nA / V * V _{act}) ^[83]	±200 nA
Input capacitance (typical) ^[84]	1600 pF	110 pF ^[85]	150 pF

Surge tracker

Voltage or current surges are detected against programmable minimum and maximum values. The surge tracker always checks against both high (positive) and low (negative) thresholds, which can be set individually. Two alarms/flags indicate if thresholds have been exceeded. These alarms/flags are latched (stored) and can be retrieved any time in the testflow.

For accuracy specifications of the voltage surge tracker see "Digital compare (high-speed mode)". Voltage threshold accuracies are specified in "Input voltage and threshold ranges (single ended)".

^[81] Available in PMU operating mode only.

^[82] Measured in high-impedance mode.

^[83] V_{act} is the actual voltage of the pin, which is either the programmed force voltage or the voltage forced by the DUT, depending on the mode the channel is in (forcing or clamping).

^[84] Force pin: The input capacitance is specified for the disconnected mode. Sense pin: The specified input capacitance is for both the connected mode and disconnected mode.

^[85] In Digital I/O operation mode, 165 pF in PMU operation mode.

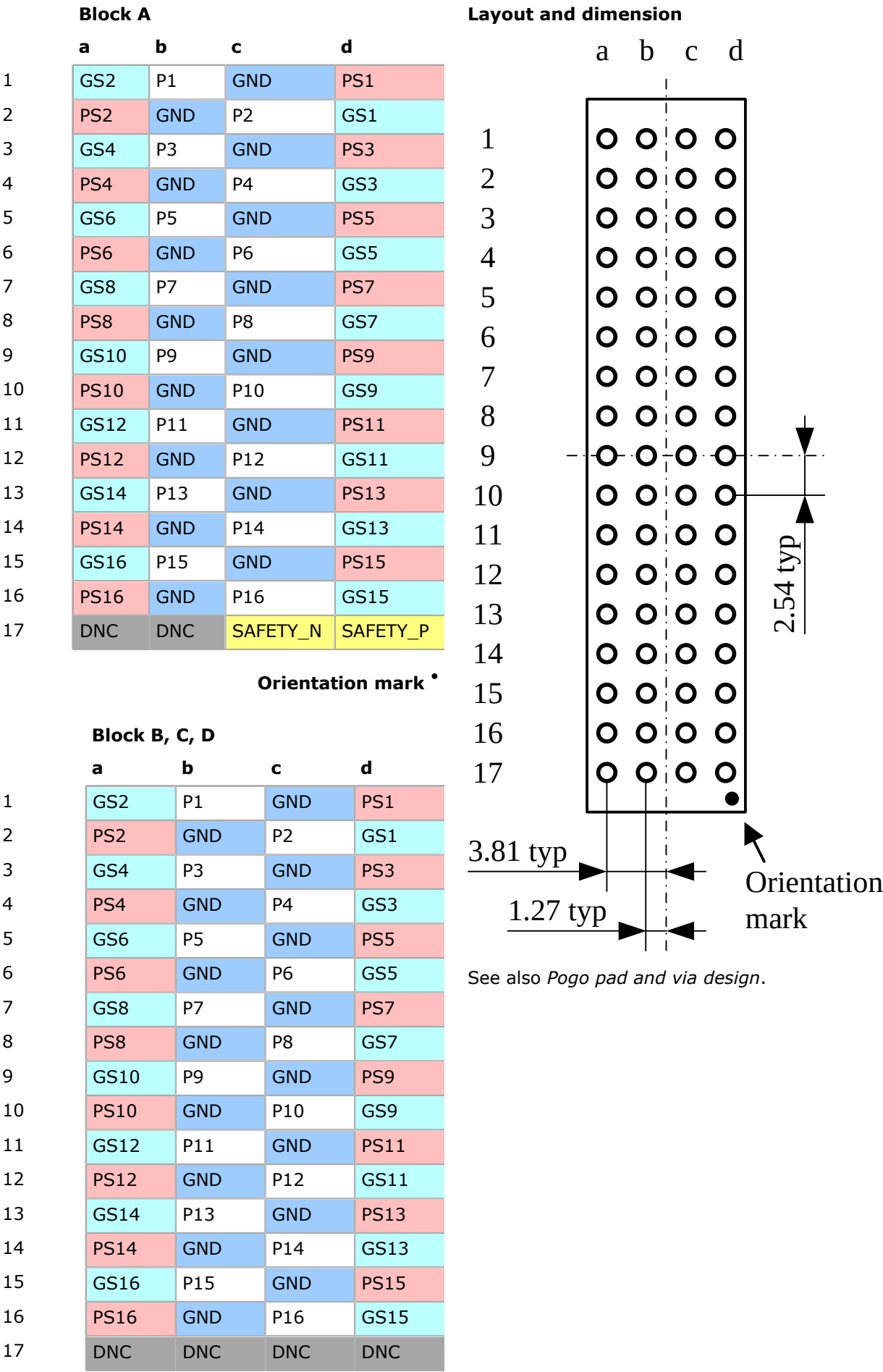
Surge tracker

For accuracy specifications of the current surge tracker see "Digital compare (high-resolution mode)". Current threshold accuracies are depending on the current measurement range selected and are specified in "Threshold accuracy (current)".

AVI64 pogo pin assignment

The E8025AVP pogo cable assembly connects the 64 channels of the DC Scale AVI64 card to the DUT board.

Pogo pin assignment



Pogo pin description

Pin	Description	DUT board connection
P#	High force line of channel #	See the <i>Pogo pin "P#"</i> section in <i>AVI64 pogo pins</i> .
PS#	High sense and digital I/O line of channel #	For more information, refer to <i>Pogo pin "PS#"</i> .
GND	Ground line of all channels	Connect to DUT board ground. For more information, refer to <i>Pogo pin "GND"</i> .
GS#	Ground sense line of channel #	See the <i>Pogo pin "GS#"</i> section in <i>AVI64 pogo pins</i> .
SAFETY_P	Control line of the safety circuit	Connect to an adequate +5.0 V supply via a safety interlock on the DUT board or the DUT interface. For more information, refer to the <i>SAFETY_P</i> section in <i>AVI64 pogo pins - Safety</i> .
SAFETY_N	Return line of the safety circuit	Connect to the return pin of the +5.0 V supply at SAFETY_P and to the DUT board ground. For more information, refer to the <i>SAFETY_N</i> section in <i>AVI64 pogo pins - Safety</i> .
DNC	Do not connect pin	These pins are internally used. Do not connect these pins on the DUT board.

Notes on the pogo pin descriptions:

- Pin: # = Channel 1 ... 16
- The DNC pins are internally connected and reserved. To ensure that adjacent high voltage pins or traces have no effect on the DNC pins, adequate measures like sufficient insulation space have to be respected.

▲ CAUTION GND is wired to several pogo pins. All GND pogo pins must be connected in parallel on the DUT board to ensure that the specifications for the card and the maximum ratings for the pogo pins are met.

DC Scale DPS128 Device Power Supply (E8023CS/ E8023CSH) - high current specifications

Configuration

	E8023CSH	E8023CS
Maximum number of channels per board	64	128
Maximum source/sink current per channel	1.0 A @ 2.5 V	1.0 A @ 2.5 V
Maximum source/sink current per channel	0.5 A @ 7 V	0.5 A @ 7 V
Maximum source/sink current per board	64 A /64 A@ 2.5 V	66 A /58 A@ 2.5 V
Voltage range	-2.5 V to +7 V	-2.5 V to +7 V

The DPS128/DPS64 card features two power domains with a dedicated power budget. Each domain is able to deliver half of the overall power budget. The power domain to channel mapping must be observed for DPS128 cards when assigning the available channels to the DUT pins. For more information refer to TDC topic DPS128/DPS64 power budget (147811).

	A#Class test head ^[86]	Compact test head ^[86]	Small test head ^[86]	Large test head ^[86]
Maximum number of boards per system	8	16	32	32
Maximum number of channels per system (E8023CSH)	512	1024	2048	2048
Maximum number of channels per system (E8023CS)	1024	2048	2048	2048

- Parallel connection (ganging) possible for any number of successive channels within a board.
- Up to 64 ganging groups of successive channels are supported within each card.
- Maximum source/sink current per board cannot be exceeded.
- Within a test head, a DPS128 card can be mixed with any other supported device power supply.
- Requires E7010E or E7010F Enhanced Diagnostics Interconnect Board.
- E8023CS only: 32 boards per system only possible if 4 pogo-block version is used (2 channels ganged to 1 channel and 2.0A @2.5V per channel). For more information about the E8023PC pogo cable see TDC topic 141738.

Supply voltage / current range specifications (high current)

Voltage force	Range	Resolution	Accuracy
	-2.5 V to 7 V	200 μ V	± 3 mV

^[86] Applies to test head with DUT Scale Interface and standard pogo block

Voltage measure	Range	Resolution	Accuracy
	-2.5 V to 7 V	200 μ V	± 2 mV ^[87] ± 1 mV ^[88]

Current force (clamp)	Range	Resolution	Accuracy
	200 mA to 1.0 A	50 μ A	-3%, +5% of range
	40 mA to 200 mA	10 μ A	-3%, +5% of range
	20 mA to 100 mA	5 μ A	-3%, +5% of range
	5 mA to 25 mA	1 μ A	-3%, +5% of range
	2.5 mA to 12.5 mA	0.5 μ A	-3%, +5% of range
	500 μ A to 2.5 mA	100 nA	-3%, +5% of range
	250 μ A to 1.25 mA	50 nA	-3%, +5% of range
	50 μ A to 250 μ A	10 nA	-3%, +5% of range
	25 μ A to 125 μ A	5 nA	-3%, +5% of range
	5 μ A to 25 μ A	1 nA	-3%, +5% of range
	2.5 μ A to 12.5 μ A	0.5 nA	-3%, +5% of range
Current force (n channels ganged)	n x 200 mA to n x 1.0 A	n x 50 μ A	-3%, +5% of total ganged current

High accuracy current force ^{[89][90]}	Range	Resolution	Accuracy
	200 mA to 1.0 A	50 μ A	$\pm$ (3 mA +0.2 % of setting)
	40 mA to 200 mA	10 μ A	$\pm$ (0.6 mA +0.2 % of setting)
	20 mA to 100 mA	5 μ A	$\pm$ (0.6 mA +0.2 % of setting)
	5 mA to 25 mA	1 μ A	$\pm$ (75 μ A +0.2 % of setting)
	2.5 mA to 12.5 mA	0.5 μ A	$\pm$ (75 μ A +0.2 % of setting)
	500 μ A to 2.5 mA	100 nA	$\pm$ (7.5 μ A +0.2 % of setting)
	250 μ A to 1.25 mA	50 nA	$\pm$ (7.5 μ A +0.2 % of setting)
	50 μ A to 250 μ A	10 nA	$\pm$ (750 nA +0.2 % of setting)
	25 μ A to 125 μ A	5 nA	$\pm$ (750 nA +0.2 % of setting)
	5 μ A to 25 μ A	1 nA	$\pm$ (120 nA +0.2 % of setting)

^[87] Requires 4 averaging samples under pattern control.

^[88] Up to 200 mA forced current, base calibration and auto calibration period 1 month

^[89] Requires the use of a suitable API for high accuracy force current mode and at least 1 Ω total additional resistive load (incl. DUT) in the Force path

^[90] The high accuracy current force mode is not supported for ganged channels.

High accuracy current force ^{[89][90]}	Range	Resolution	Accuracy
	2.5 μ A to 12.5 μ A	0.5 nA	$\pm$ (120 nA +0.2 % of setting)

Current measurement

Range	Resolution	Accuracy
$\pm$ 1.0 A	50 μ A	$\pm$ (1 mA +0.1 % of reading) ^[91]
$\pm$ 200 mA	10 μ A	$\pm$ (250 μ A +0.1 % of reading) ^[92]
$\pm$ 100 mA	5 μ A	$\pm$ (250 μ A +0.1 % of reading) ^[92]
$\pm$ 25 mA	1 μ A	$\pm$ (25 μ A +0.1 % of reading) ^[93]
$\pm$ 12.5 mA	0.5 μ A	$\pm$ (25 μ A +0.1 % of reading) ^[93]
$\pm$ 2.5 mA	100 nA	$\pm$ (2.5 μ A + 0.1 % of reading) ^[94]
$\pm$ 1.25 mA	50 nA	$\pm$ (2.5 μ A + 0.1 % of reading) ^[95]
$\pm$ 250 μ A	10 nA	$\pm$ (250 nA + 0.1 % of reading) ^[96]
$\pm$ 125 μ A	5 nA	$\pm$ (250 nA + 0.1 % of reading) ^[97]
$\pm$ 25 μ A	1 nA	$\pm$ (50 nA + 0.2 % of reading) ^[97]
$\pm$ 12.5 μ A	0.5 nA	$\pm$ (50 nA + 0.2 % of reading) ^[97]
n x ($\pm$ 1.0 A) (n channels ganged)	n x 50 μ A	$\pm$ (n x 1 mA +0.1 % of total current)

Supported "current force" and "current measurement" range combinations

In voltage force mode, current measure range can be changed without prior disconnect.

^[89] Requires the use of a suitable API for high accuracy force current mode and at least 1 Ω total additional resistive load (incl. DUT) in the Force path

^[90] The high accuracy current force mode is not supported for ganged channels.

^[91] Requires 16 averaging samples under pattern control. Maximum supported external capacitance to achieve accuracy is 220 μ F. More samples required for larger capacitances.

^[92] Requires 16 averaging samples under pattern control. Maximum supported external capacitance to achieve accuracy is 47 μ F. More samples required for larger capacitances.

^[93] Requires 16 averaging samples under pattern control. Maximum supported external capacitance to achieve accuracy is 15 μ F. More samples required for larger capacitances.

^[94] Requires 16 averaging samples under pattern control. Maximum supported external capacitance to achieve accuracy is 4.7 μ F. More samples required for larger capacitances.

^[95] Requires 16 averaging samples under pattern control. Maximum supported external capacitance to achieve accuracy is 2.2 μ F. More samples required for larger capacitances.

^[96] Requires 64 averaging samples under pattern control. Maximum supported external capacitance to achieve accuracy is 4.7 μ F. More samples required for larger capacitances.

^[97] Requires 64 averaging samples under pattern control. Maximum supported external capacitance to achieve accuracy is 2.2 μ F. More samples required for larger capacitances.

		Current force range											
Current measure range		12.5 μA	25 μA	125 μA	250 μA	1.25 mA	2.5 mA	12.5 mA	25 mA	100 mA	200 mA	1 A	
	±12.5 μA	ok		ok		ok		ok		ok		ok	
	±25 μA		ok		ok		ok		ok		ok		
	±125 μA			ok		ok		ok		ok		ok	
	±250 μA				ok		ok		ok		ok		
	±1.25 mA					ok		ok		ok		ok	
	±2.5 mA						ok		ok		ok		
	±12.5 mA							ok		ok		ok	
	±25 mA								ok		ok		
	±100 mA									ok		ok	
	±200 mA										ok		
	±1 A											ok	

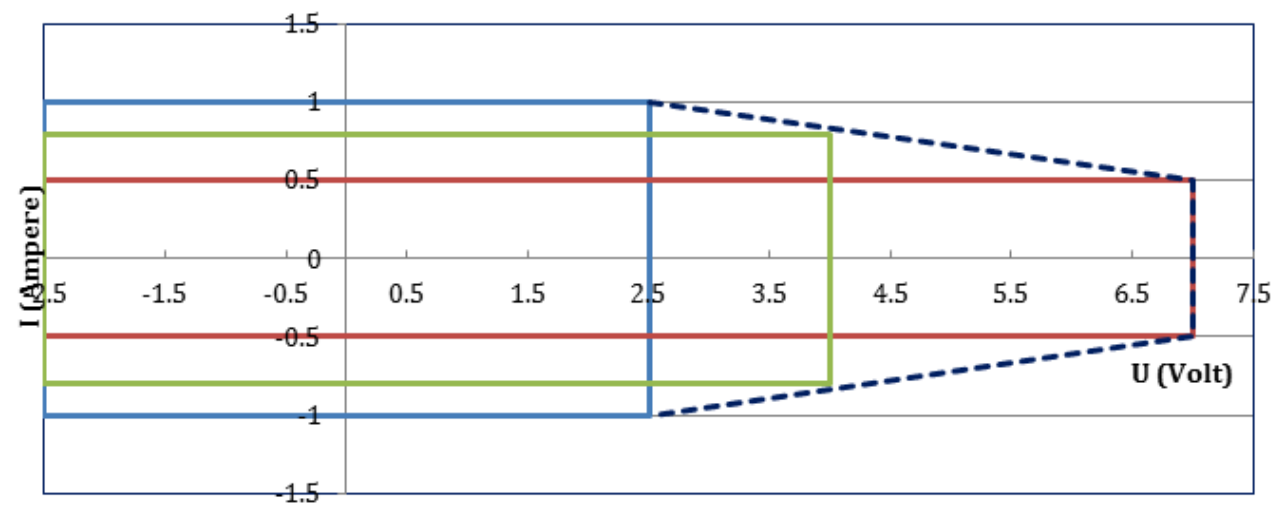
For the minimum clamp current distance, refer to "Minimum clamp current distance" section in topic "DPS128 / DPS64 functionality and features" (140638).

DPS128 power diagram high current

The programmed voltage determines the maximum current. If 7 V is programmed, the maximum current is 0.5 A (red box), if 2.5 V is programmed the maximum current is 1 A for all voltages below 2.5 V (blue box).

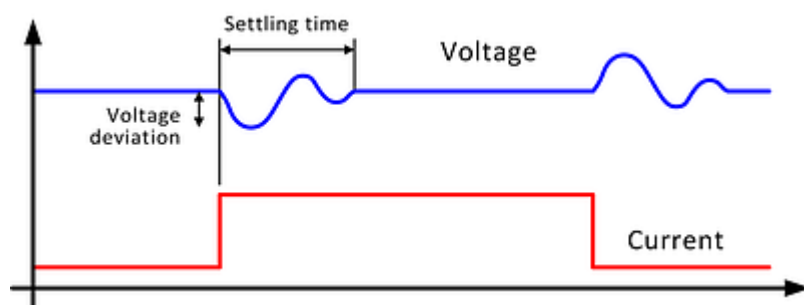
The power diagram applies to an additional voltage drop in the Force path beyond the pogo pin of ≤ 0.1 V.

Output characteristic: I versus U



For voltages between 2.5 V and 7 V (green box), the maximum current can be calculated as follows:

$$I_{\text{green}} = \frac{I_{\text{red}} - I_{\text{blue}}}{U_{\text{red}} - U_{\text{blue}}} (U_{\text{green}} - U_{\text{blue}}) + I_{\text{blue}}$$

Dynamic behavior**Explanation of terms: Response time****Settling time after loadstep**

at 2 V, settling into ± 20 mV of programmed value

Loadstep	1 μF	47 μF	100 μF
0.1 A to 1.0 A	20 μs	25 μs	30 μs
1.0 A to 0.1 A	20 μs	25 μs	30 μs

Voltage deviation after loadstep

Loadstep	1 μF	47 μF	100 μF
0.1 A to 1.0 A	320 mV	120 mV	85 mV
1.0 A to 0.1 A	500 mV	125 mV	85 mV

Settling times for DPS128 measurement ranging

Wait times are to be considered when changing the DPS128 measurement range:

- Auto-range execution time: The required wait time to allow the auto-range algorithm to finish.
- Range settling time: The required wait time to achieve the specified accuracy of the target range.

For more information, refer to the "Settling times for DPS128 measurement ranging" section in topic "DPS128 / DPS64 specific setup tasks" (140640).

Other performance characteristics

Disconnect state	
Input leakage current	< 20 nA (AC relay open)
Voltmeter mode (high impedance mode)	
Voltage range	-2.5 V to +7 V
Voltage resolution	200 μV
Voltage accuracy	± 1 mV ^[88]
Input leakage current	<30 nA
Compliance range (at disconnect state)	
Maximum external voltage at force input	-6 V to +12 V
Maximum external voltage at sense input	-7 V to +16 V

Load capacitance range

Required external capacitance for high frequency noise filtering	1nF to 10nF ^[98]
Maximum supported external capacitance with stability maintained	470 μ F per channel (n x 470 μ F up to 5 mF if ganged) ^[99]
Required external capacitance range for DPS style operation	1 - 100 μ F per channel ^[99]

Slew rate control

Slew rate control	On-, off-, change level- rate individually controlled for each channel
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DC profiling

DC profiling	Supported per channel for voltage and current profiles
Sample rate	Up to 100 ksps

Data analysis

Data analysis	raw data are stored on board without software interaction and may be uploaded for further computation. Averaging, min and max number are calculated and stored on board
Result analysis	built-in result analysis available that indicate the measurement data is pass or fail against a user specified test limit

Surge tracker

Voltage or current surges are detected against programmable min. and max. values. The surge tracker always checks against both high (positive) and low (negative) thresholds, which can be set individually. Two flags indicate if thresholds have been exceeded. These flags are latched (stored) and can be retrieved any time in the testflow.

Voltage compare mode

Minimum detectable pulse width	1 μ s
Threshold voltage accuracy	$\pm$ 10 mV typical
Input and threshold range	-2.5 V to +7 V

Current compare mode

Minimum detectable pulse width ^[100]	1 μ s
Threshold current accuracy	$\pm$ 10 % of selected current range typical
Input and threshold range	Same as current measurement range

^[98] It is recommended to place a low ESL capacitance close to the pogo location. The detailed setup depends on the configuration of the DUT interface board. For more details see topic 149015 in the DUT board design reference.

^[99] In addition to "Required external capacitance for high frequency noise filtering".

^[100] in current range 1A

Analog performance

AWG mode (16 bit DAC)	
Step rate	200 ksps (kilo samples per second)
Analog bandwidth (-3 dB, 1 V swing)	5 kHz
Total channel memory	28 MB
Digitizer mode (16 bit ADC)	
Sample rate	90 ksps (kilo samples per second)
Signal path bandwidth (-3dB)	20 kHz
Total channel memory	28 MB, 11 to 22 bytes required per sample
Trigger functions	
Trigger	sequencer#controlled (no external trigger pin required)

DC Scale DPS128 Device Power Supply (E8023CSV/ E8023CSW) - high voltage specifications

Configuration

E8023CSV	High current	High voltage
Maximum number of channels per board	64	64
Maximum source/sink current per channel	1.0 A @ 2.5 V	
Maximum source/sink current per channel	0.5 A @ 7 V	0.2 A @ -6 to +15 V
Maximum source/sink current per board	62 A/54 A @ 2.5 (only high current channels used)	12.8 A/12.8 A (only high voltage channels used)
Voltage range	-2.5 V to +7 V	-6 V to +15 V

E8023CSW	High voltage
Maximum number of channels per board	128
Maximum source/sink current per channel	0.2 A @ -6 to +15 V
Maximum source/sink current per board	13.2 A/24.4 A
Voltage range	-6 V to +15 V

The DPS128/DPS64 card features two power domains with a dedicated power budget. Each domain is able to deliver half of the overall power budget. The power domain to channel mapping must be observed for DPS128 cards when assigning the available channels to the DUT pins. For more information refer to TDC topic DPS128/DPS64 power budget (147811).

	A#Class test head ^[101]	Compact test head ^[101]	Small test head ^[101]	Large test head ^[101]
Maximum number of boards per system	8	16	16	16
Maximum number of channels per system (depends on product structure decision)	1024	2048	2048	2048

- Parallel connection (ganging) possible for any number of successive channels within a board.
- Up to 64 ganging groups of successive channels are supported within each card.
- Maximum current per board cannot be exceeded.
- Sink capability per channel according to power charts.
- Within a test head, a DPS128 card can be mixed with any other device power supply.
- Requires E7010E or E7010F Enhanced Diagnostics Interconnect Board.

^[101] Applies to test head with DUT Scale Interface and standard pogo block

Supply voltage / current range specifications (high voltage)

Voltage force	Range	Resolution	Accuracy
	-6 V to 15 V	500 μ V	± 5 mV

Voltage measure	Range	Resolution	Accuracy
	-6 V to 15 V	100 μ V	± 4 mV ^[102]

Current force (clamp)	Range	Resolution	Accuracy
Current force (n channels ganged)	20 to 200 mA	10 μ A	-3%, +5% of range
	2.5 mA to 25 mA	1 μ A	-3%, +5% of range
	250 μ A to 2.5 mA	100 nA	-3%, +5% of range
	25 μ A to 250 μ A	10 nA	-3%, +5% of range
	2.5 μ A to 25 μ A	1 nA	-3%, +5% of range
	n x 20 mA to n x 200 mA	n x 10 μ A	-3%, +5% of total ganged current

High accuracy current force ^{[103][104]}	Range	Resolution	Accuracy
	20 mA to 200 mA	10 μ A	$\pm (1 \text{ mA} + 0.2 \% \text{ of setting})$
	2.5 mA to 25 mA	1 μ A	$\pm (100 \mu\text{A} + 0.2 \% \text{ of setting})$
	250 μ A to 2.5 mA	100 nA	$\pm (10 \mu\text{A} + 0.2 \% \text{ of setting})$
	25 μ A to 25 μ A	10 nA	$\pm (1 \mu\text{A} + 0.2 \% \text{ of setting})$
	2.5 μ A to 25 μ A	1 nA	$\pm (100 \text{ nA} + 0.2 \% \text{ of setting})$

Current measurement

Range	Resolution	Accuracy
± 200 mA	10 μ A	$\pm (300 \mu\text{A} + 0.1\% \text{ of reading})$ ^[105]
± 25 mA	1 μ A	$\pm (30 \mu\text{A} + 0.1\% \text{ of reading})$ ^[106]
± 2.5 mA	100 nA	$\pm (3 \mu\text{A} + 0.1\% \text{ of reading})$ ^[107]

^[102] Requires 4 averaging samples under pattern control.

^[103] Requires the use of a suitable API for high accuracy force current mode and at least 1 Ω total additional resistive load (incl. DUT) in the Force path

^[104] The high accuracy current force mode is not supported for ganged channels.

^[105] Requires 16 averaging samples under pattern control. Maximum supported external capacitance to achieve accuracy is 47 μ F. More samples required for larger capacitances.

^[106] Requires 16 averaging samples under pattern control. Maximum supported external capacitance to achieve accuracy is 15 μ F. More samples required for larger capacitances.

^[107] Requires 16 averaging samples under pattern control. Maximum supported external capacitance to achieve accuracy is 4.7 μ F. More samples required for larger capacitances.

Range	Resolution	Accuracy
± 250 µA	10 nA	± (300 nA + 0.1% of reading) ^[108]
± 25 µA	1 nA	± (60 nA + 0.1% of reading) ^[109]
n x (± 200 mA) (n channels ganged)	n x 10 µA	± (n x 300 µA +0.1% of total current)

Supported current force and current measurement range combinations

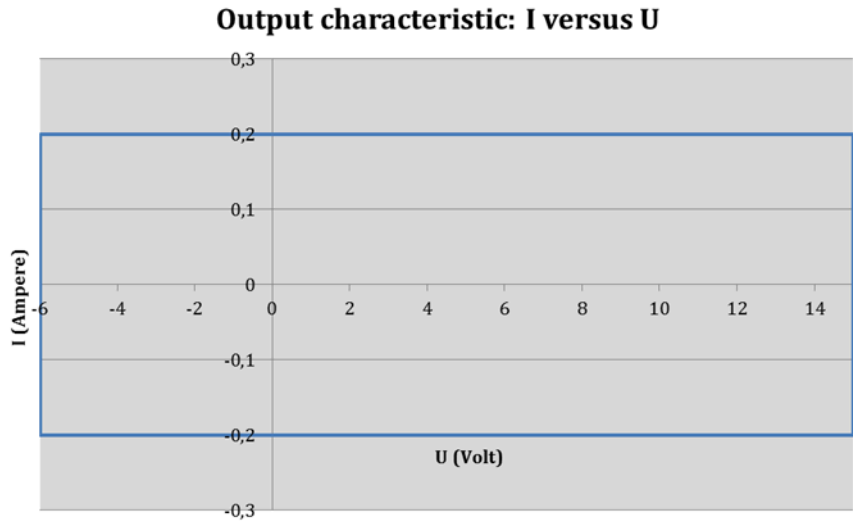
In voltage force mode, current measure range can be changed without prior disconnect.

Current measure range	Current force range					
		25 μA	250 μA	2.5 mA	25 mA	200 mA
	± 25 μA	ok	ok	ok	ok	ok
	± 250 μA	ok		ok	ok	ok
	± 2.5 mA	ok			ok	ok
	± 25 mA	ok				ok
	± 200 mA	ok				

For the minimum clamp current distance, refer to "Minimum clamp current distance" section in topic "DPS128 / DPS64 functionality and features" (140638).

DPS128 power diagram high voltage

The power diagram applies to an additional voltage drop in the Force path beyond the pogo pin of = 0.1V.



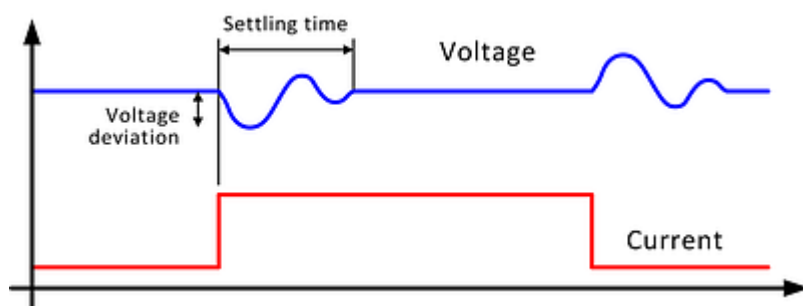
DPS128 power diagram high voltage

Dynamic behavior

Explanation of terms: Response time

^[108] Requires 64 averaging samples under pattern control. Maximum supported external capacitance to achieve accuracy is 4.7 µF. More samples required for larger capacitances.

^[109] Requires 64 averaging samples under pattern control. Maximum supported external capacitance to achieve accuracy is 2.2 µF. More samples required for larger capacitances. Applies to = 50% relative humidity.



Settling time after loadstep

at 5 V, settling into ± 20 mV of programmed value

Loadstep	100 nF	1 μ F
0.1 A to 0.2 A	12 μ s	12 μ s
0.2 A to 0.1 A	12 μ s	12 μ s

Voltage deviation after loadstep

Loadstep	100 nF	1 μ F
0.1 A to 0.2 A	190 mV	100 mV
0.2 A to 0.1 A	200 mV	110 mV

Settling times for DPS128 measurement ranging

Wait times are to be considered when changing the DPS128 measurement range:

- Auto-range execution time: The required wait time to allow the auto-range algorithm to finish.
- Range settling time: The required wait time to achieve the specified accuracy of the target range.

For more information, refer to the "Settling times for DPS128 measurement ranging" section in topic "DPS128 / DPS64 specific setup tasks" (140640).

Other performance characteristics

Disconnect state	
Input leakage current	< 40 nA (AC relay open)

Compliance range (at disconnect state)	
Maximum external voltage at force input	-11 V to +17 V
Maximum external voltage at sense input	-11 V to +17 V

Load capacitance range	
Required external capacitance for high frequency noise filtering	1nF to 10nF ^[110]
Maximum supported external capacitance with stability maintained	22 μ F per channel (47 μ F total if ganged) ^[111]

^[110] It is recommended to place a low ESL capacitance close to the pogo location. The detailed setup depends on the configuration of the DUT interface board. For more details see topic 149015 in the DUT Board Design Guide.

^[111] In addition to "Required external capacitance for high frequency noise filtering".

Load capacitance range

Required external capacitance range for DPS style operation	100 nF - 4.7 μ F per channel ^[111]
---	---

Slew rate control

Slew rate control	On-, off-, change level- rate individually controlled for each channel
-------------------	--

DC profiling

DC profiling	Supported per channel for voltage and current profiles
Sample rate	Up to 100 ksps

Data analysis

Data analysis	raw data are stored on board without software interaction and may be uploaded for further computation. Averaging, min and max number are calculated and stored on board
Result analysis	built-in result analysis available that indicate the measurement data is pass or fail against a user specified test limit

Voltmeter mode (high impedance mode)

Voltage range	-6 V to +15 V
Voltage resolution	500 μ V
Voltage accuracy	$\pm$ 4 mV
Voltage accuracy	$\pm$ 3 mV ^[112]
Input leakage current	< 40 nA

Analog performance**AWG mode (16 bit DAC)**

Step rate	200 ksps (kilo samples per second)
Analog bandwidth (-3 dB, 1 V swing)	5 kHz
Total channel memory	28 MB

Digitizer mode (18 bit ADC)

Sample rate	90 ksps (kilo samples per second)
Signal path bandwidth (-3dB)	20 kHz
Total channel memory	28 MB, 11 to 22 bytes required per sample

Trigger functions

Trigger	sequencer#controlled (no external trigger pin required)
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^[112] Base calibration and auto calibration period 1 month

DC Scale FVI16 Floating Power VI Source (E8026PV)

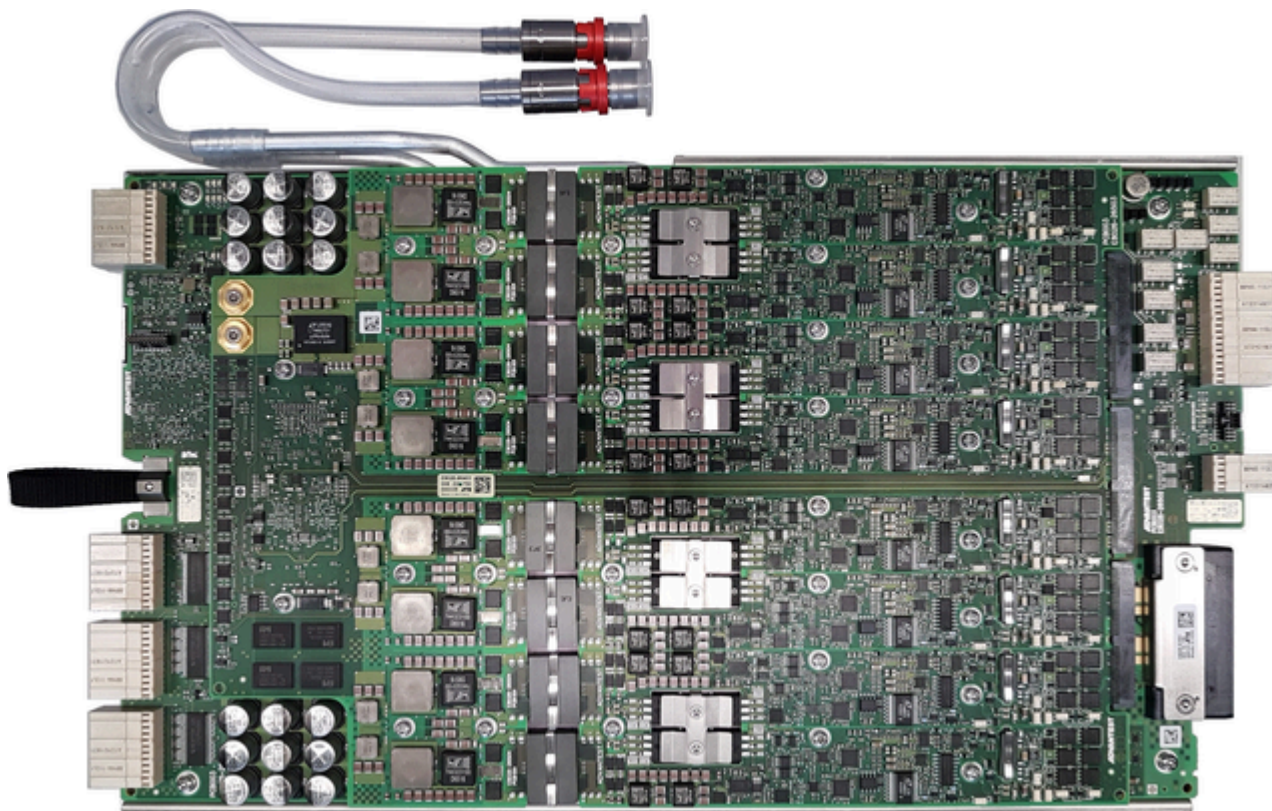
The FVI16 card is a high-current and high-voltage floating VI source that is optimized for a wide range of analog power tests. The card belongs to the DC Scale family and can reside in any channel card slot.

FVI16 provides 16 fully floating high-current and high-voltage channels. Each channel can supply voltages from -60 V to +120 V within a floating range of ± 200 V and offers pulse current capability up to ± 10 A per channel and ± 3 A continuous.

A floating differential high-voltage time measurement unit (TMU) with separate inputs is available for each channel.

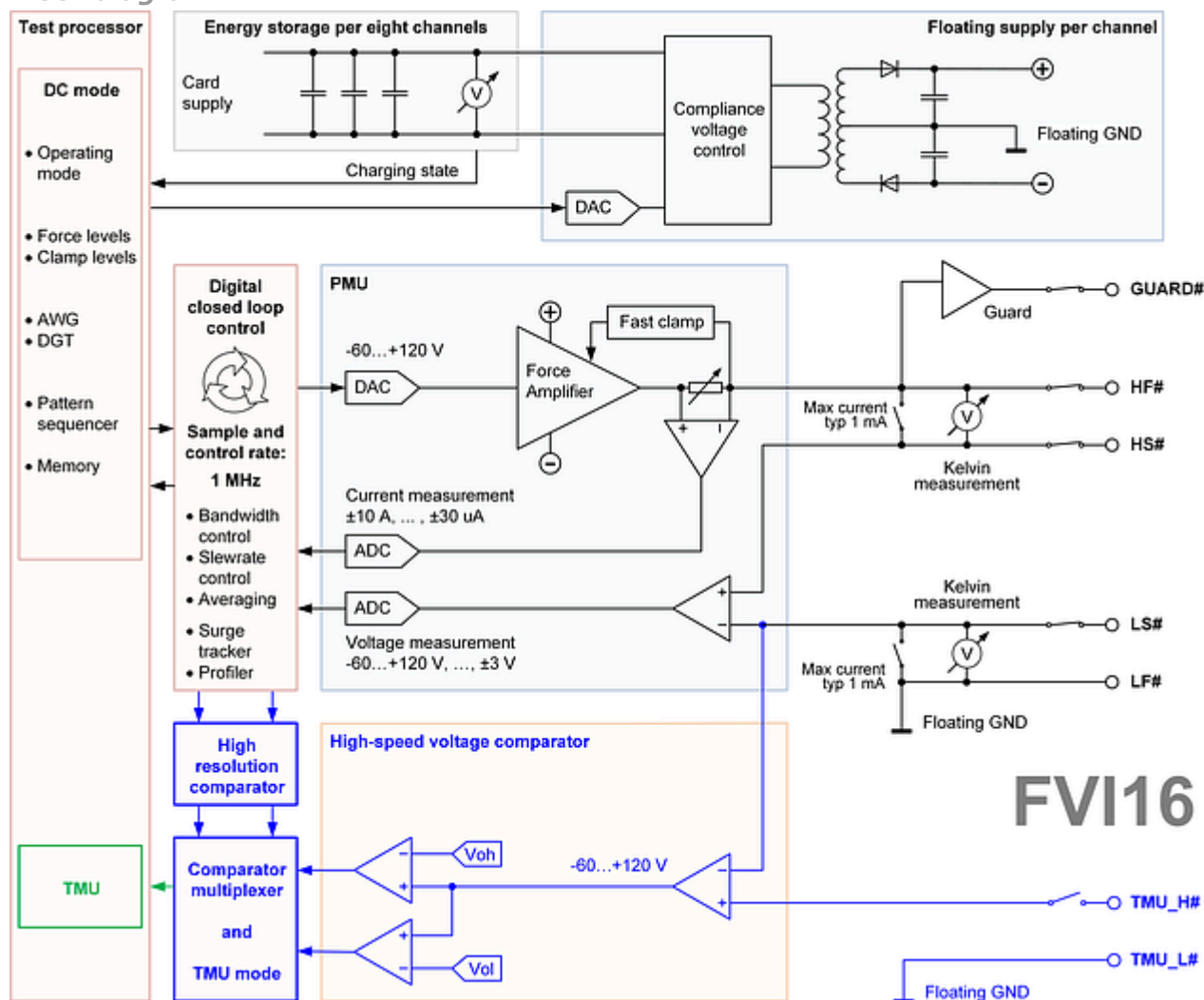
Scope of specifications

Advantest specifies and verifies the specifications at the DUT interface pogo level. Using an adequate DUT board and valid system calibration, these specifications are valid at the device under test as well. Specifications describe warranted product performance. Characteristics are included as typical values to provide additional useful information by describing typical non-warranted performance.



FVI16 block diagram and configuration

Block diagram



Configuration

FVI16				
Number of VI channels per card	16 (2 groups, 8 channels each)			
	A -Class	C-Class	S-Class	L-Class
Maximum number of cards per system	8	16	32	64
Maximum Power VI channels per system	128	256	512	1024

Other configuration data

- FVI16 is slot compatible to AVI64, DPS64, DPS128, and Pure Clock cards.
- All channels are fully floating.
- All channels have guard lines.
- FVI16 card can be mixed with any other device power supply, except PVI8.
- Diagnostics requires the Diagnostics Interconnect Board E8026DIAG to cover interconnection tests.

FVI16 voltage force and measurement specifications

Voltage force	Range	Resolution	Accuracy
High-power mode	± 3 V	> 17 bit	± (0.2 mV + 0.02 % of setting) ± (40 µV + 0.003 % of relative setting) ^[113]
	± 10 V	> 17 bit	± (1 mV + 0.02 % of setting) ± (80 µV + 0.003 % of relative setting) ^[113]
	± 30 V	> 17 bit	± (3m V + 0.02 % of setting) ± (700 µV + 0.01 % of relative setting) ^[113]
	± 60 V	> 17 bit	± (8 mV + 0.02 % of setting) ± (1 mV + 0.01 % of relative setting) ^[113]
Programmable compliance voltage	+10 V ... +65 V	> 12 bit	
High-voltage mode (max. 30 mA DC)	-60 V ... +120 V	> 17 bit	± (20 mV + 0.03 % of setting) ± (3.5 mV + 0.01 % of relative setting) ^[113]

Stacked voltage force accuracy: N times single channel voltage accuracy

Voltage measurement	Range ^[114]	Resolution	Accuracy ^[115]
High-power mode	± 3 V	25 µV	± (0.2 mV + 0.02 % of reading) ± (40 µV + 0.003 % of relative reading) ^[116]
	± 10 V	80 µV	± (1 mV + 0.02 % of reading) ± (80 µV + 0.003 % of relative reading) ^[116]
	± 30 V	250 µV	± (3 mV + 0.02 % of reading) ± (700 µV + 0.01 % of relative reading) ^[116]
	± 60 V	800 µV	± (8 mV + 0.02 % of reading) ± (1 mV + 0.01 % of relative reading) ^[116]
High-voltage mode (max. 30 mA DC)	-60 V ... +120 V	2 mV	± (20 mV + 0.03 % of reading) ± (3.5 mV + 0.01 % of relative reading) ^[116]

Stacked voltage measurement accuracy: N times single channel voltage accuracy

^[113] The relative accuracy specification describes the setting repeatability of a channel within 6 hours.
^[114] Voltage measurement range is controlled by voltage force settings (or voltage clamp, if active).
^[115] Requires averaging over 500 samples and 20 kHz measurement filter.
^[116] The relative accuracy specification describes the repeatability of measurements of a channel within 6 hours.

Current clamp	Range	Resolution	Accuracy ^[117]
High-power mode	±10 A	> 15 bit	± (10 mA + 0.2 % of setting)
	±3 A	> 15 bit	± (2.5 mA + 0.2 % of setting)
	±300 mA	> 15 bit	± (250 µA + 0.2 % of setting)
High-power mode or	±30 mA	> 15 bit	± (25 µA + 0.2 % of setting)
High-voltage mode	±3 mA	> 15 bit	± (2.5 µA + 0.2 % of setting + 50 nA/V * V _{act})
	±300 µA	> 15 bit	± (250 nA + 0.2 % of setting + 5 nA/V * V _{act})
	±30 µA	> 15 bit	± (25 nA + 0.2 % of setting + 2 nA/V * V _{act})

Ganged current clamp accuracy: N times single channel clamp accuracy

FVI16 current force and measurement specifications

Current force	Range ^[118]	Resolution	Accuracy ^[119]
High-power mode	±10 A	> 16 bit	± (2 mA + 0.25 % of setting + 0.1 mA/V * V _{act})
	±3 A	> 16 bit	± (1.5 mA + 0.15 % of setting + 33 µA/V * V _{act})
	±300 mA	> 17 bit	± (150 µA + 0.05 % of setting + 5 µA/V * V _{act})
High-power mode or High-voltage mode	±30 mA	> 17 bit	± (20 µA + 0.05 % of setting + 0.5 µA/V * V _{act})
	±3 mA	> 17 bit	± (500 nA + 0.05 % of setting + 50 nA/V * V _{act})
	±300 µA	> 17 bit	± (50 nA + 0.05 % of setting + 5 nA/V * V _{act})
	±30 µA	> 17 bit	± (10 nA + 0.1 % of setting + 1 nA/V * V _{act})

Ganged current force accuracy: N times single channel current force accuracy

Current measurement	Range ^[120]	Resolution	Accuracy ^{[119][121]}
High-power mode	±10 A	> 16 bit	± (2 mA + 0.25 % of reading + 0.1 mA/V * V _{act})
	±3 A	> 16 bit	± (1.5 mA + 0.15 % of reading + 33 µA/V * V _{act})
	±300 mA	> 17 bit	± (150 µA + 0.05 % of reading + 5 µA/V * V _{act})

^[117] V_{act} is the actual voltage of the channel, which is either the programmed force voltage or the voltage forced by the DUT, depending on the mode the FVI16 channel is in (forcing or clamping).

^[118] Current ranges can be selected independent of programmed voltage range.

^[119] V_{act} is the actual voltage of the channel, which is either the programmed force voltage or the voltage forced by the DUT, depending on the mode the FVI16 channel is in (forcing or clamping).

^[120] Current measurement range is controlled by current force settings (or current clamp, if active).

^[121] Requires averaging over 500 samples and 20 kHz measurement filter.

Current measurement	Range ^[120]	Resolution	Accuracy ^{[119][121]}
High-power mode or High-voltage mode	±30 mA	> 17 bit	± (20 µA + 0.05 % of reading + 0.5 µA/V * V _{act})
	±3 mA	> 17 bit	± (500 nA + 0.05 % of reading + 50 nA/V * V _{act})
	±300 µA	> 17 bit	± (50 nA + 0.05 % of reading + 5 nA/V * V _{act})
	±30 µA	> 17 bit	± (10 nA + 0.1 % of reading + 1 nA/V * V _{act})

Ganged current force accuracy: N times single channel current force accuracy

Voltage clamp	Range	Resolution	Accuracy
High-power mode	± 3 V	> 17 bit	± (3 mV + 0.1 % of setting)
	± 10 V	> 17 bit	± (10 mV + 0.1 % of setting)
	± 30 V	> 17 bit	± (30 mV + 0.1 % of setting)
	± 60 V	> 17 bit	± (60 mV + 0.1 % of setting)
High-voltage mode	-60 V ... +120	> 17 bit	± (120 mV + 0.1 % of setting)

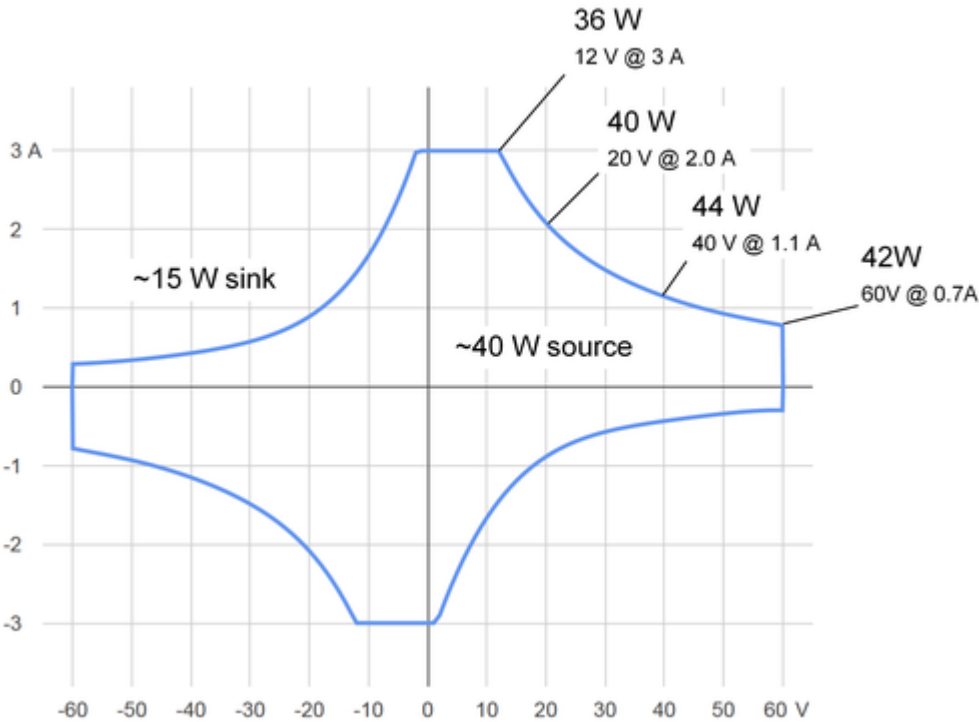
FVI16 voltage and current capability specifications

All specifications and diagrams in this section are valid at the pogo pins.

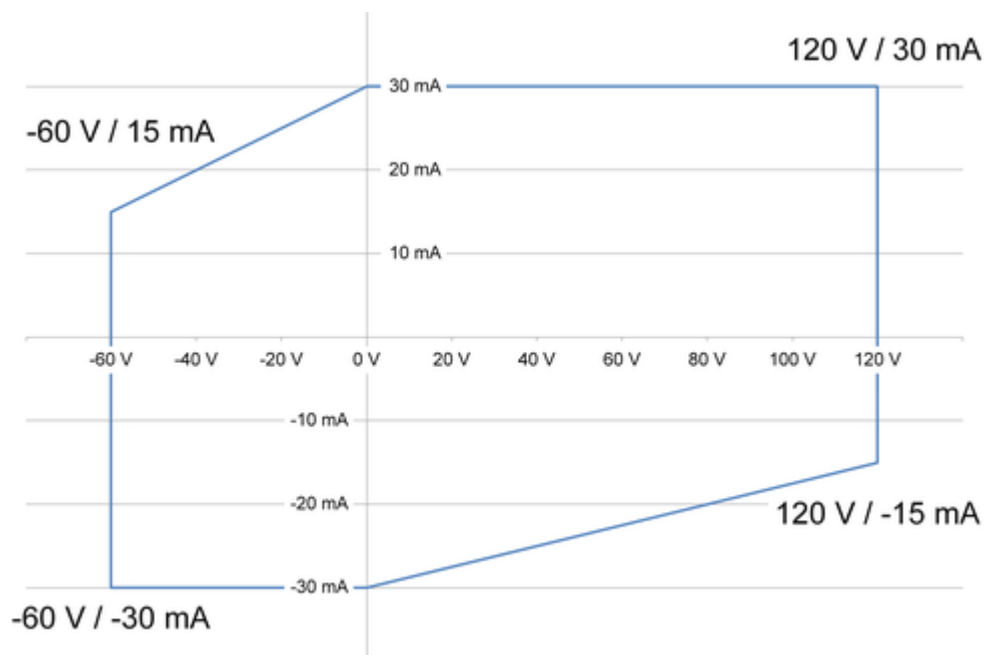
Continuous voltage and current capabilities per channel

The maximum total power budget per card is 400 W. This budget is equally distributed to two channel groups 1 ... 8 and 9 ... 16 providing 200 W per group.

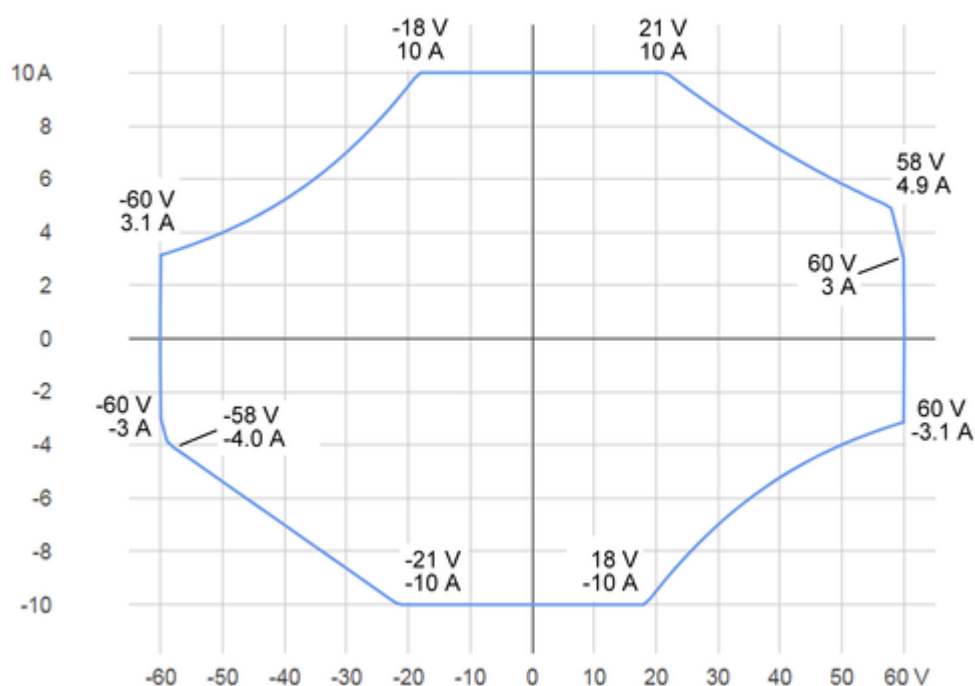
High-power mode:



^[120] Current measurement range is controlled by current force settings (or current clamp, if active).
^[121] Requires averaging over 500 samples and 20 kHz measurement filter.

High-voltage mode:**Maximum pulse power per channel (~250W)**

Pulse 0.5ms, 10% duty cycle

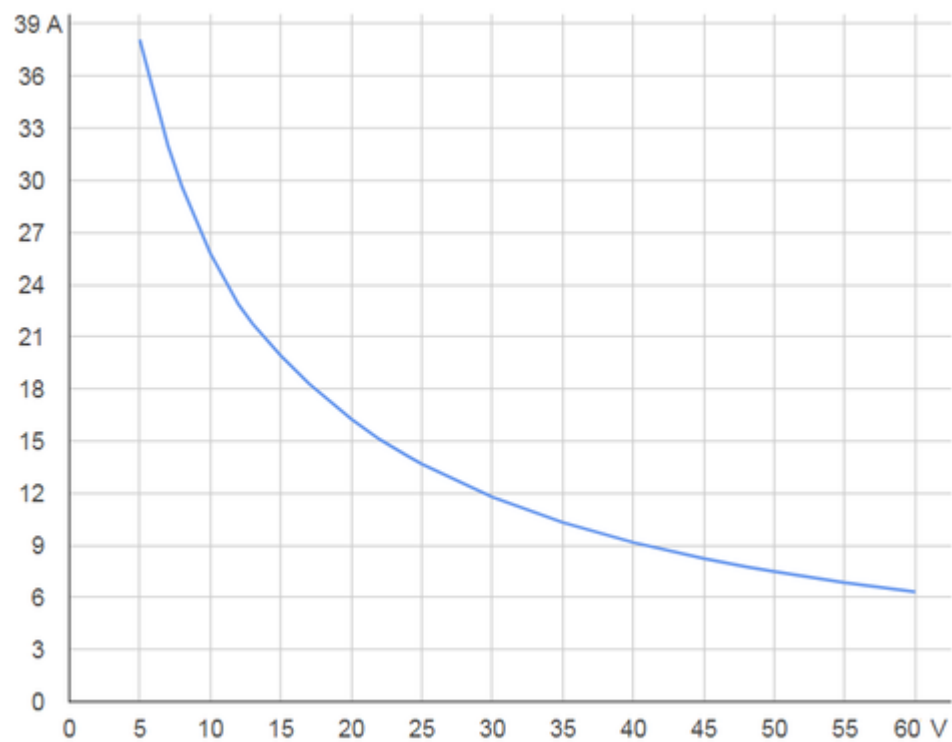


The diagram above shows the pulse power rating in high-power mode for 0.5 ms pulse width and 10 % duty cycle:

- X-axis: Voltage in [V]
- Y-axis: Current in [A]

Typical maximum continuous voltage and current capability per card

The following diagram shows the typical maximum continuous voltage and current capability per card at pogo interface with all 16 channels ganged.



FVI16 digitizer specifications

Per-pin voltage digitizer ^[122]	Range	Resolution	Accuracy ^[123]	INL typ.
High-power mode	±3 V	25 µV	± (1 mV + 0.02 % of reading)	± 3 LSB
	±10 V	80 µV	± (5 mV + 0.02 % of reading)	± 3 LSB
	±30 V	250 µV	± (15 mV + 0.02 % of reading)	± 6 LSB
	±60 V	800 µV	± (20 mV + 0.02 % of reading)	± 6 LSB
High-voltage mode	-60 V ... +120\ 2 mV		± (40 mV + 0.03 % of reading)	± 8 LSB
Input impedance (typical)	> 1 GΩ			
Input capacitance (typical)	200 pF			
Sampling rate	> 0 ... 1 Msps (Million samples per second, ADC sample clock 1 MHz)			
Analog bandwidth (typical, -3 dB)	250 kHz			
Programmable measurement filter	100 Hz ... 250 kHz			
Waveform capture memory per pin	not exceeding total available memory size per channel (107 MByte)			

^[122] Voltage and current digitizer are available for every channel and can be used in parallel. Software selectable range settings are the same as for VI modes.

^[123] Condition: Single sample and measurement filter set to 20 kHz

Per-pin current digitizer ^[122]				
Mode	Measurement filter	Range	Resolution	Accuracy ^[124]
High-power	1 kHz	±30 µA	> 17 bit	± (50 nA + 0.1 % of reading + 2 nA/V * V _{act})
High-voltage				± (100 nA + 0.1 % of reading + 2 nA/V * V _{act})
High-power	1 kHz	±300 µA	> 17 bit	± (250 nA + 0.05 % of reading + 5 nA/V * V _{act})
High-voltage				
High-power	1 kHz	±3 mA	> 17 bit	± (2.5 µA + 0.05 % of reading + 50 nA/V * V _{act})
High-voltage				
High-power	20 kHz	±30 mA	> 17 bit	± (100 µA + 0.05 % of reading)
High-voltage				
High-power	20 kHz	±300 mA	> 17 bit	± (750 µA + 0.05 % of reading)
		±3 A	> 17 bit	± (7.5 mA + 0.15 % of reading)
		±10 A	> 16 bit	± (15 mA + 0.25 % of reading)
Sampling rate		> 0 ... 1 Msps (Million samples per second, ADC sample clock 1MHz)		
Programmable measurement filter		100 Hz ... 250 kHz		
Waveform capture memory per pin		not exceeding total available memory size per channel (107 MByte)		

FVI16 AWG specifications

The FVI16 output can be sequencer#controlled via events (defined in pattern, stored in test processor memory) to generate arbitrary waveforms.

Per-pin voltage AWG	
Voltage ranges	±3 V, ±10 V, ±30 V, ±60 V, -60 V ... +120 V
Resolution	> 17 bit
Accuracy	See voltage force accuracy of ranges (after settling)
Maximum current	See voltage / current capabilities (VI ratings)
Sampling rate	> 0 ... 1 Msps (Million samples per second)
Analog bandwidth (typical)	50 kHz
Waveform memory (per pin) ^[125]	Up to 1 Msamples (Million samples)
INL (typical)	
	±3 V range: ±3 LSB
	±10 V range: ±3 LSB
	±30 V range: ±6 LSB
	±60 V range: ±6 LSB
	-60 V ... +120 V range: ±8 LSB

^[124] V_{act} is the actual voltage of the channel, which is either the programmed force voltage or the voltage forced by the DUT, depending on the mode the FVI16 channel is in (forcing or clamping).

^[125] Waveform memory shared between all AWG modes and DC test points.

Per-pin current AWG

Current ranges	$\pm 30 \mu\text{A}$, $\pm 300 \mu\text{A}$, $\pm 3 \text{ mA}$, $\pm 30 \text{ mA}$, $\pm 300 \text{ mA}$, $\pm 3 \text{ A}$, $\pm 10 \text{ A}$
Resolution	> 17 bit
Accuracy	See current force accuracy of ranges (after settling)
Maximum voltage	See voltage / current capabilities (VI ratings)
Sampling rate	> 0 ... 1 Msps (Million samples per second)
Analog bandwidth (typical)	50 kHz
Waveform memory (per pin) ^[125]	Up to 1 Msamples (Million samples)

FVI16 TMU specifications**Timing Measurement Unit (TMU)**

Timing measurements are done using the comparator functionality available per channel (see comparator specifications above).

The TMU allows for time delay / skew measurements ($V_{OH} / I_{CH} \# V_{OH} / I_{CH}$, $V_{OL} / I_{CL} \# V_{OL} / I_{CL}$) also across different instrument types (i.e. PinScale1600), frequency / pulse width measurements, and rise / fall time measurements ($V_{OL} / I_{CL} \# V_{OH} / I_{CH}$, $V_{OH} / I_{CH} \# V_{OL} / I_{CL}$).

Timing Measurement Unit (TMU)^[126]

DC / AC Performance	See comparators
Accuracy for rise / fall time measurements (single shot)	$\pm 4 \text{ ns}$
Accuracy for skew to other FVI16 TMU channel (single shot)	$\pm 4 \text{ ns}$
Accuracy for single time stamp to ARM (time zero)	$\pm 2 \text{ ns}$

The TMU can be used with all compare modes as defined below. The following combinations are supported.

Compare modes	VI Comparator	Inputs
High-speed voltage	n.a.	TMU_H# and TMU_L#
High-resolution voltage	supported	HS# and LS#
High-resolution current	supported	HF#

High-speed (HS) comparator inputs

Programmable thresholds	V_{OL} , V_{OH}
Input voltage and threshold ranges	$-60 \text{ V} \leq V_{OL}$, $V_{OH} \leq +120 \text{ V}$
Threshold accuracy	$\pm (200 \text{ mV} + 0.1 \% \text{ of setting})$
TMU_H safe input voltage	Clamped to programmed VI output compliance voltage of same channel. Input voltage max 20 V over programmed compliance range of same channel.
TMU_L safe input voltage	Keep at the same voltage level as LS pin of the channel
Threshold hysteresis	25 mV (typical)
Programmable compare event hold-off time	0.0 ... 20.475 μs Default hold-off time: 100 ns

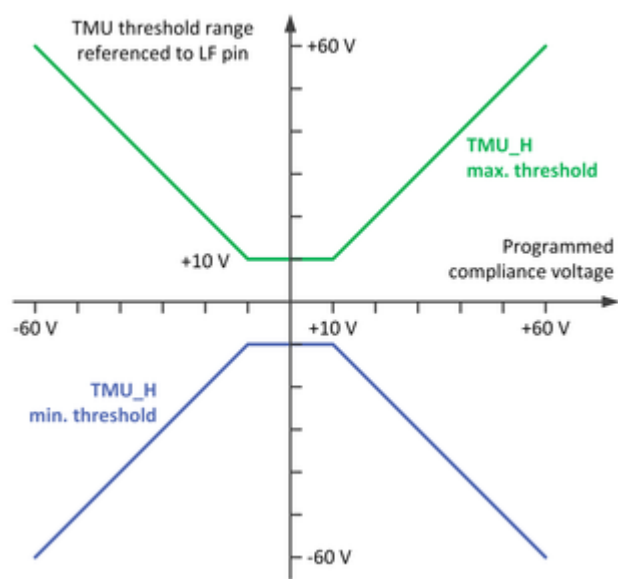
^[126] Condition: Compliance voltage set to 65V

High-speed (HS) comparator inputs

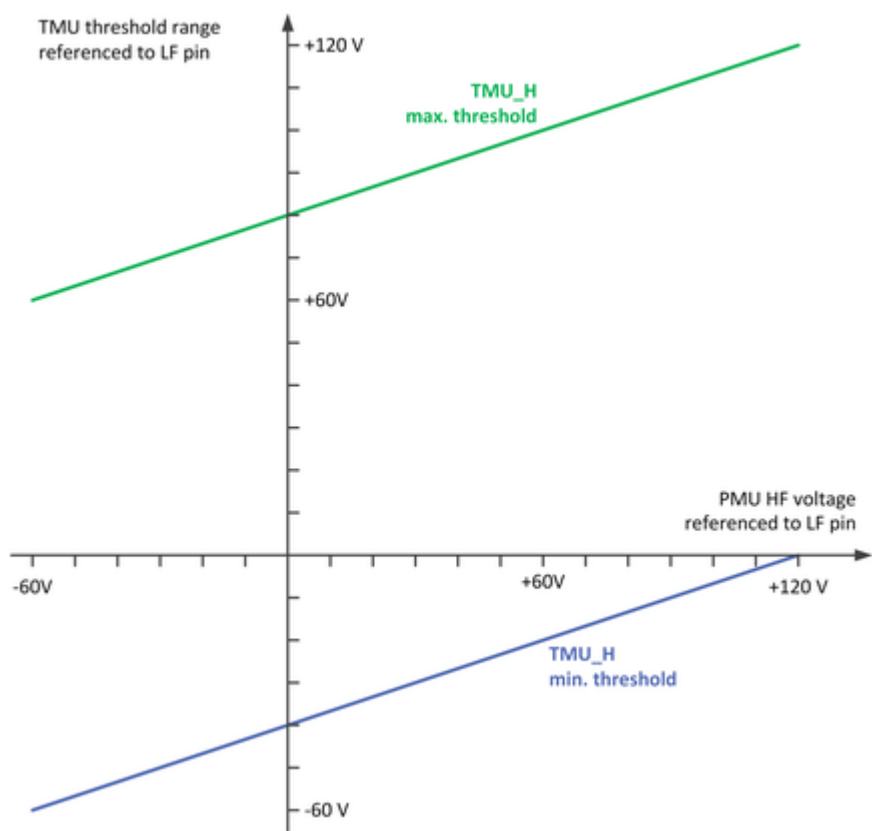
Input impedance (typical)	>1 MΩ
Signal path bandwidth	45 MHz (typical)
Compare timing resolution	1 ps
PMU dependency	See diagram below

High-speed (HS) comparator threshold range

High-power (±60 V) mode:



High-voltage (-60 ... 120 V) mode:



High-resolution TMU mode (Voltage comparator via HS/LS and current comparator via HF)

Programmable dual thresholds		$V_{OL}, V_{OH}, I_{CL}, I_{CH}$ (with programmable hysteresis)
Threshold ranges (voltage) ^[127]		$\pm 3\text{ V}, \pm 10\text{ V}, \pm 30\text{ V}, \pm 60\text{ V}, -60\text{ V} \dots +120\text{ V}$
Threshold accuracy (voltage)		
	$\pm 3\text{ V}$ range:	$\pm (2\text{ mV} + 0.02\text{ \% of setting})$
	$\pm 10\text{ V}$ range:	$\pm (5\text{ mV} + 0.02\text{ \% of setting})$
	$\pm 30\text{ V}$ range:	$\pm (15\text{ mV} + 0.02\text{ \% of setting})$
	$\pm 60\text{ V}$ range:	$\pm (40\text{ mV} + 0.02\text{ \% of setting})$
	$-60\text{ V} \dots +120\text{ V}$ range:	$\pm (100\text{ mV} + 0.03\text{ \% of setting}) @ 250\text{ kHz}$
Threshold hysteresis (voltage)		Programmable within full range
Threshold ranges (current) ^[128]		$\pm 30\text{ }\mu\text{A}, \pm 300\text{ }\mu\text{A}, \pm 3\text{ mA}, \pm 30\text{ mA}, \pm 300\text{ mA}, \pm 3\text{ A}, \pm 10\text{ A}$
Threshold accuracy (current) ^[129]		
High-current mode	$\pm 30\text{ }\mu\text{A}$ range:	$\pm (300\text{ nA} + 0.1\text{ \% of setting} + 1\text{ nA/V} * V_{act})$
High-voltage mode		$\pm (1.5\text{ }\mu\text{A} + 0.1\text{ \% of setting} + 1\text{ nA/V} * V_{act})$
High-current mode	$\pm 300\text{ }\mu\text{A}$ range:	$\pm (3\text{ }\mu\text{A} + 0.05\text{ \% of setting} + 5\text{ nA/V} * V_{act})$
High-voltage mode		$\pm (15\text{ }\mu\text{A} + 0.05\text{ \% of setting} + 5\text{ nA/V} * V_{act})$
High-current mode	$\pm 3\text{ mA}$ range:	$\pm (20\text{ }\mu\text{A} + 0.05\text{ \% of setting} + 50\text{ nA/V} * V_{act})$
High-voltage mode		$\pm (100\text{ }\mu\text{A} + 0.05\text{ \% of setting} + 50\text{ nA/V} * V_{act})$
High-current mode	$\pm 30\text{ mA}$ range:	$\pm (100\text{ }\mu\text{A} + 0.05\text{ \% of setting} + 0.5\text{ }\mu\text{A/V} * V_{act})$
High-voltage mode		$\pm (200\text{ }\mu\text{A} + 0.05\text{ \% of setting} + 0.5\text{ }\mu\text{A/V} * V_{act})$
High-current mode	$\pm 300\text{ mA}$ range:	$\pm (1\text{ mA} + 0.05\text{ \% of setting} + 5\text{ }\mu\text{A/V} * V_{act})$
	$\pm 3\text{ A}$ range:	$\pm (10\text{ mA} + 0.15\text{ \% of setting} + 33\text{ }\mu\text{A/V} * V_{act})$
	$\pm 10\text{ A}$ range:	$\pm (30\text{ mA} + 0.25\text{ \% of setting} + 0.1\text{ mA/V} * V_{act})$
Threshold hysteresis (current)		Programmable within full range
Compare timing accuracy		$\pm 500\text{ ns}$ ^[130]
Compare timing resolution		5 ns (interpolated) ^[131]
Signal path bandwidth		250 kHz (typical)

High-speed TMU mode Typical apps cases	Range	Resolution	Accuracy (typ.) ^[132]
Rise time (single shot measurement)	$20 \text{ ns} \dots 100 \text{ ms}$	$< 1 \text{ ns}$	$\pm 4 \text{ ns}$
Fall time (single shot measurement)	$20 \text{ ns} \dots 100 \text{ ms}$	$< 1 \text{ ns}$	$\pm 4 \text{ ns}$
Rise time (repetitive)	$0 \text{ ns} \dots 100 \text{ ms}$	$< 1 \text{ ns}$	$\pm 4 \text{ ns}$

^[127] Voltage measure range is used in compare mode.

^[128] Current measure range is used in compare mode.

^[129] V_{act} is the actual voltage of the channel, which is either the programmed force voltage or the voltage forced by the DUT, depending on the mode the channel is in (forcing or clamping).

^[130] Additional fixed channel delay in high resolution mode of $3 \mu\text{s}$.

^[131] Sampling rate = $1 \mu\text{s}$

^[132] Condition: Slew rate $<$ tracking slew rate

High-speed TMU mode Typical apps cases	Range	Resolution	Accuracy (typ.) ^[132]
Fall time (repetitive)	0 ns ... 100 ms	< 1 ns	±4 ns
Pulse-width (single shot)	20 ns ... 100 ms	< 1 ns	±4 ns
Pulse-width (time delta of repetitive pulse)	0 ns ... 100 ms	< 1 ns	±4 ns
Frequency (= multiple period duration measure)	0 ... 50 MHz	< 1 ns	±4 ns
Delay to digital	0ns ... 100 ms	< 1 ns	±4 ns ^[133]
Delay to other FVI16 channel	0ns ... 100 ms	< 1 ns	±4 ns ^[133]

FVI16 channel stacking and ganging specifications

Channel stacking	
Maximum channel floating range	±200 V to system ground ^[134]
Maximum stacked channels	3 channels; any channel combination within same card
Typical source stacks	+200 V: AVI64 + FVI16 = 80 V + 120 V = 200 V (applicative) +180 V: 3 x FVI16 = 3 x 60 V = 180 V (native, SW support) -180 V: 3 x FVI16 = 3 x -60 V = -180 V (native, SW support)
Native SW supported voltage ranges for stacking	±60 V
Native SW supported current ranges for stacking	±30 µA, ±300 µA, ±3 mA, ±30 mA, ±300 mA, ±3 A and ±10 A
Software support	Stacked setup represented as single channel Flexible stack / unstack within same test program
Hardware support	Routing on DUT board Layout need to have sufficient high voltage creepage distance
Channel ganging	
Maximum ganged source channels	16 channels per card with integrated software support; any channel combination within same card > 16 channels, user need to take care of individual source handling
Maximum ganged current	±155 A pulsed per card for full 16 channel setup, ±97 A pulsed per card for a 10 channel setup
Native SW supported voltage ranges for ganging	±3 V, ±10 V, ±30 V and ±60 V
Native SW supported current ranges for ganging	±10 A

^[132] Condition: Slew rate < tracking slew rate

^[133] Requires application specific calibration of the path

^[134] The voltage at any of the channel pins (HF#, HS#, LF#, LS#, TMU_H#, TMU_L#) must not exceed the floating range.

Channel ganging

Software support	Ganged setup represented as single channel Flexible gang / ungang within same test program
Hardware support	Wiring of ganged channels on DUT board; layout needs to have sufficient current capability

FVI16 VI slew rate and bandwidth control specifications

Digital feedback loop enables flexible adjustments of slew rate and bandwidth of force signals to optimize the adaption upon different load conditions.

Programmable slew rate for voltage

Mode	Range	Voltage slew rate		
		Min	Max	Default
High-power	± 3 V	63 $\mu\text{V}/\mu\text{s}$	1 $\text{V}/\mu\text{s}$	0.5 $\text{V}/\mu\text{s}$
	± 10 V	210 $\mu\text{V}/\mu\text{s}$	1 $\text{V}/\mu\text{s}$	0.5 $\text{V}/\mu\text{s}$
	± 30 V	625 $\mu\text{V}/\mu\text{s}$	1 $\text{V}/\mu\text{s}$	0.5 $\text{V}/\mu\text{s}$
	± 60 V	2.1 $\text{mV}/\mu\text{s}$	1 $\text{V}/\mu\text{s}$	0.5 $\text{V}/\mu\text{s}$
High-voltage	-60 V ... +120 V	2.4 $\text{mV}/\mu\text{s}$	0.2 $\text{V}/\mu\text{s}$	0.1 $\text{V}/\mu\text{s}$

Programmable slew rate for current

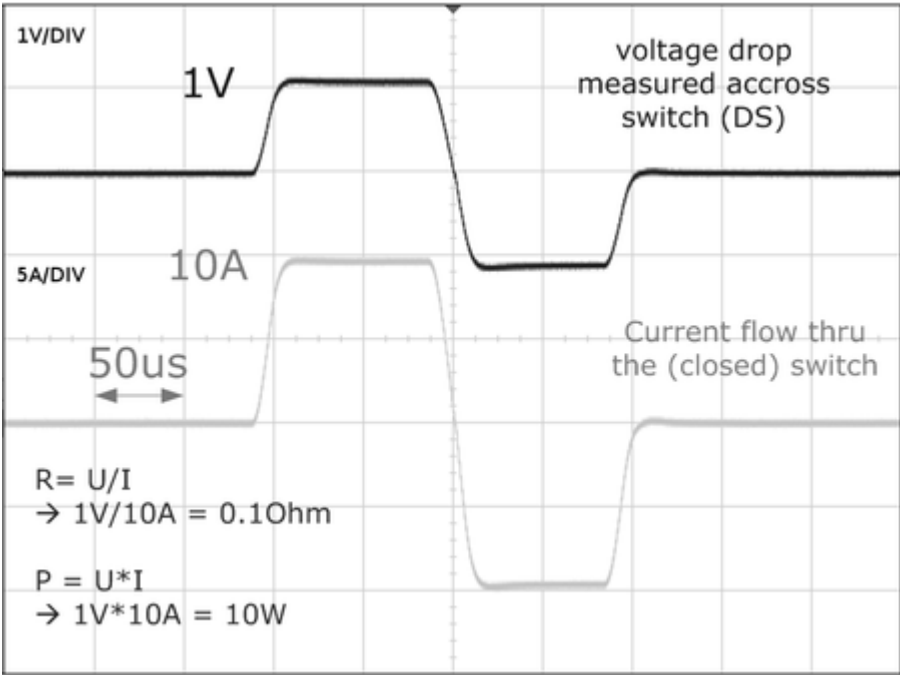
Mode	Range	Current slew rate		
		Min	Max	Default
High-power	± 30 μA	540 $\text{pA}/\mu\text{s}$	3 $\mu\text{A}/\mu\text{s}$	3 $\mu\text{A}/\mu\text{s}$
	± 300 μA	5.4 $\text{nA}/\mu\text{s}$	30 $\mu\text{A}/\mu\text{s}$	30 $\mu\text{A}/\mu\text{s}$
	± 3 mA	54 $\text{nA}/\mu\text{s}$	300 $\mu\text{A}/\mu\text{s}$	300 $\mu\text{A}/\mu\text{s}$
	± 30 mA	540 $\text{nA}/\mu\text{s}$	3 $\text{mA}/\mu\text{s}$	3 $\text{mA}/\mu\text{s}$
	± 300 mA	6.9 $\mu\text{A}/\mu\text{s}$	30 $\text{mA}/\mu\text{s}$	30 $\text{mA}/\mu\text{s}$
	± 3 A	94 $\mu\text{A}/\mu\text{s}$	300 $\text{mA}/\mu\text{s}$	300 $\text{mA}/\mu\text{s}$
	± 10 A	517 $\mu\text{A}/\mu\text{s}$	1 $\text{A}/\mu\text{s}$	1 $\text{A}/\mu\text{s}$
High-voltage	± 30 μA	540 $\text{pA}/\mu\text{s}$	3 $\mu\text{A}/\mu\text{s}$	3 $\mu\text{A}/\mu\text{s}$
	± 300 μA	5.4 $\text{nA}/\mu\text{s}$	30 $\mu\text{A}/\mu\text{s}$	30 $\mu\text{A}/\mu\text{s}$
	± 3 mA	54 $\text{nA}/\mu\text{s}$	300 $\mu\text{A}/\mu\text{s}$	300 $\mu\text{A}/\mu\text{s}$
	± 30 mA	540 $\text{nA}/\mu\text{s}$	3 $\text{mA}/\mu\text{s}$	3 $\text{mA}/\mu\text{s}$

Programmable bandwidth for voltage and current

Mode	Range	Voltage filter		
		Min	Max	Default
High-power	Up to ± 60 V	100 Hz	50 kHz	10 kHz
High-voltage	-60 V ... +120 V	100 Hz	5 kHz	5 kHz

Mode	Range	Current filter		
		Min	Max	Default
High-power	Up to ±60 V All current ranges	100 Hz	50 kHz	10 kHz
High-voltage	-60 V ... +120 V All current ranges	100 Hz	50 kHz	50 kHz

Typical settling times (for example, 10 A High-power RDSON Test)



Condition	
R_{DUT} :	0.1 Ω
I_{DUT} :	10 A
Force mode:	CFVM
Sequence:	1. Pulse to slew up 2. Settling to spec 3. Measure voltage

Expectation	
Pulse time including settling time:	< 50 μ s
Stable measurement at:	< 40 μ s
Maximum overshoot:	< 5 %

FVI16 load capacitance and measurement filter specifications

The digital feedback loop offers flexible adjustments to the available load capacity and offers also programmable measurement filters.

Programmable load capacitance

Mode	Range	Load capacitance		
		Min	Max	Default
High-power	±30 μ A	0 nF	0.66 nF	0 nF
	±300 μ A	0 nF	6.6 nF	0 nF
	±3 mA	0 nF	66 nF	0 nF
	±30 mA	0 nF	0.66 μ F	0 nF
	±300 mA	0 nF	6.6 μ F	0 nF
	±3 A	0 nF	33 μ F	0 nF
	±10 A	0 nF	100 μ F	0 nF
High-voltage	±30 μ A	0 nF	0.66 nF	0 nF
	±300 μ A	0 nF	6.6 nF	0 nF
	±3 mA	0 nF	66 nF	0 nF
	±30 mA	0 nF	0.66 μ F	0 nF

Programmable measurement filter for voltage and current

Mode	Range	Measurement filter		
		Min	Max	Default
High-power	All voltage ranges and all current ranges	100 Hz	250 kHz	250 kHz
High-voltage	All current ranges	100 Hz	250 kHz	250 kHz

One common filter setting applies for both voltage and current measurement.

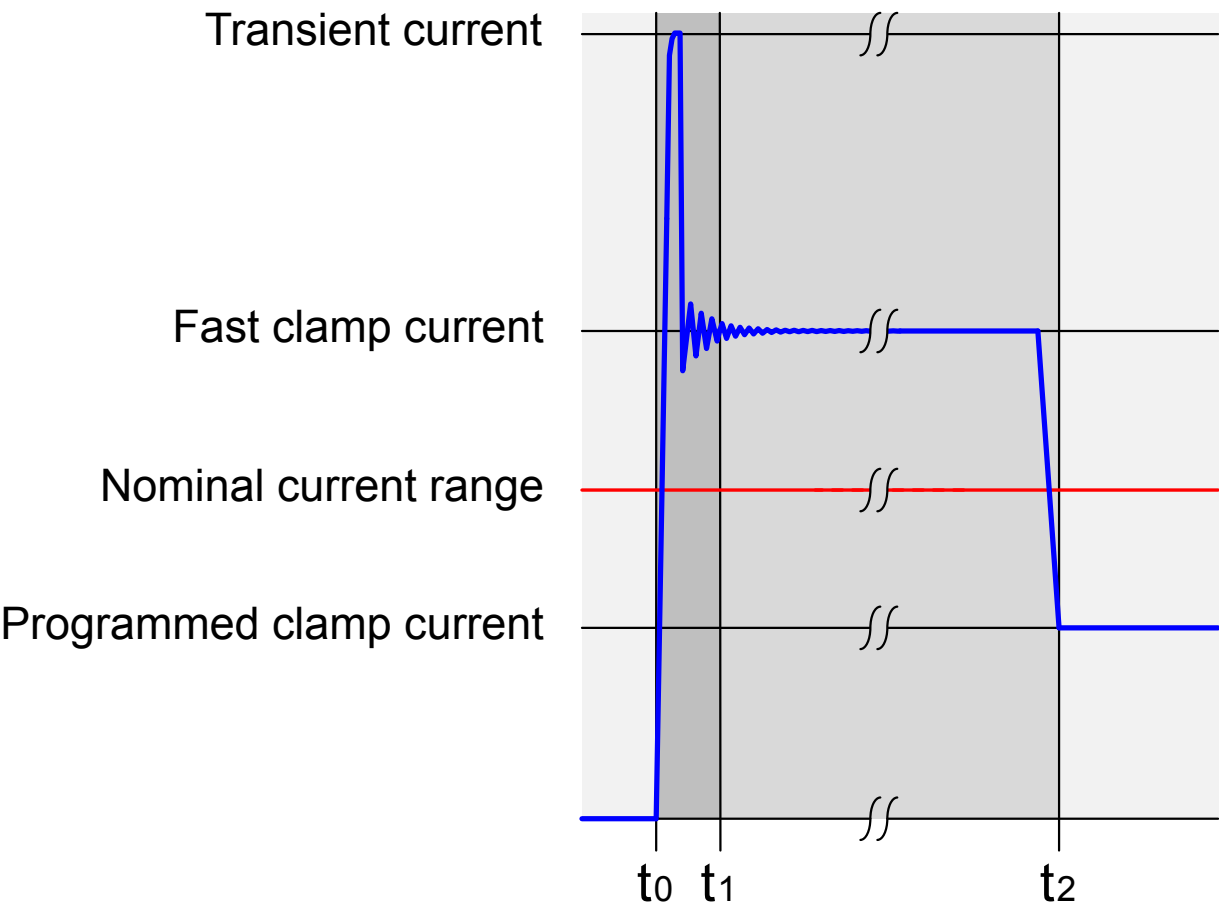
Typical measurement filter settings

Filter	Tau	Settling time (6 Tau)	Application
250 kHz	0.64 μ s	3.8 μ s	for fastest measurement results
50 kHz	3.18 μ s	19.1 μ s	for fast measurement results
20 kHz	7.96 μ s	47.8 μ s	for accurate measurements results
5 kHz	31.8 μ s	191.0 μ s	for measurements in the high-voltage mode

FVI16 fast current clamp specifications

A sudden short between HF# and LF# pins causes high current flow exceeding current limit settings for a limited time: Current spike transients.

The fast current clamp is a circuitry within the FVI16 channel to minimize the current spike energy by reducing the current and by reducing the spike duration. In a second stage the control loop reduces current flow to the programmed current clamp settings.

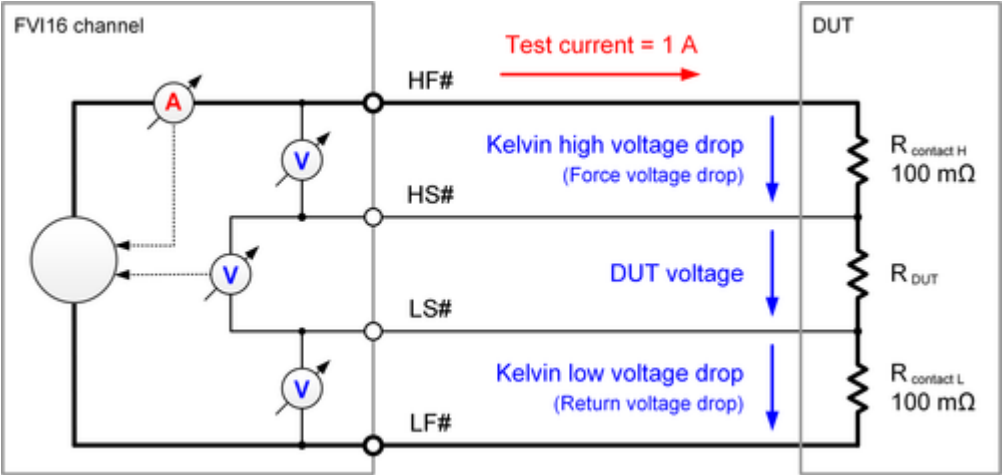


High-power mode	Transient time (typ.)		Typical fast clamp current t1 ... t2
±3 V, ±10 V, ±30 V, ±6 t1	t2		
±10 A	< 2 μs	< 20 μs	17 A
±3 A	< 2 μs	< 40 μs	5 A
±300 mA	< 2 μs	< 45 μs	600 mA
±30 mA	< 2 μs	< 80 μs	75 mA
±3 mA	< 2 μs	< 100 μs	20 mA
±300 μA	< 2 μs	< 100 μs	12 mA
±30 μA	< 2 μs	< 100 μs	< 10 mA

High-voltage mode	Transient time (typ.)		Typical fast clamp current t1 ... t2
-60 V ... +120 V	t1	t2	
±30 mA	< 2 μs	< 125 μs	75 mA
±3 mA	< 2 μs	< 250 μs	20 mA
±300 μA	< 2 μs	< 250 μs	12 mA
±30 μA	< 2 μs	< 250 μs	< 10 mA

FVI16 Kelvin measurement unit specifications

The Kelvin measurement unit is measuring the voltage drop of the force lines between HF# - HS# and LF# - LS#. A user defined test current (for example, 1 A, ~10 % of the maximum range) has to be forced through the DUT in order to measure the $I \cdot R$ voltage drop.



High force Kelvin drop voltage (Force drop voltage)

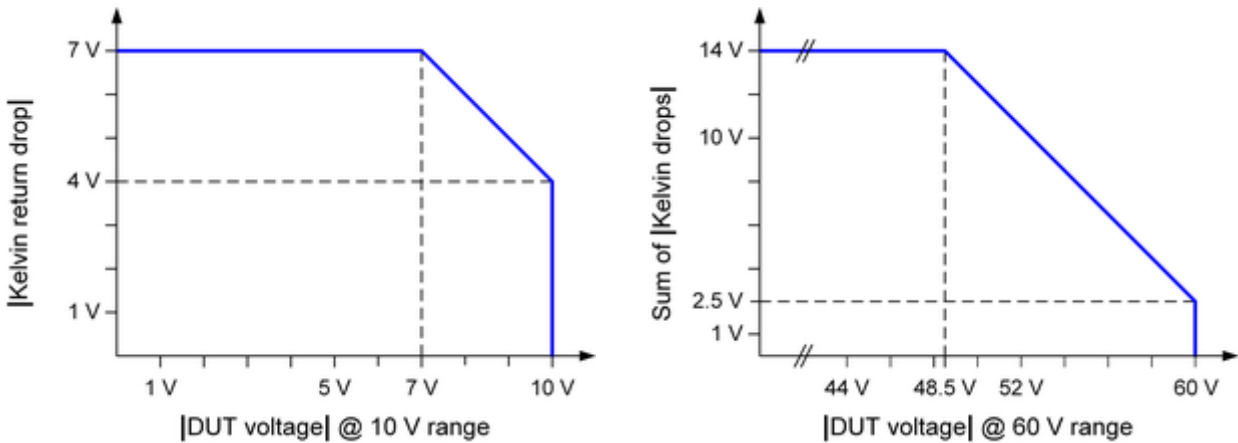
Voltage range	Condition	Programmable Accuracy range	Example and comment
±3 V		±7 V ± (20 mV * I _{force} /A + 10 mV)	e.g. @ 1 A = ±30 mV e.g. @ 10 A = ±210 mV
±10 V		±7 V ± (20 mV * I _{force} /A + 10 mV)	e.g. @ 1 A = ±30 mV e.g @ 10 A = ±210 mV
±30 V		±7 V ± (20 mV * I _{force} /A + 10 mV)	e.g. @ 1 A = ±30 mV e.g @ 10 A = ±210 mV
±60 V	V _{force} ≤ 48.5 V	±7 V ± (20 mV * I _{force} /A + 10 mV)	e.g. @ 1 A = ±30 mV
	V _{force} > 48.5 V	< 7 V see right diagram below	e.g @ 10 A = ±210 mV V _{force} ≠ V _{clamp} for I _{force} operation
-60 V ... + 120		±1 V ±12 mV	

Low force Kelvin drop voltage (Return drop voltage)

Voltage range	Condition	Programmable Accuracy range	Example and comment
±3 V		±7 V ± (20 mV * I _{force} /A + 10 mV)	e.g. @ 1 A = ±30 mV e.g. @ 10 A = ±210 mV
±10 V	Kelvin return threshold + V _{force} ≤ 14 V	±7 V ± (20 mV * I _{force} /A + 10 mV)	e.g. @ 1 A = ±30 mV e.g @ 10 A = ±210 mV
	Kelvin return threshold + V _{force} > 14 V	< 7 V see left diagram below	V _{force} ≠ V _{clamp} for I _{force} operation, a kelvin low flag is set in case of violation

Voltage range	Condition	Programmable Accuracy range	Example and comment
±30 V		±7 V	$\pm (20 \text{ mV} * I_{\text{force}}/\text{A} + 10 \text{ mV})$ e.g. @ 1 A = ±30 mV e.g @ 10 A = ±210 mV
±60 V	$ V_{\text{force}} \leq 48.5 \text{ V}$ $ V_{\text{force}} > 48.5 \text{ V}$	±7 V < 7 V see right diagram below	$\pm (20 \text{ mV} * I_{\text{force}}/\text{A} + 10 \text{ mV})$ e.g. @ 1 A = ±30 mV e.g @ 10 A = ±210 mV $V_{\text{force}} \neq \text{clamp}$ for I_{force} operation
-60 V ... + 120		N/A	N/A

Maximum Kelvin drop voltage vs. DUT voltage



For the diagram above, you must also consider an internal voltage drop, which is defined by 1.0 Ω multiplied by the programmed force or clamp current.

Automatic disconnect for weak force to sense connection.

To protect probe card needles and test sockets, the VI source output gets automatically disconnected in case a weak force to sense connection leads to a Kelvin voltage drop exceeding the programmed threshold voltage and the VI is set to the 3 A or 10 A range^[135].

Example:

Configuration

- R_{contact} = 2.0 Ω
- Test current = 1.0 A
- Programmed Kelvin threshold limit = 1.0 V

Voltage drop

$R_{\text{contact}} * \text{Test current} = 2.0 \text{ } \Omega * 1.0 \text{ A} = 2.0 \text{ V}$

Result

The voltage drop exceeds the programmed threshold of 1.0 V and the FVI16 channel gets disconnected.

^[135] SmarTest 7.5.2.1 and higher. With lower SmarTest versions, the channel is disconnected in all current ranges.

FVI16 other specifications

Interface characteristics		
Compliance range if disconnected	HF, LF, HS, LS	±200 V
Input leakage current at disconnect state (typical)	HF, LF, HS, LS	< 20 nA
Lumped capacitance at disconnect state (typical)	LF to system GND	< 50 nF
	HS to LS	< 100 pF
	HS to LF	< 200 pF
	HF to LF	< 5 nF
Surge tracker		
Trigger	Voltage surges are detected against programmable minimum and maximum values. The surge tracker always checks against both high (positive) and low (negative) thresholds, which can be set individually. Two flags indicate if thresholds have been exceeded. These flags are latched (stored) and can be retrieved any time in the testflow.	
Minimum detectable pulse width	4 μs (typical)	
Threshold voltage accuracy	See voltage digitizer specification (single shot)	
Input and threshold range	Same as the measure voltage range (max. -60 ... 120 V)	
DC profiling		
DC profiling	Supported per channel for voltage and current profiles. Profiling of the charge status of the on-board energy storages is supported.	
Maximum sample rate	1 Msps The maximum sample rate is available per channel group 1 ... 8 and 9 ... 16 and is shared by all channels of the group:	
	Channels	Maximum sample rate
	1	1 Msps
	2	500 ksps
	3	333 ksps
	...	...
	Profiling the current of ganging groups is supported with a maximum sample rate of 125 ksps.	

FVI16 pogo pin assignment

The E8026PVP pogo cable assembly connects the sixteen floating channels of the DC Scale FVI16 card to the DUT board.

Pogo pin assignment

Block A					Block B				
	a	b	c	d		a	b	c	d
1	DNC	HF1	LF1	HS1	1	DNC	HF5	LF5	HS5
2	TMU_H1	LF1	HF1	LS1	2	TMU_H5	LF5	HF5	LS5
3	TMU_L1	HF1	LF1	GUARD1	3	TMU_L5	HF5	LF5	GUARD5
4	Separation				4	Separation			
5	DNC	HF2	LF2	HS2	5	DNC	HF6	LF6	HS6
6	TMU_H2	LF2	HF2	LS2	6	TMU_H6	LF6	HF6	LS6
7	TMU_L2	HF2	LF2	GUARD2	7	TMU_L6	HF6	LF6	GUARD6
8	Separation				8	Separation			
9	DNC	HF3	LF3	HS3	9	DNC	HF7	LF7	HS7
10	TMU_H3	LF3	HF3	LS3	10	TMU_H7	LF7	HF7	LS7
11	TMU_L3	HF3	LF3	GUARD3	11	TMU_L7	HF7	LF7	GUARD7
12	Separation				12	Separation			
13	DNC	HF4	LF4	HS4	13	DNC	HF8	LF8	HS8
14	TMU_H4	LF4	HF4	LS4	14	TMU_H8	LF8	HF8	LS8
15	TMU_L4	HF4	LF4	GUARD4	15	TMU_L8	HF8	LF8	GUARD8
16	Separation				16	Separation			
17	DNC	DNC	SAFETY1_GND	SAFETY1_5V	17	DNC	DNC	DNC	DNC
Orientation mark °					Orientation mark °				

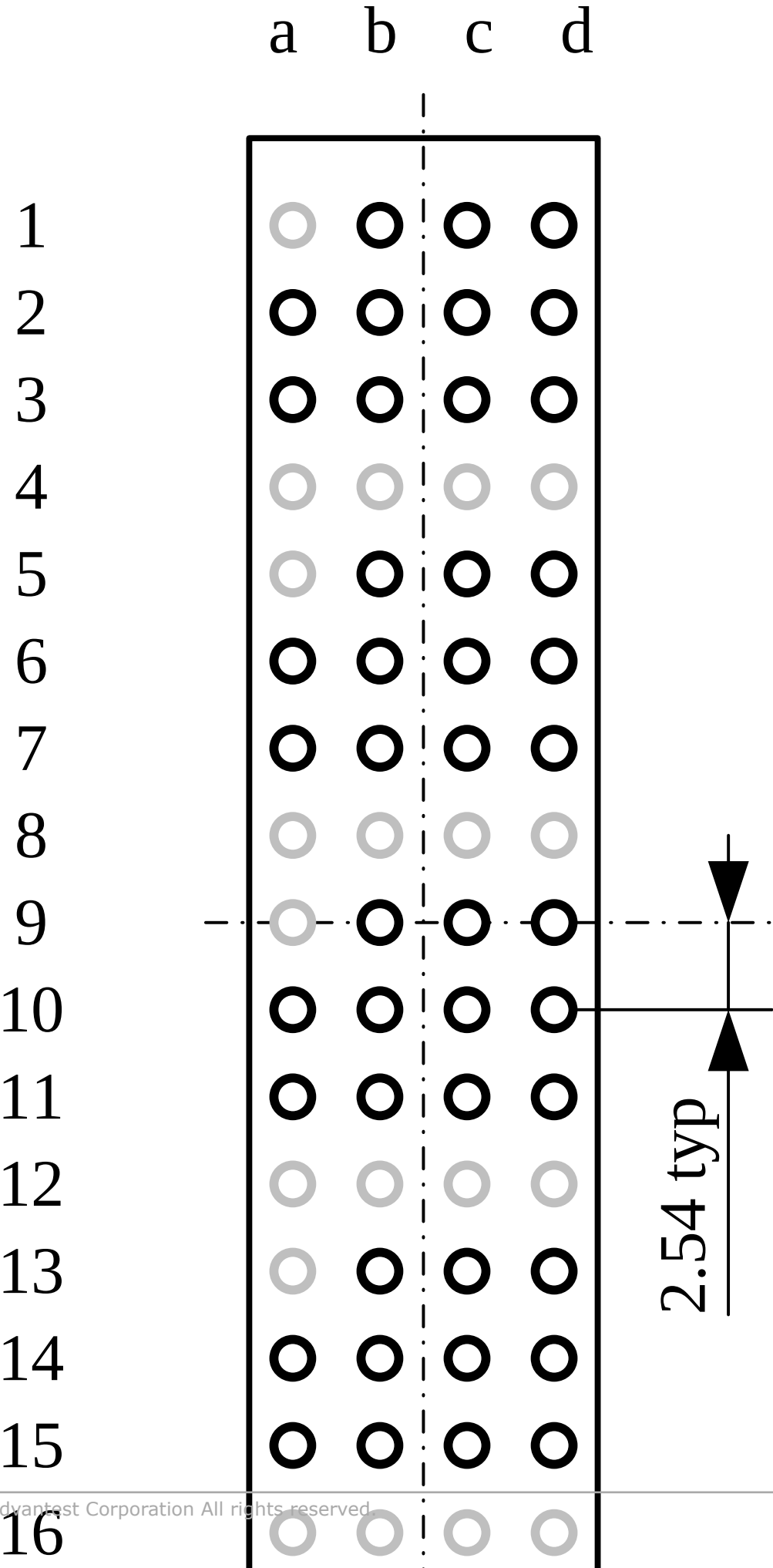
Block C				
	a	b	c	d
1	DNC	HF9	LF9	HS9
2	TMU_H9	LF9	HF9	LS9
3	TMU_L9	HF9	LF9	GUARD9
4	Separation			
5	DNC	HF10	LF10	HS10
6	TMU_H10	LF10	HF10	LS10
7	TMU_L10	HF10	LF10	GUARD10
8	Separation			
9	DNC	HF11	LF11	HS11
10	TMU_H11	LF11	HF11	LS11
11	TMU_L11	HF11	LF11	GUARD11
12	Separation			
13	DNC	HF12	LF12	HS12
14	TMU_H12	LF12	HF12	LS12
15	TMU_L12	HF12	LF12	GUARD12
16	Separation			
17	DNC	DNC	SAFETY2_GND	SAFETY2_5V

Orientation mark °

Block D				
	a	b	c	d
1	DNC	HF13	LF13	HS13
2	TMU_H13	LF13	HF13	LS13
3	TMU_L13	HF13	LF13	GUARD13
4	Separation			
5	DNC	HF14	LF14	HS14
6	TMU_H14	LF14	HF14	LS14
7	TMU_L14	HF14	LF14	GUARD14
8	Separation			
9	DNC	HF15	LF15	HS15
10	TMU_H15	LF15	HF15	LS15
11	TMU_L15	HF15	LF15	GUARD15
12	Separation			
13	DNC	HF16	LF16	HS16
14	TMU_H16	LF16	HF16	LS16
15	TMU_L16	HF16	LF16	GUARD16
16	Separation			
17	DNC	DNC	DNC	DNC

Orientation mark °

Layout and dimension



Pogo pin description

Pin	Description	DUT board connection
HF#	High force line of the channel	See the <i>Pogo pins "HF#" and "LF#" section in FVI16 pogo pins.</i>
LF#	Low force line of the channel	
HS#	High sense line of the channel	For more information, refer to <i>Pogo pins "HS#" and "LS#".</i>
LS#	Low sense line of the channel	
GUARD#	GUARD line of the HF# and HS# line	For more information, refer to <i>Pogo pin "GUARD#".</i>
TMU_H#	TMU high input	For more information, refer to <i>FVI16 pogo pins - TMU.</i>
TMU_L#	TMU low input	
SAFETY1_5V	Control line of the safety circuit for channel 1 ... 8	Connect to an adequate +5.0 V supply via a safety interlock on the DUT board or the DUT interface. For more information, refer to the <i>SAFETY1_5V and SAFETY2_5V pins section in FVI16 pogo pins - Safety.</i>
SAFETY2_5V	Control line of the safety circuit for channel 9 ... 16	
SAFETY1_GND	Return line of the safety circuit for channel 1 ... 8	Connect to the return pin of the +5.0 V supply at SAFETY_P and to the DUT board ground. For more information, refer to the <i>SAFETY1_GND and SAFETY2_GND pins section in FVI16 pogo pins - Safety.</i>
SAFETY2_GND	Return line of the safety circuit for channel 9 ... 16	
DNC	Do not connect pin	These pins are internally used and reserved. Do not connect these pins on the DUT board.
Separation	Separation row between the channels	To ensure that adjacent high voltage pins or traces have no effect on the DNC pins and Separation rows, adequate measures like sufficient insulation space have to be respected.

Notes on the pogo pin descriptions:

- Pin: # = Channel 1 ... 16
- The DNC pins are internally connected and reserved. To ensure that adjacent high voltage pins or traces have no effect on the DNC pins, adequate measures like sufficient insulation space have to be respected.

⚠ CAUTION The force lines "HF#" and "LF#" are each wired to three pogo pins. These three pogo pins must be connected in parallel on the DUT board to ensure that the specifications for the card and the maximum ratings for the pogo pins are met.

DC Scale UHC4T Device Power Supply (E8024UH)

UHC4T is a device power supply card for the V93000 ATE platform.

Scope of specifications

Advantest specifies and verifies the specifications at the DUT interface pogo level. Using an adequate DUT board and valid system calibration, these specifications are valid at the device under test as well. Specifications describe warranted product performance. Characteristics are included as typical values to provide additional useful information by describing typical non-warranted performance.

Configuration

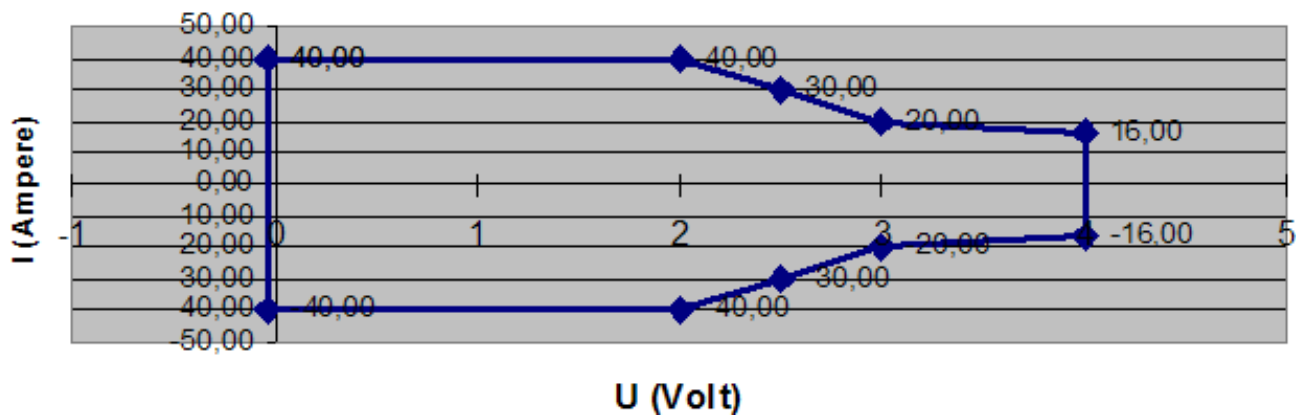
Maximum number of channels per board	4
Maximum source/sink current per channel	40 A @ 2.0 V
Maximum source/sink current per channel	30 A @ 2.5 V
Maximum source/sink current per channel	20 A @ 3.0 V
Maximum source/sink current per channel	16 A @ 4.0 V
Maximum source/sink current per board	160 A @ 2.0 V
Maximum source/sink current per system	Maximum number of boards x 160 A
Maximum voltage	0 V to +4 V

Channel and board maxima

	A test head ^[136]	Compact test head ^[136]	Small test head ^[136]	Large test head
Maximum number of boards per system	8	16	32	64
Maximum number of channels per system	32	64	128	256

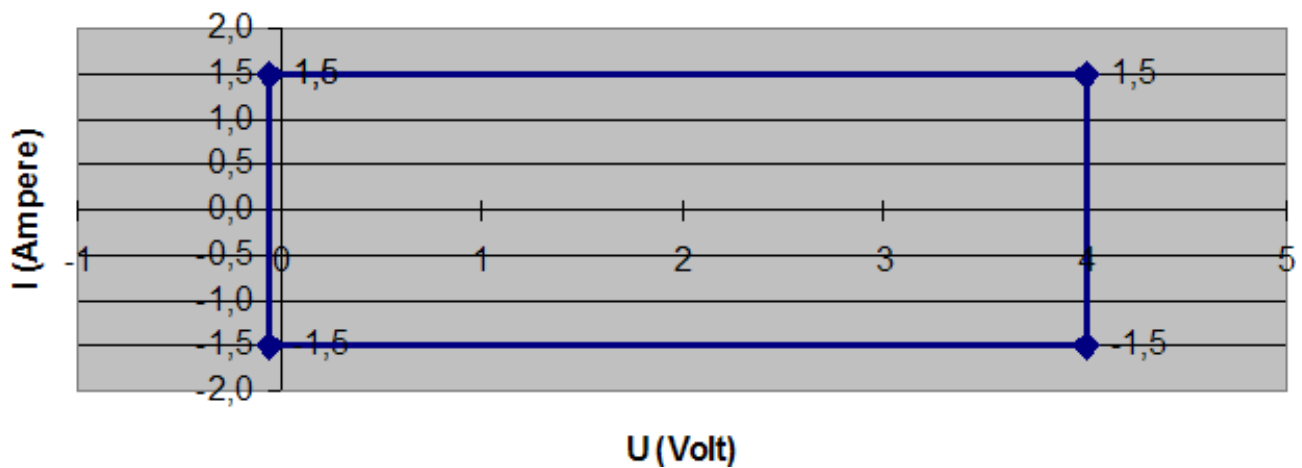
^[136] Applies to test head with Small DUT Interface and DUT Scale Interface

Output characteristic: I versus U High power mode



UHC4T power diagram high power mode

Output characteristic: I versus U Precision mode



UHC4T power diagram precision mode

- The E8024UH UHC4T requires SmarTest 7.3.2 or higher
- Parallel connection (ganging) possible for any number of successive channels within a card and any number of E8024UH cards within a test head.
- The maximum current per board cannot be exceeded.
- The UHC4T requires a universal DC/DC. An adjacent card must be a UHC4T again (located in a pair of uneven/even card cage slots).
- Static sink power on a board or both boards that are connected to the same DC/DC can be larger than the source power within these limits for each channel of a board:
 - $\leq 0.6 \text{ V} : \leq 40 \text{ A}$
 - $\leq 1.0 \text{ V} : \leq 7 \text{ A}$
 - $\leq 2.5 \text{ V} : \leq 4 \text{ A}$
 - $\leq 4.0 \text{ V} : \leq 2 \text{ A}$
- Dynamic sink power (dis-charge) is possible for supported capacitance.

- Either the high-power mode or the precision mode of a channel is active for force and measure. During the range transition from 1.5 to 10/40 A, both modes are active to avoid glitches.

On the pogo block, four DUT_CAP_ON lines (DUT_CAP_ON_1 through DUT_CAP_ON_4) are available to switch the big blocking capacitance on the DUT board on or off.

If the 40 A range is selected, the signal level is 0 V. If the 1.5 A range is selected, the signal level is 5 V.

- The UHC4T card requires no Enhanced Diagnostics Interconnect Board.

Supply voltage and current range specifications

	Range	Resolution	Accuracy
Voltage force	0 to 4.0 V	200 μ V	± 2.5 mV
Voltage measure	0 to 4.0 V	200 μ V	± 2 mV ^[137]
Current force (clamp)	10 to 40 A	2 mA	$\pm (100 \text{ mA} + 0.5\% \text{ of setting})$
	0.5 to 10 A	2 mA	$\pm (50 \text{ mA} + 0.2\% \text{ of setting})$
	15 mA to 1.5 A	100 μ A	± 10 mA
Current force (n channels ganged)	n x 0.5 A to n x 40 A	n x 2 mA	$\pm (n \times 100 \text{ mA} + 0.5\% \text{ of total ganged current})$
Current measurement	± 40 A	2 mA	$\pm (50 \text{ mA} + 0.5\% \text{ of reading})$ ^[138]
	± 10 A	2 mA	$\pm (20 \text{ mA} + 0.2\% \text{ of reading})$ ^[138]
	± 1.5 A	100 μ A	$\pm (1.5 \text{ mA} + 0.1 \% \text{ of reading})$ ^[139]
	± 50 mA	2.5 μ A	$\pm (50 \text{ \muA} + 0.1 \% \text{ of reading})$ ^[140]
	± 2 mA	100 nA	$\pm (2 \text{ \muA} + 0.1 \% \text{ of reading})$ ^[141]
Current measurement (n channels ganged)	n x 40 A	n x 2 mA	$\pm (n \times 50 \text{ mA} + 0.5\% \text{ of total current})$

Fast unganging is supported, resulting in single-channel accuracy for Idd-measurements. This includes DUT capacitance switching using the four DUT_CAP_ON lines.

^[137] Requires 8 averaging samples under pattern control.

^[138] Requires 64 averaging samples under pattern control.

^[139] Requires 256 averaging samples under pattern control.

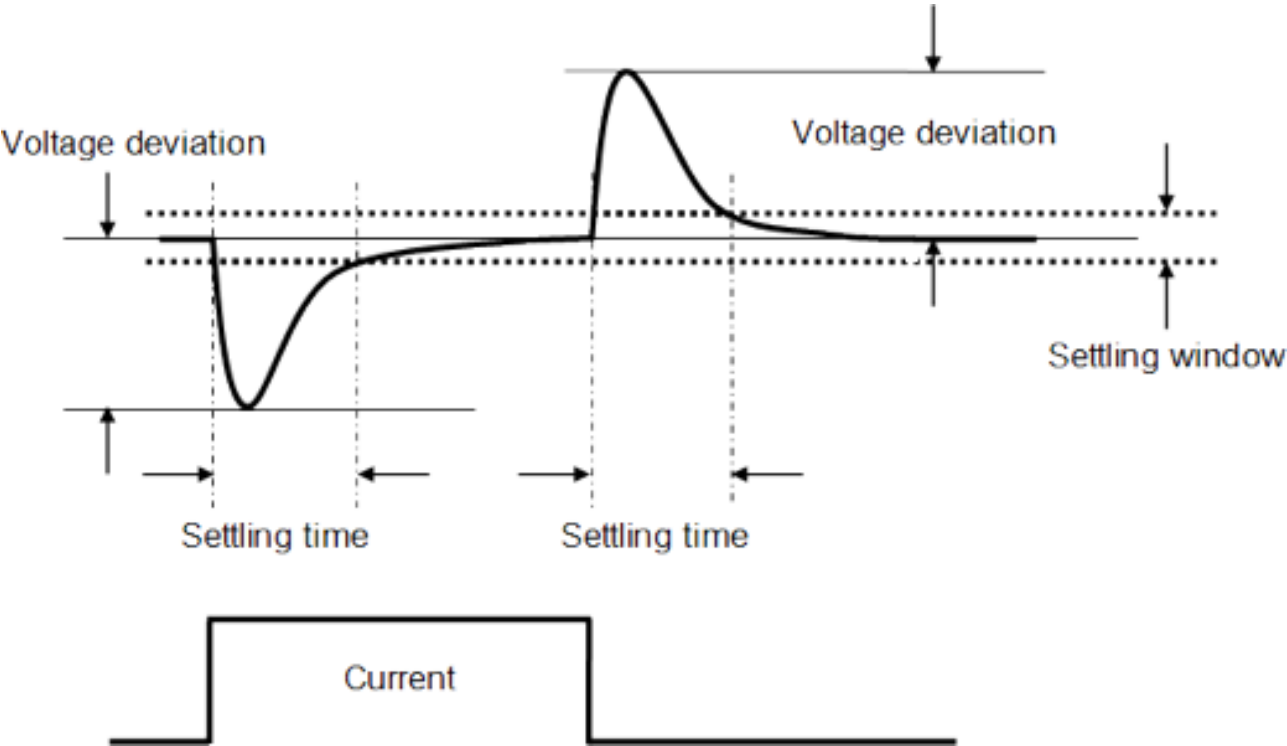
^[140] Requires 256 averaging samples under pattern control. Maximum supported external capacitance to achieve accuracy is 100 μ F.

^[141] Requires 256 averaging samples under pattern control. Maximum supported external capacitance to achieve accuracy is 10 μ F.

Supported "Current force" and "Current measurement" range combinations

		Current force range, 10/40 A high-power mode	Current force range, 1.5 A precision mode
Current measurement range, high-power mode	10/40A	yes	no
Current measurement range, precision mode	1.5 A	no	yes
	50 mA	no	yes
	2 mA	no	yes

Explanation of terms



Explanation of terms

Settling time after load step (dI/dt = 2.5 A/μs)

At 2 V, settling into ±20 mV of programmed value, single-channel operation:

Load step	270 μF (bulk)	540 μF (bulk)
10 A to 30 A	0 μs	0 μs
30 A to 10 A	0 μs	0 μs

At 2 V, settling into ±10 mV of programmed value, single-channel operation:

Load step	270 μF (bulk)	540 μF (bulk)
10 A to 30 A	15 μs	15 μs
30 A to 10 A	18 μs	18 μs

Settling time after load step ($dI/dt = 40 \text{ A}/\mu\text{s}$)

At 2 V, settling into $\pm 20 \text{ mV}$ of programmed value, single-channel operation:

Load step	1080 μF (bulk), 100 μF (ceramic)	2160 μF (bulk), 200 μF (ceramic)
10 A to 30 A	7 μs	7 μs
30 A to 10 A	7 μs	6 μs

At 2 V, settling into $\pm 10 \text{ mV}$ of programmed value, single-channel operation:

Load step	1080 μF (bulk), 100 μF (ceramic)	2160 μF (bulk), 200 μF (ceramic)
10 A to 30 A	9 μs	8 μs
30 A to 10 A	10 μs	7 μs

Settling time after load step ($dI/dt = 400 \text{ A}/\mu\text{s}$)

At 2 V, settling into $\pm 20 \text{ mV}$ of programmed value, ganging four channels on one board:

Load step	4 mF (bulk), 200 μF (ceramic)	8 mF (bulk), 400 μF (ceramic)
40 A to 120 A	4 μs	5 μs
120 A to 40 A	9 μs	4 μs

At 2 V, settling into $\pm 10 \text{ mV}$ of programmed value, ganging four channels on one board:

Load step	4 mF (bulk), 400 μF (ceramic)	8 mF (bulk), 400 μF (ceramic)
40 A to 120 A	10 μs	7 μs
120 A to 40 A	13 μs	10 μs

Voltage deviation after load step ($dI/dt = 2.5 \text{ A}/\mu\text{s}$)

At 2 V, single-channel operation:

Load step	270 μF (bulk)	540 μF (bulk)
10 A to 30 A	18 mV	18 mV
30 A to 10 A	18 mV	18 mV

Voltage deviation after load step ($dI/dt = 40 \text{ A}/\mu\text{s}$)

At 2 V, single-channel operation:

Load step	1080 μF (bulk), 100 μF (ceramic)	2160 μF (bulk), 200 μF (ceramic)
10 A to 30 A	53 mV	35 mV
30 A to 10 A	55 mV	35 mV

Voltage deviation after load step ($dI/dt = 400 \text{ A}/\mu\text{s}$)

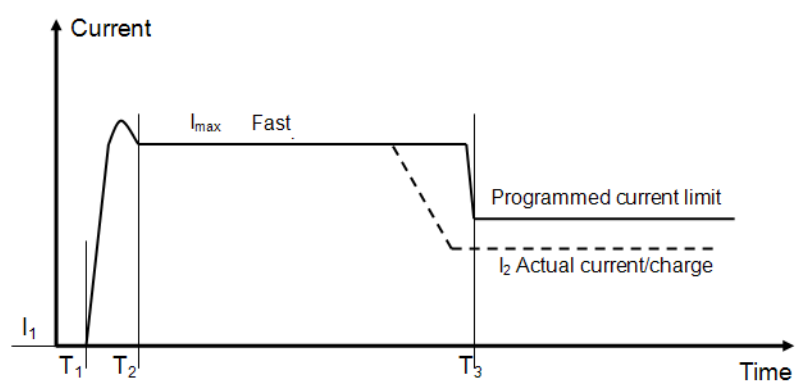
At 2 V, ganging four channels on one board:

Load step	4 mF (bulk), 200 µF (ceramic)	8 mF (bulk), 400 µF (ceramic)
40 A to 120 A	50 mV	35 mV
120 A to 40 A	80 mV	60 mV

Modes and warnings

Fast precharge	
Maximum supported current	40 A (independent of actual voltage setting) is forced for a programmable time (0-10 ms), independent of actual current limit settings, if the 10/40 A current force range is selected
Programmable timing resolution	1 µs
Precharge start	When DUT is connected
Mode selection	Explicitly enabled or disabled with a firmware command (default condition is disabled) or implicitly with a programmed time in an equation

- The fast precharge settings are identical for all channels of a gang group
- This mode is not enabled for voltage changes



At T_1 current starts to flow after connect was initiated (HIZ or LOZ connect)
 $T_2 - T_1$ is settling time
At T_2 I_{max} flows for fast precharge
 $T_3 - T_1$ is fast precharge time (programmable with 1 µs resolution between 0 - 10 ms)
At T_3 current is reduced to I_2 which is equal to or smaller than programmed current limit. If the programmed fast precharge time is too long, current drops to actual load before T_3 is reached

Fast precharge mode

Peak mode	
Peak current	Peak current > 40 A for a few microseconds
Mode selection	Enabled, no control flag or alarm is set

Current limit

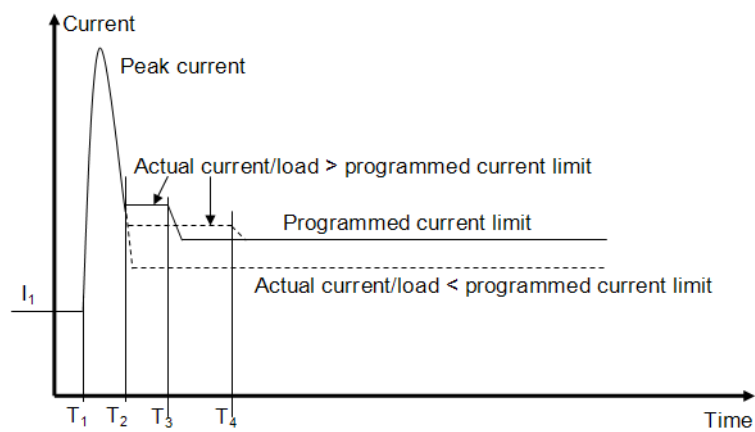
Typical times to limit the current at a current limit of 10 A (one channel, no ganging, 1080 µF)

Load step	Response time
0 A to 40 A	600 µs

Load step	Response time
8 A to 24 A	1.3 ms
8 A to 16 A	3.2 ms

- Activation time of current limit depends on the actual load (overdrive of comparator)

Load change (constant current mode selected)



At T_1 load change occurs
Peak current is delivered for few microseconds
At T_2 current is reduced to actual current/load, over current flag is triggered. Flag is reset, if actual current $\leq$ programmed current limit
 $T_{3,4} - T_2$ is current limit activation time which depends on overdrive of limit comparator
At $T_{3,4}$ current limit becomes active and current is limited to programmed current limit

Constant current mode

Constant current mode	
Constant current mode	Maximum allowed current is equal to actual programmed current limit
Disconnect mode	
Disconnect mode	If current exceeds the set current limit, the channel is disconnected
Disconnect delay time	Programmable (0 - 65 ms, timing resolution 1 μ s)
Mode selection	Explicitly enabled or disabled with a firmware command (default condition is disabled) or implicitly with a programmed time in an equation

- Time delay counter is triggered by the overcurrent flag.
- The settings are identical for all channels of a gang group.
- This mode is selectable in all current force ranges.
- The safety line is not activated (same behavior as HC-DPS).

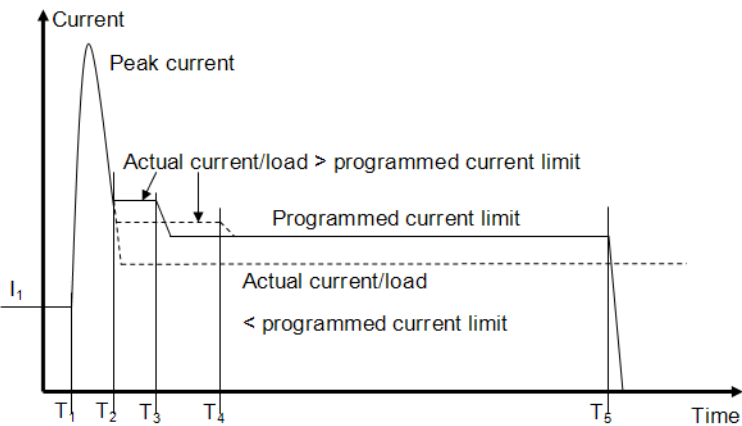
Voltage drop protection	
Allowed voltage drop	Individually programmable for force and return path between 0 - 500 mV, programming resolution is 1 mV
Total allowed voltage drop	500 mV

Voltage drop protection

Mode selection	Always on, in all current ranges, except for the time the fast precharge mode is on
Default voltage limit	250 mV for both the force and return path

- Force voltage at output of UHC4T is continuously compared to voltage at charge. If the difference is above the programmed limit, the output channel is disconnected. The voltage drop can be measured with the UHC4T.
- There are separate comparators for both the force and the return line on both the source and the sink path (four in total).
- The safety line is not activated in case of alarm (same behavior as HC-DPS).

Load change (disconnect mode selected)



At T₁ load change occurs, peak current is delivered for few microseconds
At T₂ current is reduced to actual current /load
T_{3,4} - T₂ is current limit activation time which depends on overdrive of limit comparator
At T_{3,4} current limit becomes active and current is limited to programmed current limit
Over current flag is triggered, if actual current/load > programmed current limit
Flag is reset, if actual current < programmed current limit
At T₅ the channel is disconnected (HIZ or LOZ)
T₅ - T_{3,4} is disconnect time (programmable with 1 μ s resolution between 0 - 65 ms)

Disconnect mode

Other protection mechanisms (one line per board)

Safety line	Connected to DUT board via pogo block. Internally pulled high to +5V from DUT board with a resistor of 46.4k. When pulled low, all channels of UHC are disconnected. Can be used to simultaneously disconnect all UHC4T channels by an external alarm switch.
+5 V from DUT board	If not applied, all UHC4T channels are disconnected. Can be used to detect improperly connected DUT board.

Other performance characteristics

Disconnect state

Input leakage current	< 100 nA @ ≤ 3.4 V (AC relay open)
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Voltmeter mode (high impedance mode)

Voltage range	0 V to +4 V
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Voltmeter mode (high impedance mode)

Voltage resolution	200 μ V
Voltage accuracy	± 2 mV
Input leakage current	< 100 nA @ ≤ 3.4 V (device connected to force and sense lines)

Differential mode (using force sense and ground sense)

Voltage range	Same as voltage measure
Voltage resolution	Same as voltage measure
Voltage accuracy	Same as voltage measure

High speed differential sampling mode (14 bit)^[142]

Number of differential inputs	1 per channel, individually addressable and usable within a gang chain
Sample rate	10 Msps (megasamples per second)
Voltage window	-2.5 V to +4 V
Differential voltage range	0 V to $ 3 $ V
Voltage resolution	400 μ V
DC accuracy	2 mV
Signal path bandwidth (-3 dB)	5 MHz
Total channel memory	224 MB

AWG mode (16 bit DAC)

Step rate	1 Msps (megasamples per second)
Analog bandwidth (-3 dB, 3 V swing)	3 kHz
Total channel memory	224 MB
Vbump function	Sequencer#controlled, programmable

Digitizer mode (16 bit ADC)

Sample rate	475 ksps (kilosamples per second)
Signal path bandwidth (-3 dB)	25 kHz
Total channel memory	224 MB

^[142] Not supported on SmarTest 8.

Trigger functions

Trigger	Sequencer#controlled (no external trigger pin required)
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Data analysis

Data analysis	Raw data is stored on board without software interaction and may be uploaded for further computation. Averaging, min and max number are calculated and stored on board.
Result analysis	Built-in result analysis available that indicates the measurement data is pass or fail against a user specified test limit.

Compliance range (at disconnect state)

Maximum external voltage at force input	-3 V to +10 V
Maximum external voltage at sense input	-3 V to +10 V
Maximum external voltage at sampling input	-3 V to +10 V
Maximum external voltage at safety line	-3 V to +5.5 V
Maximum external voltage at +5 V DUT signal	-3 V to +5.5 V

Settling time after re-programming (settling within given band of programmed step or within accuracy band, 2 x 270 μ F capacitive charge)

Voltage force (0-2 V step)	200 μ s ($\pm$ 1% band)
Current force / measure (0-10 A step)	40 μ s (+0 -20% band) 4 ms ($\pm$ 1% band)

Charge capacitance range

Minimum required external capacitance	270 μ F per channel for high power mode 1 μ F per channel for precision mode
Maximum supported external capacitance with stability maintained	10 mF per channel
Recommended external capacitance	1 - 2 mF, see also the explanation for the voltage/ current specifications, above

Blocking caps (examples, both types are required on DUT board)

Tantalum (example for bulk capacitance)

Kemet	T530D477M2R5A(1)E010 - 470 μ F/2.5V
Panasonic	OSCON 4SVPC560MX - 560 μ F/4V

Ceramic (example)

TDK	C1608X5R1E105K (1 μ F/25V/0603/X5R-Ceramic), array of 25 each
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Ceramic (example)	
TDK	C1608X7R1H104K (100nF/50V/0603/X7R-Ceramic), array of 25 each
Power ramp at probe	
Probe card protection	User-programmable sequence for both shorts and open-check, using the precision mode
Thermal control	
Current monitor	Current monitor output per channel provides voltage proportional to the output current, 1 V per 10 A current, reference potential is GSn- line (Ground Sense). Note Sense-lines are sensitive to noise. It it is recommended to use the GND-plane close to the connection of the GSn-line.
Typical accuracy	±5 mV ± 1% of reading

DC Scale XPS256 / XPS128 / XPS128+HV Universal DPS/VI Card



Hardware support in SmarTest

XPS256 variant		SmarTest 7	SmarTest 8
XPS256	(EK751HC)	7.10.1 and higher	8.3.4 and higher
XPS128	(EK751HCH)	not supported	8.5.3 and higher
XPS128+HV	(EK751HCV)	not supported	8.5.4 and higher

XPS256 overview

General specifications		
Maximum channel count		
	XPS256	256 channels per card
	XPS128	128 channels per card
	XPS128+HV	128 channels per card
Maximum source current per channel		
	XPS256 / XPS128	1.0 A @ -2.5 V ... +3.0 V
		0.6 A @ -2.5 V ... +7.0 V

General specifications

	XPS128+HV	1.0 A @ -2.5 V ... +3.0 V 0.6 A @ -2.5 V ... +7.0 V 0.25 A @ -10.0 V ... +15.0 V 0.15 A @ -1.0 V ... +24.0 V
Maximum sink current per channel		
	XPS256 / XPS128	1.0 A @ -2.0 V ... +3.0 V 0.6 A @ -2.5 V ... +7.0 V
	XPS128+HV	1.0 A @ -2.0 V ... +3.0 V 0.6 A @ -2.5 V ... +7.0 V 0.25 A @ -10.0 V ... +15.0 V 0.15 A @ -1.0 V ... +24.0 V
Maximum source current per card		
	XPS256	256 A @ -2.5 V ... +2.0 V ^[143]
	XPS128 / XPS128+HV	128 A @ -2.5 V ... +2.0 V ^[143]
Maximum sink current per card		
	XPS256	230 A @ -2.1 V ... +4.0 V ^[143]
	XPS128 / XPS128+HV	115 A @ -2.1 V ... +4.0 V ^[143]
Voltage range		
	XPS256 / XPS128	-2.5 V ... +7.0 V
	XPS128+HV	-10.0 V ... +24.0 V

For more details on maximum current/voltage capabilities, refer to the graphs in section "Maximum continuous voltage and current capability per card" (topic 350737).

Calibration, operation, environmental

Warm-up time	60 minutes
Basic maintenance period	6 months
Base calibration period (traceable calibration)	6 months
Calibration period (system auto adjustment) ^[144]	3 months
Operating temperature range	15 °C ... 30 °C (59 °F ... 86 °F)
Temperature range for guaranteed specifications	20 °C ... 30 °C (68 °F ... 86 °F)
Maximum humidity at 30 °C	< 70 % R. H., non-condensing
Storage temperature range (without water)	-40 °C ... +70 °C (-40 °F ... 158 °F)

^[143] Combined voltage drop on force and GND line not exceeding ± 0.2 V.

^[144] Valid at ambient temperature within ± 2.5 K of calibration temperature.

System configuration		CX	SX	LX
Maximum number of cards per system		9	18	27
Maximum number of channels per system	XPS256	2304 ^[145]	4096 ^[145]	4096 ^[145]
	XPS128 / XPS128+HV	1152	2304 ^[145]	3456 ^[145]

Other specifications

- Maximum source/sink current per card cannot be exceeded.
- Requires E7010F, Rev. 11 or higher or EK855AD Enhanced Diagnostics Interconnect Board.

XPS256 voltage specifications

Voltage force	Range	Resolution	Accuracy ^[146] ^[149]
XPS256 / XPS128	-0.5 V ... +1.5 V	10 μ V	$\pm$ (150 μ V + 0.02 % of setting)
	-2.5 V ... +7.0 V	50 μ V	$\pm$ (400 μ V + 0.02 % of setting)
XPS128+HV	-0.5 V ... +1.5 V	10 μ V	$\pm$ (150 μ V + 0.02 % of setting)
	-2.5 V ... +7.0 V	50 μ V	$\pm$ (400 μ V + 0.02 % of setting)
	-10.0 V ... +15.0 V	150 μ V	$\pm$ (1 mV + 0.02 % of setting)
	-1.0 V ... +24.0 V	150 μ V	$\pm$ (1 mV + 0.02 % of setting)

Voltage measurement	Range	Resolution	Accuracy ^[146] ^[148]
XPS256 / XPS128	-0.5 V ... +1.5 V	10 μ V	$\pm$ (150 μ V + 0.02 % of value)
	-2.5 V ... +7.0 V	50 μ V	$\pm$ (400 μ V + 0.02 % of value)
XPS128+HV	-0.5 V ... +1.5 V	10 μ V	$\pm$ (150 μ V + 0.02 % of value)
	-2.5 V ... +7.0 V	50 μ V	$\pm$ (400 μ V + 0.02 % of value)
	-10.0 V ... +15.0 V	150 μ V	$\pm$ (1 mV + 0.02% of value)
	-1.0 V ... +24.0 V	150 μ V	$\pm$ (1 mV + 0.02% of value)

Voltage clamp	Range	Resolution	Accuracy ^[147]
XPS256 / XPS128	-0.5 V ... +1.5 V	10 μ V	$\pm$ (500 μ V + 0.1 % of setting)
	-2.5 V ... +7.0 V	50 μ V	$\pm$ (500 μ V + 0.1 % of setting)
XPS128+HV	-0.5 V ... +1.5 V	10 μ V	$\pm$ (500 μ V + 0.1 % of setting)
	-2.5 V ... +7.0 V	50 μ V	$\pm$ (500 μ V + 0.1 % of setting)
	-10.0 V ... +15.0 V	150 μ V	$\pm$ (3 mV + 0.1 % of setting)
	-1.0 V ... +24.0 V	150 μ V	$\pm$ (3 mV + 0.1 % of setting)

^[145] Maximum 2048 channels with HD type pogo cable assembly.

^[146] Measurement conditions VBw = 30 kHz, IBw = 10 kHz, needs open loop protection = "none", needs SmarTest 8.5.1 / SmarTest 7.10.3 or later.

XPS256 current specifications

Current force	Range	Resolution	Accuracy ^[151] ^[154]
	± 1.0 A	10 µA	± (1 mA + 0.15 % of setting + 1 nA/V * V _{act})
	± 100 mA	1 µA	± (100 µA + 0.15 % of setting + 1 nA/V * V _{act})
	± 10 mA	100 nA	± (10 µA + 0.15 % of setting + 1 nA/V * V _{act})
	± 1 mA	10 nA	± (1 µA + 0.15 % of setting + 1 nA/V * V _{act})
	± 100 µA	1 nA	± (250 nA + 0.15 % of setting + 1 nA/V * V _{act})
	± 10 µA	100 pA	± (50 nA + 0.15% of setting + 1 nA/V * V _{act})
Current force (n channels ganged)	± n * 1.0 A	n * 10 µA	± (n * 1 mA + 0.15 % of setting + n * 1 nA/V * V _{act})

Current measurement	Range	Resolution	Accuracy ^[152] ^[155] ^[154]
	± 1.0 A	10 µA	± (1 mA + 0.1 % of value + 1 nA/V * V _{act})
	± 100 mA	1 µA	± (100 µA + 0.1 % of value + 1 nA/V * V _{act})
	± 10 mA	100 nA	± (10 µA + 0.1 % of value + 1 nA/V * V _{act})
	± 1 mA	10 nA	± (1 µA + 0.1 % of value + 1 nA/V * V _{act})
	± 100 µA	1 nA	± (250 nA + 0.1 % of value + 1 nA/V * V _{act}) ^[156]
	± 10 µA	100 pA	± (50 nA + 0.1 % of value + 1 nA/V * V _{act}) ^[156]
Current measure (n channels ganged)	± n * 1.0 A	n * 10 µA	± (n * 1 mA + 0.1 % of value + n * 1 nA/V * V _{act})

Current clamp	Range	Resolution	Accuracy ^[153]
	10 mA ... 1.0 A	10 µA	± (2 mA + 0.2 % of setting)
	1 mA ... 100 mA	1 µA	± (200 µA + 0.2 % of setting)
	100 µA ... 10 mA	100 nA	± (20 µA + 0.2 % of setting)
	10 µA ... 1 mA	10 nA	± (2 µA + 0.2 % of setting)
	1 µA ... 100 µA	1 nA	± (400 nA + 0.2 % of setting)
	100 nA ... 10 µA	100 pA	± (70 nA + 0.2 % of setting)
Current clamp (n channels ganged)	n * (10 mA ... 1.0 A)	n * 10 µA	± (n * 2 mA + 0.2 % of setting)

^[147] Measurement conditions VBw = 30 kHz, IBw = 10 kHz.
^[148] Requires 40 samples (20 µs).
^[149] Max cap load < 1 mF / channel

Current clamp	Range	Resolution	Accuracy ^[153]
Clamp response time (typical)	< 35 μ s ^[150]		

Hardware current limiter			
Short circuit current	< 1.8 A		

XPS256 voltage and current capability specifications

Continuous voltage and current capabilities per channel
XPS256 / XPS128 / XPS128+HV

The following diagram shows the capability for a single channel in the 1.0 A high-current range for voltages up to 7 V.

[150] At Ibandwidth = 50 kHz, into programmed value $\pm$ 10 % of Irange, loadstep mode standard

[151] Measurement conditions VBw = 30 kHz, IBw = 10 kHz, needs open loop protection = "none", needs SmarTest 8.5.1 / SmarTest 7.10.3 or later.

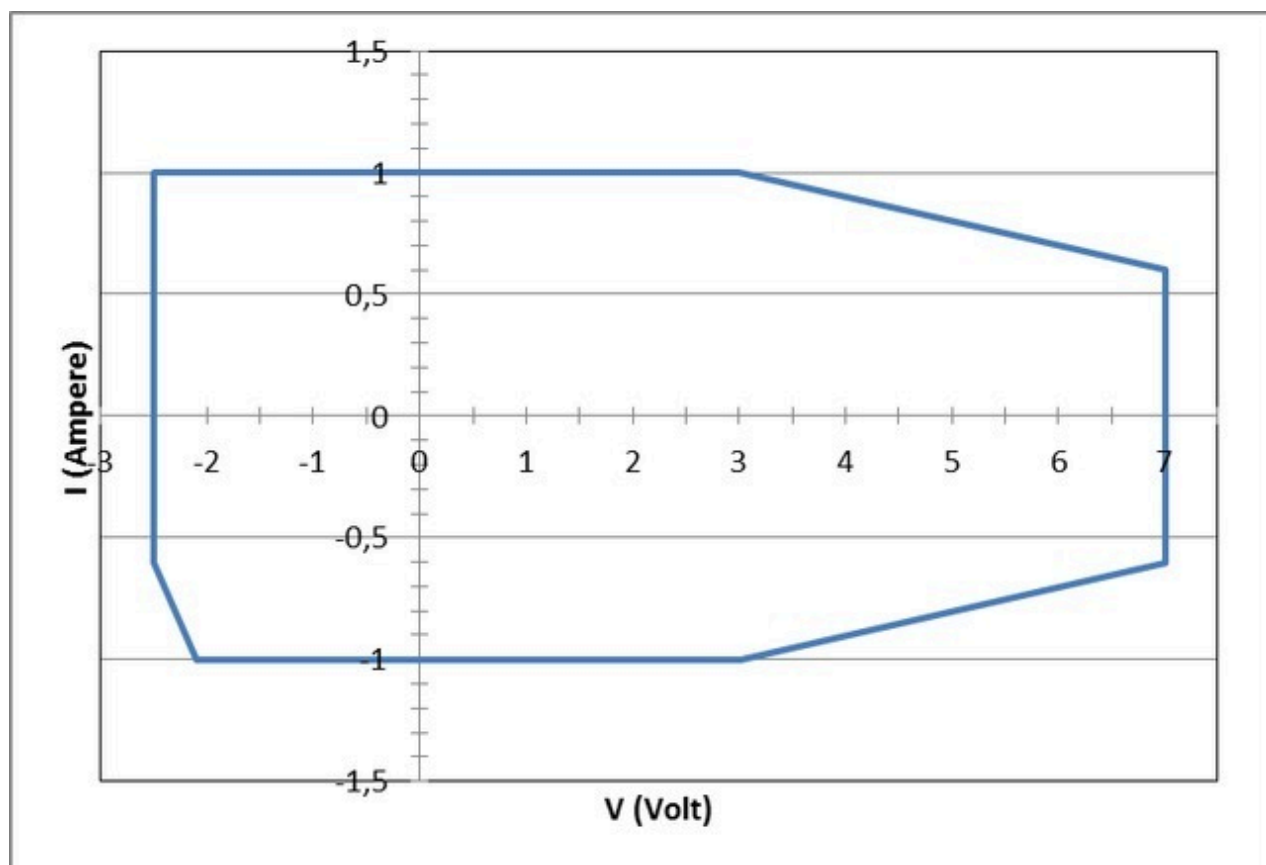
[152] Measurement conditions VBw = 30 kHz, IBw = 10 kHz, measure bandwidth = 30 kHz, needs open loop protection = "none", needs SmarTest 8.5.1 / SmarTest 7.10.3 or later.

[153] Measurement conditions VBw = 30 kHz, IBw = 10 kHz

[154] V_{act} is the actual voltage of the channel, which is either the programmed force voltage or the voltage forced by the DUT, depending on the mode the channel is in (forcing or clamping).

[155] Requires 160 samples (80 μ s).

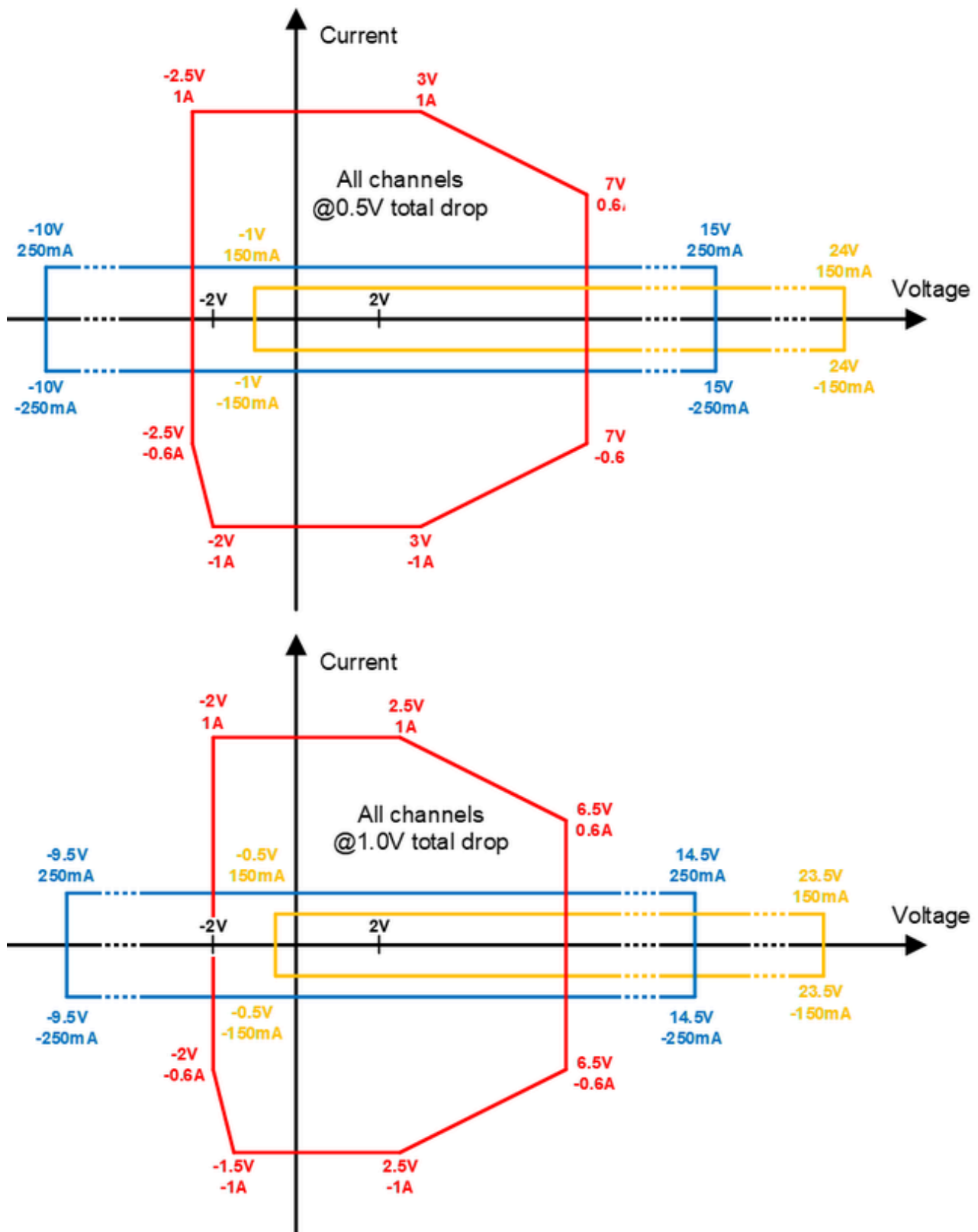
[156] Requires 640 samples (320 μ s).



The power diagram applies to an additional voltage drop in the force and return path beyond the pogo pin of total ≤ 0.5 V.

XPS128+HV

The following diagrams apply to the 1.0 A high-current and high-voltage ranges of XPS128+HV.

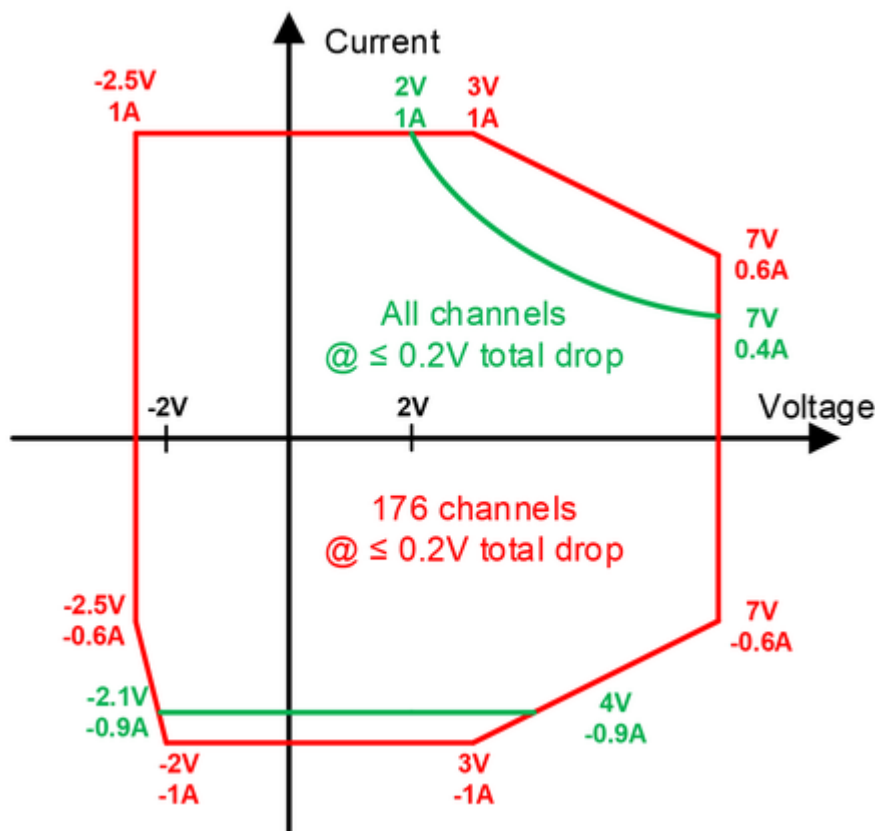


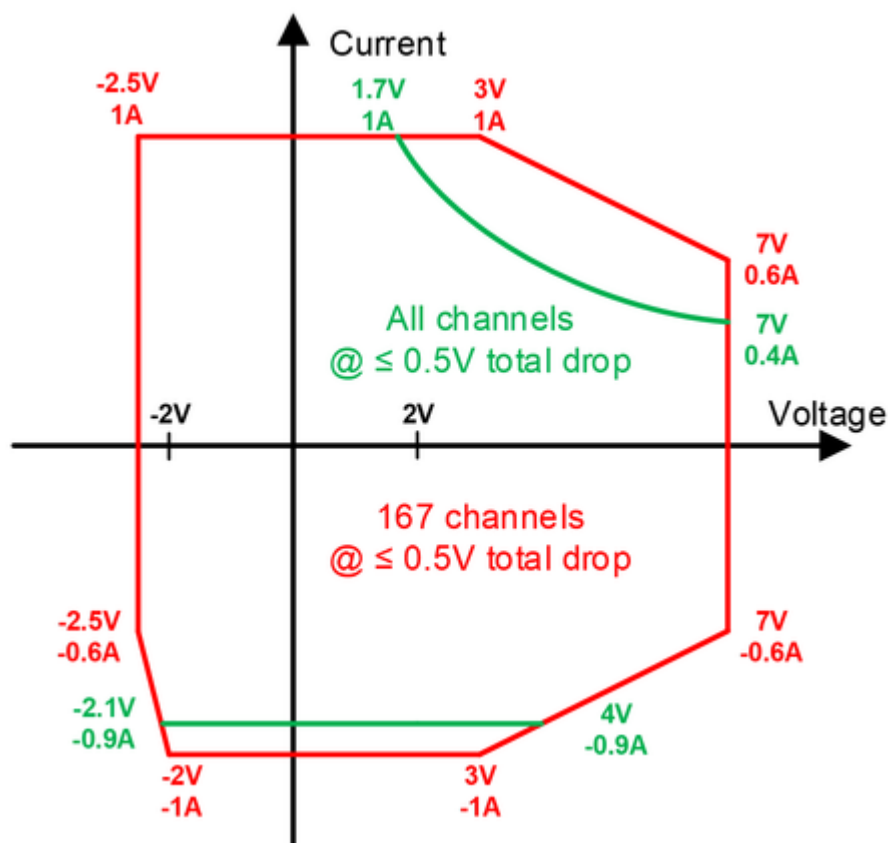
The upper diagram applies to an additional voltage drop in the force and return path beyond the pogo pin of total ≤ 0.5 V while the lower diagram applies to an additional voltage of total ≤ 1.0 V.

Maximum continuous voltage and current capability per card

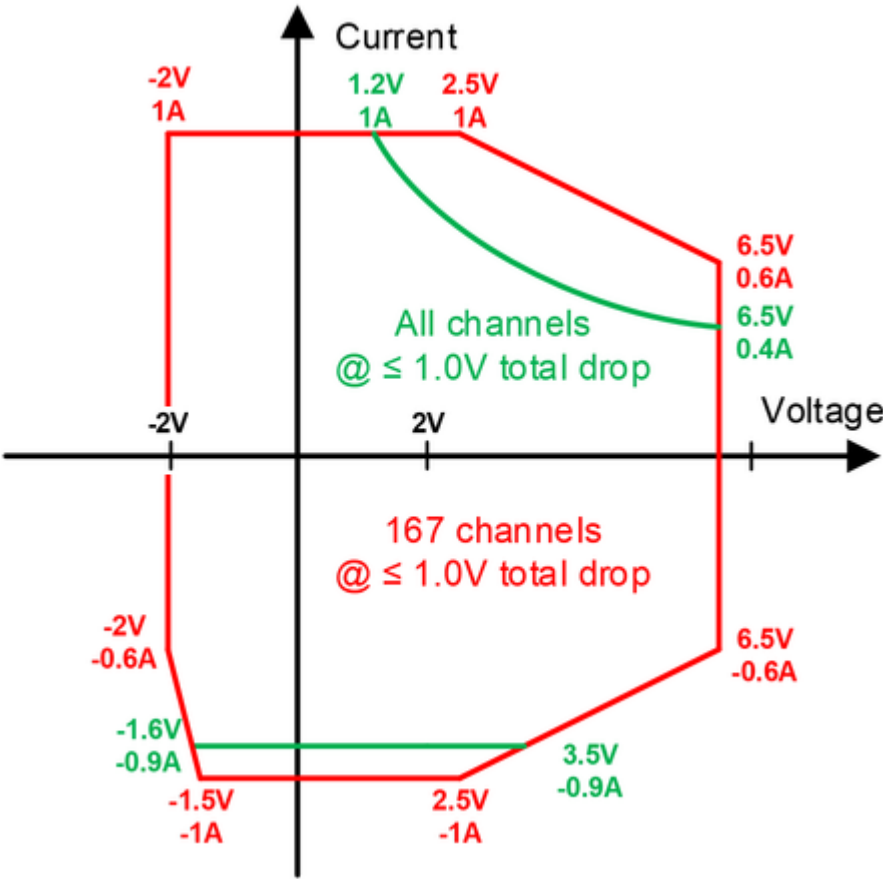
The following diagrams show the capability per card for a different number of active channels and different maximum total voltage drop on the force lines P and GND from the pogo pins to the DUT.

Note that for XPS128 and XPS128+HV, the channel count limitation does not apply. The two variants support the maximum continuous voltage and current capability for all 128 channels as defined by the red lines.

All channels / 176 channels @ 0.2 V maximum total voltage drop for force and return path**All channels / 167 channels @ 0.5 V maximum total voltage drop for force and return path**

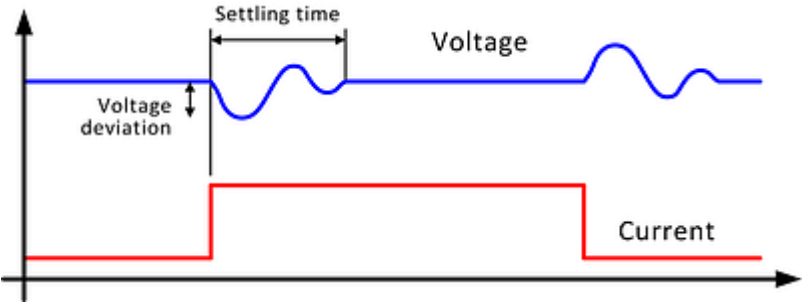


All channels / 167 channels @ 1.0 V maximum total voltage drop for force and return path



XPS256 load step response specifications

Explanation of terms:



Load step response ^[157]	Regulation mode	Load step	100 μF ^[161]	47 μF ^[161]	10 μF ^[158]
Settling time ^[159]	Fast	50 % ^[162]	20 μs	15 μs	10 μs
		80 % ^[163]	20 μs	15 μs	10 μs
	Standard	50 % ^[162]	50 μs	30 μs	20 μs
		80 % ^[163]	55 μs	35 μs	25 μs

^[157] Typical at pogo pin, in $n * \pm 1.0\text{ A}$ range, clamp setting $n * 1\text{ A}$ ($n = \text{number of channels ganged}$), only valid for $-0.5\text{ V} \dots +1.5\text{ V}$, $-2.5\text{ V} \dots +7.0\text{ V}$ ranges

^[158] Voltage bandwidth: 30 kHz

^[159] At 1 V, settling into $\pm 10\text{ mV}$ of programmed value, any number of channels ganged

Load step response ^[157]	Regulation mode	Load step	100 μ F ^[161]	47 μ F ^[161]	10 μ F ^[158]
Maximum voltage deviation ^[160]	Fast	50 % ^[162]	30 mV	40 mV	85 mV
		80 % ^[163]	55 mV	70 mV	120 mV
	Standard	50 % ^[162]	60 mV	65 mV	110 mV
		80 % ^[163]	90 mV	110 mV	140 mV

XPS256 AWG, digitizer, and comparator specifications

Source waveform mode (AWG)

Regulated mode:

Max. step rate	2 Msps (million samples per second) ^[167]
Analog bandwidth (-3 dB, 1 V swing)	30 kHz (typ.)
Maximum number of samples	14 Msamples (million samples) ^[164] ^[165]
Accuracy	see Voltage force / Current force specifications

Measure waveform mode (Digitizer)

Voltage sampling:

Sample rate	2 Msps (million samples per second) ^[167]
Signal path bandwidth (-3 dB)	500 kHz (typ.)
Maximum number of samples	22 Msamples (million samples) ^[169]
Accuracy	see voltage measure specifications ^[166]

Current sampling:

Sample rate	2 Msps (million samples per second) ^[167]
Signal path bandwidth (-3 dB)	500 kHz (typ.)
Maximum number of samples	22 Msamples (million samples) ^[169]
Accuracy	see current measure specifications ^[166]

^[157] Typical at pogo pin, in $n * \pm 1.0$ A range, clamp setting $n * 1$ A (n = number of channels ganged), only valid for -0.5 V ... +1.5 V, -2.5 V ... +7.0 V ranges

^[158] Voltage bandwidth: 30 kHz

^[160] At 1 V, any number of channels ganged

^[161] Voltage bandwidth: 50 kHz

^[162] Verified at 25 % to 75 % load step

^[163] Verified at 10 % to 90 % load step

^[164] Requires SmarTest 8, 1 Msamples max. with SmarTest 7

^[165] 64 bits per sample (256 bits for SmarTest 7) in total channel memory (896 Mbit per channel), channel memory will be shared with other DC actions and ganged channels within a group of 8 channels

Voltage compare mode ^[168]

Minimum detectable pulse width	1 μ s (typ.)
Threshold voltage accuracy	Same as voltage measure (1 kHz filter)
Input and threshold range	Same as voltage measure

Current compare mode ^[168]

Minimum detectable pulse width	1 μ s (typ.)
Threshold current accuracy	Same as current measure (1 kHz filter)
Input and threshold range	Same as current measure

XPS256 other performance characteristics**Load capacitance range**

Minimum required external capacitance with stability maintained	0 nF ^[170]
Maximum supported external capacitance with stability maintained	1000 μ F per channel ^[170] (n * 1000 μ F if ganged)

Output noise

Output noise (spurious up to 20 MHz, typical) ^[171]	100 μ V rms ^[172] 600 μ V rms ^[173]
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Slew rate control (Voltage)

On-, off-, change level-rate individually controlled for each channel

	1.5 V range	7.0 V range	15.0 V range	24.0 V range
Minimum programmable slew rate	10 μ V/ μ s	45 μ V/ μ s	120 μ V/ μ s	120 μ V/ μ s

^[166] With 20 kHz digital filter

^[167] Requires SmarTest 8, 1.5 Msps max. with SmarTest 7

^[168] Requires SmarTest 8

^[169] 40 bits per sample (max. 11 Msamples, 80 bits per sample for SmarTest 7) in total channel memory (896 Mbit per channel), channel memory will be shared with other DC actions (max 16M per action)

^[170] In standard regulation mode, Vbandwidth = 10 kHz

^[171] For single channel, in 1A range

^[172] At 100 μ F per channel capacitive load

^[173] At 0 nF capacitive load

Slew rate control (Voltage)

Maximum programmable slew rate	2 V/μs	2 V/μs	2 V/μs	2 V/μs
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Slew rate control (Current)

Change level-rate individually controlled for each channel ^[174]

Range	Min	Max
± 1.0 A	10.0 μA/μs	200.0 mA/μs
± 100 mA	1.0 μA/μs	20.0 mA/μs
± 10 mA	100.0 nA/μs	2.0 mA/μs
± 1 mA	10.0 nA/μs	200.0 μA/μs
± 100 μA	1.0 nA/μs	20.0 μA/μs
± 10 μA	100.0 pA/μs	2.0 μA/μs

Disconnect mode

Disconnect mode If current reaches set current limit, the channel is disconnected

Disconnect delay time ^[175]	Range	Resolution
	0 μs ... 127 μs	1 μs
	> 127 μs ... 1.27 ms	10 μs
	> 1.27 ms ... 12.7 ms	100 μs
	> 12.7 ms ... 127 ms	1 ms
	> 127 ms ... 1.27 s	10 ms

Disconnect state

Max. input leakage current (AC relay open, typical) ± (10 nA + 1 nA/V) force and sense connected
± (1 nA + 1 nA/V) only sense connected

Voltmeter mode (high impedance mode)

Voltage range	Same as voltage measure
Voltage resolution	Same as voltage measure
Voltage accuracy	Same as voltage measure
Max. input leakage current (typical)	± (1 nA + 1 nA / V)
Max. input voltage (sense / ground sense)	-0.5 V ... +1.5 V in 1.5 V range -2.5 V ... +7.0 V in 7.0 V range -10.0 V ... +15.0 V in 15.0 V range -1.0 V ... +24.0 V in 24.0 V range

^[174] Min/max values valid for single channel, for $n \times$ ganging, numbers must be multiplied by n

^[175] Time delay counter is triggered by the overcurrent flag. The settings are identical for all channels of a gang group.

Disconnect state

Max. differential voltage (sense to ground sense)	-0.5 V ... +1.5 V in 1.5 V range
	-2.5 V ... +7.0 V in 7.0 V range
	-10.0 V ... +15.0 V in 15.0 V range
	-1.0 V ... +24.0 V in 24.0 V range

Compliance range (at disconnect state)

Min/max external voltage at force input	-4.5 V ... +7.5 V in 1.5 V range
	-4.5 V ... +7.5 V in 7.0 V range
	-13.5 V ... +18.5 V in 15.0 V range
	-4.5 V ... +27.5 V in 24.0 V range
Min/max external voltage at sense input	-4.5 V ... +7.5 V in 1.5 V range
	-4.5 V ... +7.5 V in 7.0 V range
	-13.5 V ... +18.5 V in 15.0 V range
	-4.5 V ... +27.5 V in 24.0 V range

Overvoltage protection

Maximum voltage range	-80 V ... +80 V
Maximum energy of voltage spike	20 mJ
Maximum current	1.0 A

Voltage drop protection / measurement

Allowed voltage drop	Individually programmable for force and return path between 0 and 500 mV, programming resolution is 1 mV Two level sets can be programmed
Total allowed voltage drop	1000 mV ^[176]
Voltage drop measurement accuracy (relative)	± 4 mV ^[177]
Voltage drop measurement accuracy (absolute)	± (20 mV + 50 mV / A)
Voltage drop measurement resolution	900 µV

^[176] For the maximum total output voltage (Force voltage + total voltage drop), refer to the "Continuous voltage and current capabilities per channel" section and to the "All channels / 167 channels @ 1.0 V maximum total voltage drop for force and return path" diagram in topic "XPS256 voltage and current capability specifications" (350737).

^[177] Delta measurements on same channel at same operating conditions (i.e. force current, same auto-calibration). Requires 40 averages.

DC profiling

DC profiling	Supported per channel for simultaneous voltage, current, and voltage drop profiling ^[178]
Maximum sample rate	2 Msps (million samples per second) ^[179]
Maximum number of samples	176 Msamples (million samples) ^[180]
Input signal bandwidth	500 kHz (typ.)

Surge tracker

Trigger	Voltage or current surges are detected against programmable min. and max. values. The surge tracker always checks against both high (positive) and low (negative) thresholds, which can be set individually. Two flags indicate if thresholds have been exceeded. These flags are latched (stored) and can be retrieved any time in the testflow.
Input signal bandwidth	500 kHz (typ.)
Threshold voltage accuracy (typical)	± 10 mV in 1.5 V range ± 10 mV in 7.0 V range ± 30 mV in 15.0 V range ± 30 mV in 24.0 V range
Threshold current accuracy (typical)	± 10 % of selected current range

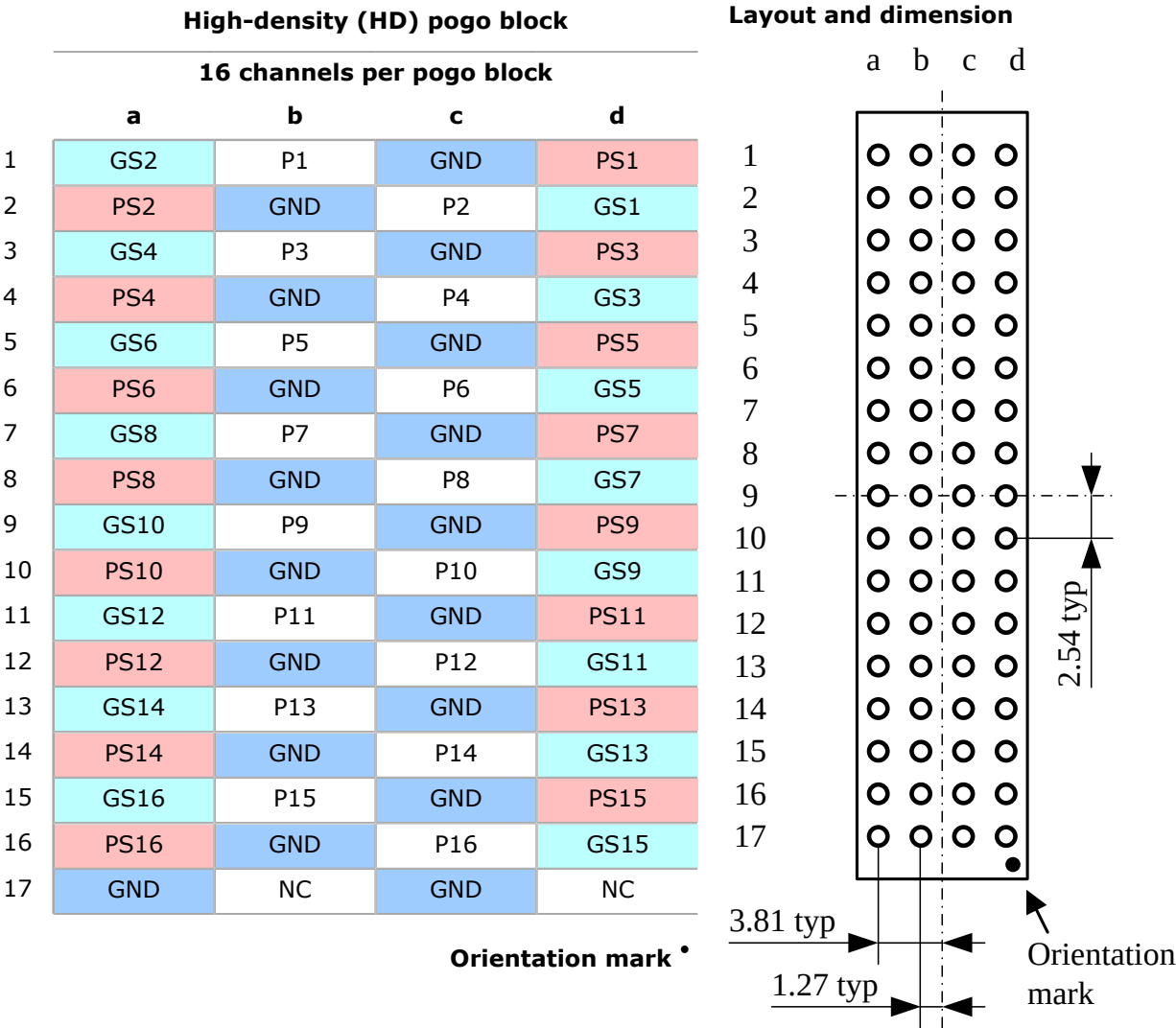
^[178] Simultaneous profiling requires SmarTest 7.10.4 or higher or SmarTest 8.4.2 or higher.

^[179] Requires SmarTest 7.10.5 / 8.4.2 or higher, valid for profiling one parameter only

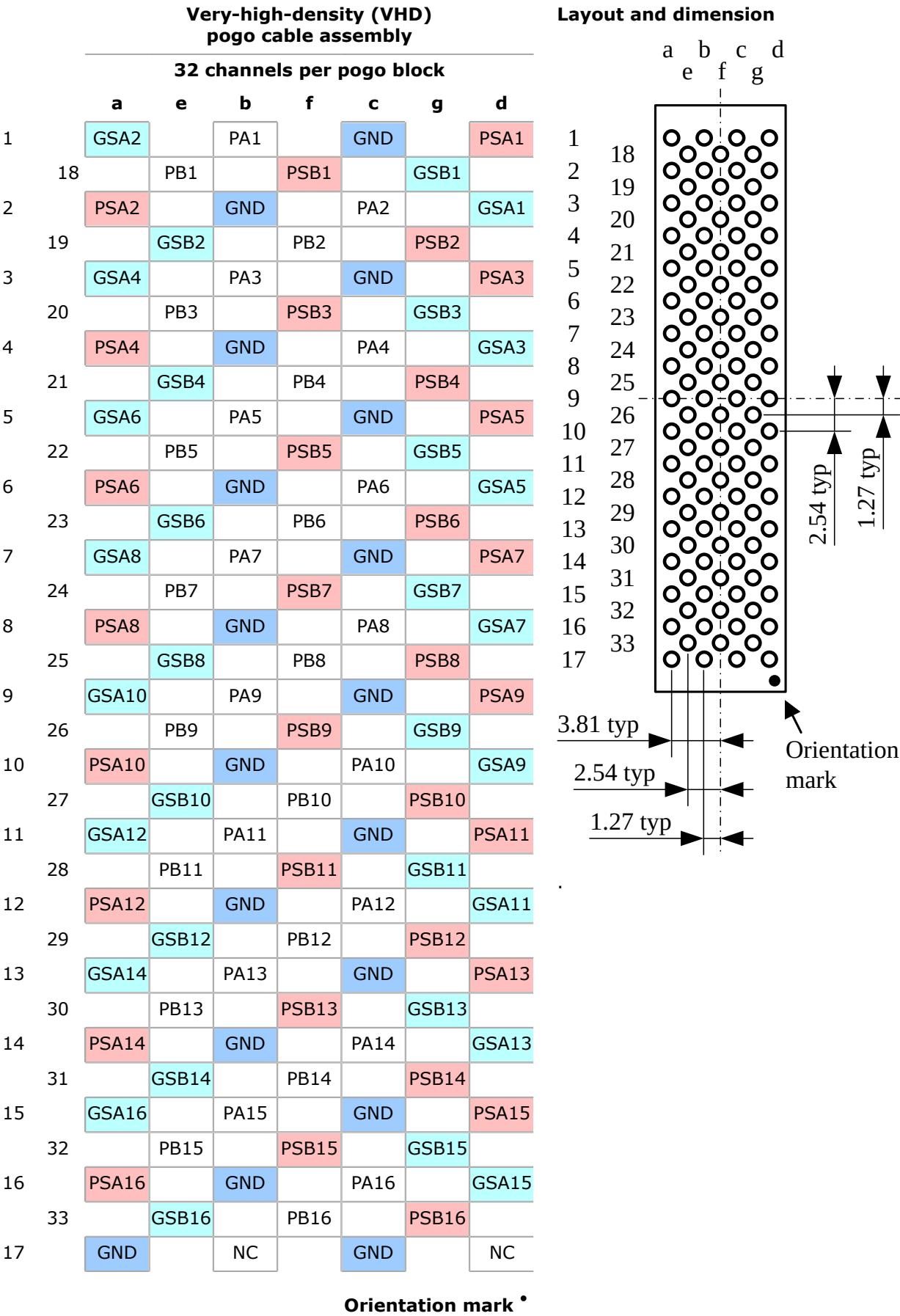
^[180] When profiling on one channel within any group of 8 channels. 22 Msamples when profiling on all 8 channels within a group of 8 channels.

XPS256 pogo block specifications

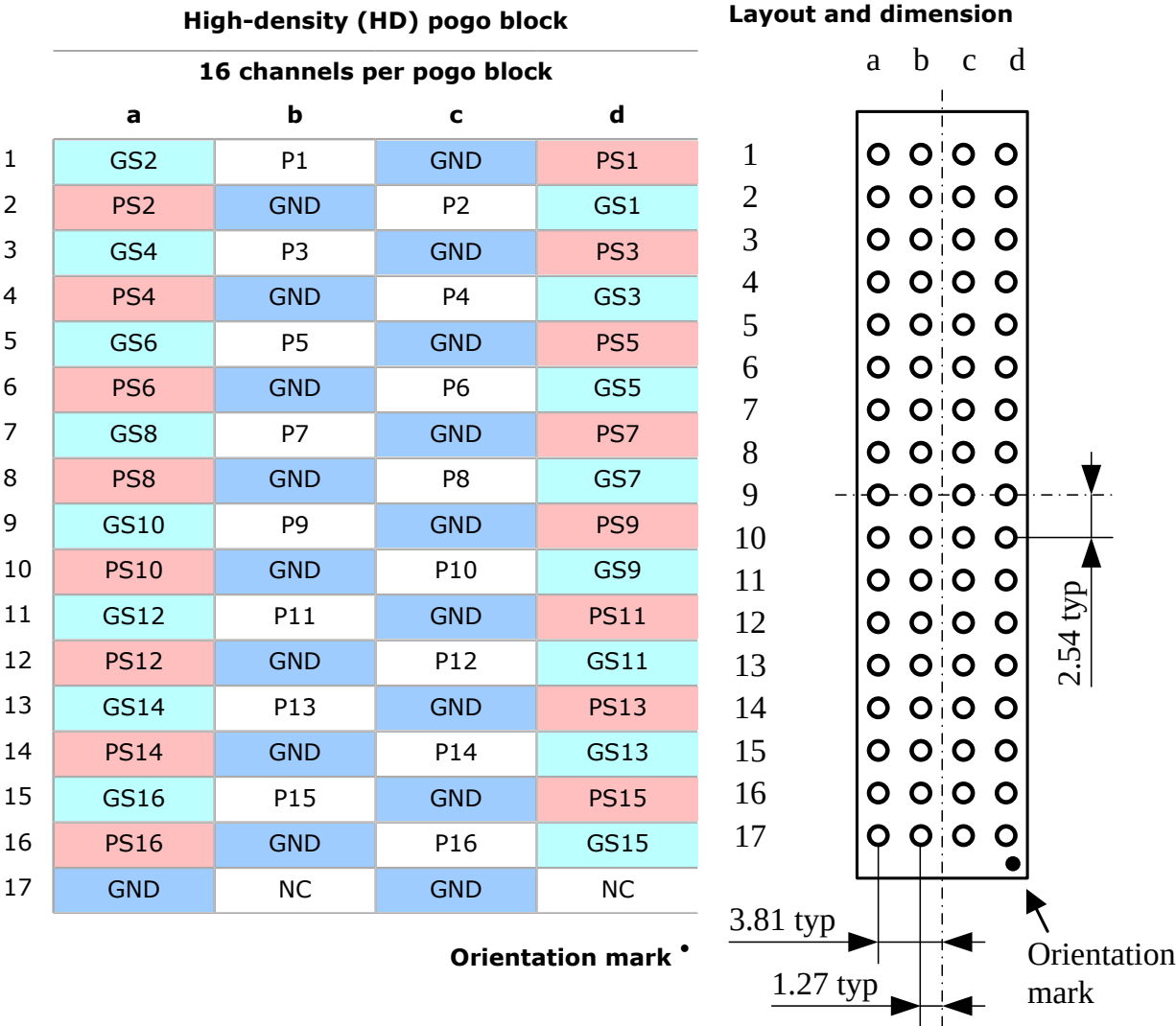
EK751HDA - High Density Pogo Cable for EK751 (16 x 16 channels)



EK751VHDA - Very High Density Pogo Cable for EK751 (8 x 32 channels)



EK751HDB - High Density Pogo Cable for EK751 (8 x 16 channels)



EK751VHDB - Very High Density Pogo Cable for EK751 (4 x 32 channels)

